

Chapter 26 — Infections of the Urinary Tract

- **Urinary tract infections (UTIs)** are common, vary widely in presentation and sequelae, and are **one of the commonest reasons for prescribing antibiotics**.
- Most cause morbidity and **rarely become life-threatening**; in certain populations, **undiagnosed and untreated UTI carries significant mortality**.
- Infection follows when **bacterial virulence rises or host defence falls** → inoculation → colonisation → infection. Careful diagnosis and treatment resolve most episodes.
- **Urine is not sterile**; bacteriuria may even be **protective against UTI**.
- With **multidrug-resistant (MDR) organisms** spreading, **clinical evaluation plus urinalysis and culture must precede antibiotics**; guidelines should govern **both agent selection and duration**; education of practitioners and patients is paramount.

DEFINITIONS

- **UTI** — an **inflammatory response of the urothelium to bacterial invasion**, usually with bacteriuria and pyuria.
- **Bacteriuria** — bacteria in the urine; usually indicates colonisation or infection, but **bacteria can sit in the urothelium without bacteriuria**, and bacteriuria may be **contamination** at collection.
- **Pyuria** — white blood cells (WBCs) in the urine; indicates infection and/or **inflammation from bacteria, stones, foreign body or other causes**.

BACTERIURIA VERSUS PYURIA

Finding	Usual meaning
Bacteriuria without pyuria	Colonisation without overt infection
Pyuria without bacteriuria (sterile pyuria)	Warrants further evaluation

- **Cystitis** — dysuria, frequency, urgency, ± suprapubic pain. The same symptoms occur with **urethral or vaginal infection** and **non-infective conditions**: interstitial cystitis/painful bladder syndrome (IC/PBS), **bladder carcinoma**, calculi.
- **Acute pyelonephritis** — chills, fever, flank pain **plus bacteriuria and pyuria**; this combination is reasonably specific for acute bacterial renal infection.
 - Diagnose on **clinical presentation, not on radiological perinephric inflammation alone**.
 - Presentation is **atypical in spinal cord injury (SCI)** — flank symptoms may be absent.
- **Chronic pyelonephritis** — **atrophic, fibrotic kidney** with morphological, radiological or functional renal disease; may be post-infective but **often not linked to current infection**.
 - Infective scar: **focal, coarse cortical scar overlying a calyx, almost always with calyceal distortion** — radiologically **identical to scarring from vesicoureteral reflux (VUR) or calyceal stone disease**.
 - Less often: **generalised cortical thinning** (atrophic pyelonephritis) → small kidney resembling **post-obstructive atrophy**.
 - **Fig. 26.1** (excretory urogram): **focal coarse scarring with blunted calyces**, one calyx reaching the capsule, in a young adult with **febrile infections and VUR in infancy** — reinfections later stopped on prophylaxis, with normal blood pressure and creatinine.
 - **Fig. 26.2**: **uniform, regular cortical atrophy** in spina bifida with neurogenic bladder and childhood febrile bacteriuria — reflux of bacteria into virtually all nephrons; this uncommon pattern is **characteristic of obstruction with superimposed infection**.

- **Xanthogranulomatous pyelonephritis (XGP)** — rare chronic form, **usually with stones**; granulomatous replacement of parenchyma → **ipsilateral loss of function**.

UNCOMPLICATED VERSUS COMPLICATED UTI

	Uncomplicated UTI	Complicated UTI
Host	Healthy, non-pregnant female	Compromised host, and/or virulent or resistant organism
Tract	Structurally and functionally normal; no obstruction	Structurally or functionally abnormal
Organism	Usually eradicated by short oral course	Frequently MDR
Typical patient	Women with isolated or recurrent cystitis	See Box 26.1

- **US Food and Drug Administration (FDA) 2018 definition of complicated UTI** — pyuria + documented pathogen in **urine or blood culture** + local/systemic features (**fever >38 °C**, chills, malaise, flank/back pain, costovertebral angle pain or tenderness) **in the presence of a functional or anatomical abnormality or catheterisation**.
- **Functional abnormalities**: renal disease impairing **urinary concentration** (tubulointerstitial: **post-obstructive**, sickle cell, lithium, chronic tubulointerstitial nephritis, medullary cystic kidney disease); bladder outlet obstruction (BOO), neurogenic bladder (NGB), detrusor underactivity.
- **Anatomical abnormalities**: benign prostatic hyperplasia (BPH); sites of residual urine (calyceal, urethral or bladder diverticula); extrinsic or intrinsic **ureteric obstruction**.
- The **European Section of Infections in Urology classification** combines clinical presentation, severity, host and pathogen risk factors — useful for practice and trial design.

BOX 26.1 FACTORS THAT SUGGEST A COMPLICATED URINARY TRACT INFECTION

- **Functional or anatomical abnormality** of the tract; **male sex**; **pregnancy**; **elderly**; **diabetes**; **immunosuppression**
- **Childhood UTI**; recent antimicrobial use; indwelling catheter; instrumentation; hospital-acquired infection
- **Symptoms >7 days at presentation**
- *Data from Schaeffer, 2002.* Pattern: every item either **impairs host defence or selects a resistant organism**. Trap: **male sex alone** makes a UTI complicated.

INFECTIONS DEFINED BY THEIR RELATIONSHIP TO OTHER UTIS

Term	Definition	Key discriminator
First / isolated	No prior UTI, or one remote UTI	—
Unresolved	Failed to respond to therapy ; same organism, similar resistance	Never cleared
Recurrent — reinfection	New event ; bacteria reintroduced	Cleared, then new inoculation
Recurrent — bacterial persistence	Same organism re-emerging from a focus in the tract (infected stone, prostate); "relapse" used interchangeably	Cleared from urine but focus remains

- These distinctions **determine the type and extent of evaluation and treatment**.
- **Antimicrobial prophylaxis (AP)** — prevention of **reinfection**; implies bacteria were **eradicated before** prophylaxis began. **Surgical AP** — before and for a limited time after a procedure (American Urological Association [AUA] 2019 statement).

- **Antimicrobial suppression** — prevents growth of a **focus of persistence that cannot be eradicated**. Contrast: **prophylaxis presumes a sterile tract; suppression presumes an ineradicable focus**.
- **Domiciliary (outpatient)** — acquired outside hospital/institution; **nosocomial (health care-associated)** — acquired while hospitalised or institutionalised.
- **Catheter-associated UTI (CAUTI)** — in patients with indwelling drainage catheters; **the commonest cause of secondary bacteraemia**; raised morbidity and mortality. Risk factors: **prolonged catheterisation, female sex, advanced age, diabetes**.

KEY POINTS: DEFINITIONS

- Bacteriuria and pyuria are **not synonymous with UTI**.
- UTIs are classified by **presumed site of origin**.
- **Uncomplicated** = healthy host, normal tract; **complicated** = factors raising acquisition and lowering treatment efficacy.

Concept — *Every definition here separates finding from disease: bacteriuria, pyuria and scarring are observations, whereas "infection", "complicated" and "persistence" are judgements about host, tract and organism — and it is the judgement, not the finding, that dictates evaluation and treatment.*

INCIDENCE AND EPIDEMIOLOGY

- UTI complaints are **among the commonest primary diagnoses in women attending emergency departments (USA)**; millions of office and emergency visits annually.
- **Almost half of all women have a UTI in their lifetime**; nearly a third have needed antibiotics for one **by age 24**; about one in ten adult women has **one UTI per year**.
- Self-reported cystitis and pyelonephritis admissions are **several-fold higher in women than men**.
- **Costs are large**, and **resistant organisms cost more to treat** than susceptible ones. A European survey of women with recurrent UTI (GESPRIT) documented repeated doctor visits, sick leave and limited-activity days each year.
- **Bacteriuria rises with institutionalisation, hospitalisation and comorbidity**. Roughly one in ten hospitalised urological patients worldwide develops a complicated UTI during admission.
- **CAUTIs are the commonest nosocomial infection**, constituting **about 80% of nosocomial UTIs**; they account for a large share of complicated UTIs, most being hospital or health care-acquired.
- Incidence is also raised in **pregnancy, SCI, diabetes mellitus (DM), multiple sclerosis (MS), human immunodeficiency virus (HIV)/acquired immunodeficiency syndrome (AIDS)** and **organ transplant recipients**.
- **Probability of recurrence rises with the number of previous infections and falls with the interval since the last infection**; established historically and confirmed in modern cohorts.
 - Highest in **young, sexually active women**: about a quarter have a second infection within 6 months of a first.
 - Women with **≥2 previous UTIs** carry **two to five times** the 1-year recurrence risk of those with 0–1.
 - **Postmenopausal women** with prior UTI are similarly at risk; even with daily trimethoprim-sulfamethoxazole (TMP-SMX) or lactobacilli prophylaxis most recur within a year.
 - Recurrent bacteriuria segregates into a **small subgroup with very high recurrence** and a majority with low recurrence.
- **About 70% of recurrences are reinfection with a different organism**.

- **No association** between **uncomplicated recurrent UTI** and **renal scarring, hypertension or progressive azotaemia** has been established.

KEY POINTS: INCIDENCE AND EPIDEMIOLOGY

- UTI is the **commonest bacterial infection**, with a large health care cost.
- Bacteriuria rises with **institutionalisation/hospitalisation** and comorbidities that **immunosuppress or alter lower tract function** (e.g. catheterisation).
- Probability of recurrent UTI (rUTI) **rises with the number of previous infections**.

Concept — *The best predictor of the next UTI is the last one: recurrence clusters in a susceptible minority and is mostly reinfection rather than relapse, which is why the chapter later locates the problem in the host's introital and bladder biology rather than in a hidden focus.*

PATHOGENESIS

- UTI results from **uropathogen–host interaction: virulence factors, inoculum size and inadequacy of host defence** determine infection and the extent of colonisation and damage.
- **Governing asymmetry: strong host → only highly virulent bacteria infect; compromised host → bacteria with minimal virulence suffice.**

Routes of Infection

Ascending Route

- **Most bacteria enter from the bowel and skin reservoir via the urethra into the bladder; adherence to introital and urothelial mucosa is central.**
- Risk factors: **faecal soiling of the perineum, spermicide use, intermittent or indwelling catheters.**
- **Most pyelonephritis is retrograde ascent** bladder → ureter → pelvis → parenchyma.
 - Reflux probably is **not required**; **cystitis-related oedema** may distort the vesicoureteric junction enough to permit reflux (Fig. 26.3).
 - Ascent is aided by anything that **impairs ureteric peristalsis: Gram-negative bacteria and endotoxins, pregnancy, ureteric obstruction, high lower-tract pressure** (all antiperistaltic).
 - Entry to parenchyma is via **collecting ducts at the papillary tips** → ascent in collecting tubules; hastened by **raised intrapelvic pressure** (obstruction, VUR), especially with **intrarenal reflux**.

FIG. 26.3 BACTERIAL ADHERENCE IN ASCENDING UTI (MODIFIED FROM GLOVER ET AL., 2014)

- Sequence: **periurethral contamination by bowel-colonising uropathogenic *E. coli* → type 1-fimbriated *E. coli* selected in bladder** (high colony-forming units [CFU], low oxygen), invasion and intracellular multiplication → ascent → **P fimbriae bind renal tubular cells → haemolysin damages epithelium, cytokine induction → crosses tubular epithelium → bacteraemia.**

Haematogenous Route

- **Uncommon in normal individuals**; kidney occasionally seeded by ***Staphylococcus aureus* bacteraemia** (e.g. oral source) or ***Candida* fungaemia**. **Obstruction enhances** haematogenous infection experimentally.

Lymphatic Route

- Direct lymphatic extension from adjacent organs (**severe bowel infection, retroperitoneal abscess**) — **rare**; little role in most UTIs.

ROUTES OF INFECTION COMPARED

Route	Frequency	Typical organism/setting
Ascending	Commonest	Bowel/skin flora; <i>E. coli</i>
Haematogenous	Uncommon	<i>S. aureus</i> , <i>Candida</i> ; obstructed kidney
Lymphatic	Rare	Severe bowel infection, retroperitoneal abscess

Urinary Pathogens

- Most UTIs are due to facultative anaerobes from bowel flora; *Staphylococcus epidermidis* and *Candida albicans* come from vaginal or perineal skin flora.
- *Escherichia coli* is by far the commonest cause — about 85% of community-acquired and 50% of hospital-acquired UTIs.
- Remaining community infections: other **Enterobacteriaceae** (*Proteus*, *Klebsiella*) and Gram-positive *Enterococcus faecalis* and *Staphylococcus saprophyticus*.
- *S. saprophyticus* causes about 10% of symptomatic lower UTIs in young, sexually active women; rare in men and the elderly (age shapes the organism).
- **Nosocomial organisms:** *E. coli*, *Klebsiella*, *Enterobacter*, *Citrobacter*, *Serratia*, *Pseudomonas aeruginosa*, *Providencia*, *E. faecalis*, *S. epidermidis*.
- **Less common:** *Gardnerella vaginalis*, *Mycoplasma*, *Ureaplasma urealyticum* (some in catheterised patients). *U. urealyticum* and *Mycoplasma hominis* found in women with **chronic symptoms and negative standard cultures** — **targeted treatment improved symptoms**.
- **Emerging uropathogens** (identified by better culture and microbiome work):

EMERGING UROPATHOGENS

Organism	Features
<i>Aerococcus urinae</i>	Mistaken for <i>Staphylococcus</i> / <i>Streptococcus</i> / <i>Enterococcus</i> ; older, comorbid adults, some urological malignancy
<i>Raoultella planticola</i>	Gram-negative rod; immunocompromised
<i>Myroides odoratimimus</i>	Gram-negative environmental (soil) rod; immunocompromised; extensive drug resistance

- *E. coli* phylogeny: extraintestinal strains grouped by multiplex polymerase chain reaction (PCR); **most uropathogenic *E. coli* (UPEC) fall in the B2 group**.
- **Sequence type ST131 (serotype O25b:H4)** — rapidly emerging cause of **MDR infection**; first noted for **extended-spectrum β -lactamases (ESBLs)**; **fluoroquinolone resistance is its hallmark phenotype**.

FIG. 26.4 RELEVANT BACTERIA FOR UROLOGICAL INFECTIONS (FROM GRABE ET AL., EUROPEAN ASSOCIATION OF UROLOGY [EAU] GUIDELINES 2015; ANAEROBES NOT CONSIDERED)

Group	Members
Obligate intracellular	<i>Chlamydia trachomatis</i>
No cell wall	<i>Mycoplasma hominis</i> , <i>M. genitalium</i> ; <i>Ureaplasma urealyticum</i>
Spirochaetes	<i>Treponema pallidum</i>
Gram-negative rods — Enterobacteriaceae	<i>Escherichia</i> , <i>Klebsiella</i> , <i>Citrobacter</i> , <i>Proteus</i> , <i>Serratia</i> , <i>Providencia</i> , <i>Enterobacter</i> , <i>Pantoea</i> , <i>Hafnia</i> ($\pm$ <i>Salmonella</i> , <i>Shigella</i>)
Gram-negative rods — non-fermenters	<i>Pseudomonas</i> , <i>Acinetobacter</i> , <i>Xanthomonas</i> , <i>Burkholderia</i>
Gram-negative cocci / coccobacilli	<i>Neisseria gonorrhoeae</i> ; <i>Haemophilus</i> , <i>Gardnerella vaginalis</i>

Group	Members
Gram-positive cocci — streptococci	α -haemolytic (<i>S. viridans</i>); β -haemolytic (<i>S. pyogenes</i> group A, <i>S. agalactiae</i> group B); non-haemolytic → <i>Enterococcus faecalis</i> , <i>E. faecium</i>
Gram-positive cocci — staphylococci	<i>S. aureus</i> ; <i>S. epidermidis</i> group; <i>S. saprophyticus</i> group
Gram-positive rods	<i>Corynebacterium urealyticum</i> ; <i>Listeria</i> , <i>Bacilli</i>
Acid-fast (Ziehl–Neelsen positive)	<i>Mycobacterium tuberculosis</i>

- Purpose: a map of **which taxonomic group each uropathogen sits in**, which predicts **stain, nitrite production and antibiotic class**. Trap: *Enterococcus* is a non-haemolytic streptococcus-group Gram-positive coccus, not an Enterobacteriaceae.

FIG. 26.5 PREVALENCE OF UROPATHOGENIC SPECIES IN UTI ENTITIES (WAGENLEHNER ET AL., 2020)

- *E. coli* dominates every entity but its share falls from uncomplicated cystitis (about three-quarters) to complicated cystitis, pyelonephritis and urosepsis (under half).
- In complicated entities the slack is taken by *Klebsiella*, *P. aeruginosa*, *Enterobacter*, *Proteus*, *Acinetobacter*, *Citrobacter*, *Enterococcus*, *S. aureus*, coagulase-negative staphylococci (CoNS) and fungi.
- *S. saprophyticus* appears only in uncomplicated cystitis.

Fastidious Organisms

Anaerobes in the Urinary Tract

- Symptomatic anaerobic UTI is uncommon, though distal urethra, perineum and vagina are colonised; anaerobes are a frequent midstream contaminant but rare in suprapubic aspirates.
- Suspect them when persistent irritative symptoms, cocci or Gram-negative rods on microscopy of centrifuged urine, and routine aerobic culture is sterile.
- Common in enteric fistulae and suppurative genitourinary (GU) infection — the large majority of scrotal, prostatic and perinephric abscesses in men contain anaerobes: *Bacteroides* (incl. *B. fragilis*), *Fusobacterium*, anaerobic cocci, *Clostridium perfringens*.

Mycobacterium tuberculosis and Other Nontuberculous Mycobacteria

- Urinary tuberculosis (UTB) presents with haematuria (gross or micro) and/or sterile pyuria, storage symptoms, dysuria, $\pm$ fever and weakness.
- Diagnosis is delayed by nonspecific symptoms → risk of renal impairment and failure.
- PCR for *M. tuberculosis* has replaced acid-fast staining as the ideal diagnostic method.

Chlamydia

- Not grown in routine aerobic culture; implicated in GU infection.

Bacterial Virulence Factors

- UPEC resides in bowel flora and infects not by chance but by virulence factors that allow adherence to perineum and urethra, migration, and an inflammatory response in the urothelium; the same factors appear in rUTI strains.
- Many determinants lie on about 20 UPEC-specific pathogenicity-associated islands (PAIs), which enlarge the genome >20% over commensal strains. Phylogroups are defined by PAIs and their virulence factors (adhesins, toxins, surface polysaccharides, flagella, iron acquisition).

Early Events in Uropathogenic *Escherichia coli* Pathogenesis

- **Sequence:** colonisation of **periurethral/vaginal tissue and urethra** → **ascent into bladder** → **adherence to surface urothelium** (engaging epithelial defence) → **biofilm** → **invasion and replication as intracellular bacterial communities (IBCs)**, which can form **quiescent intracellular reservoirs (QIRs)** dormant in underlying urothelium → in some, **renal colonisation, tissue damage and sepsis**.

FIG. 26.6 UPEC MOLECULAR PATHOGENESIS (MODIFIED FROM CROXEN AND FINLAY, 2010)

- **Type 1 pili** bind **uropilins Ia and IIIa** → signalling (partly unknown) → **invasion and apoptosis**.
- Type 1 pili binding **$\alpha 3 \beta 1$ integrins** → internalisation into **superficial facet (umbrella) cells** → IBCs ("pods").
- **Exfoliation** exposes deeper transitional cells → further invasion → **QIRs implicated in recurrence**. In the kidney, **haemolysin (HlyA)** and **secreted autotransporter toxin (Sat)** cause tissue damage with neutrophil influx.

Bacterial Adherence

- **Adherence to vaginal and urothelial cells is an essential first step;** governed by **bacterial adhesins, epithelial receptivity and the fluid bathing both**; it helps determine **organism, host and site**.
- **Adhesins** are **fimbrial (pili)** or **afimbrial**. A piliated cell carries **100–400 pili**, each **5–10 nm wide, up to 2 μ m long**, built from **pilin** subunits. Pili are defined by **haemagglutination of specific erythrocytes**.

TYPE 1 VERSUS P PILI

	Type 1 (mannose-sensitive)	P (mannose-resistant)
Structure / gene	Helical rod of FimA with adhesin FimH ; fim operon — constitutive	pap operon , located on a PAI with other virulence factors
Haemagglutination	Guinea pig erythrocytes; inhibited by mannose (MSHA)	Not inhibited by mannose
Distribution	Pathogenic and non-pathogenic <i>E. coli</i>	Most UPEC causing upper tract infection
Receptor	Mannosylated uropilins on bladder umbrella cells	Glycolipid receptors in kidney
Tropism	Bladder colonisation (cystitis)	Kidney — "P" = pyelonephritis

- **Adherence correlates with severity:** strains from **acute pyelonephritis** are **highly adhesive**; those from **asymptomatic bacteriuria (ASB)** or **healthy faeces** are **poorly adhesive**.
- **P pili frequency falls stepwise:** **pyelonephritis (most, about 90%) > cystitis > ASB > faecal isolates** (the classic graded correlation).

Phase Variation of Bacterial Pili in Vivo

- **Phase variation** — rapid switching between **piliated and non-piliated phases** with environment (e.g. broth vs solid medium); **occurs in vivo** (immunofluorescence of fresh urine shows mixed populations; eFig. 26.7, online only). Strains from different urogenital sites differ in piliation.
- **Sequence of advantage:** **type 1 pili help initial bladder adherence** → **become a liability in suspension** (enhance **apoptosis, phagocytosis and neutrophil killing**) → **P pili take over in the kidney** via glycolipid receptors.

Epithelial Cell Receptivity

Vaginal Cells

- ***E. coli* adheres more avidly to vaginal cells of women with recurrent cystitis** — the first demonstrable **biological difference in UTI-susceptible women** (Stamey's work).
- The increased adherence is **also present on buccal cells** → receptor differences are **not vagina-specific** and may be **genetic**.

- Certain **Lewis blood-group phenotypes** correlate with rUTI in women.
- Conclusion: **increased epithelial receptivity (introital, urethral, buccal) characterises susceptible women and may be inherited.**

Bladder Cells

- Bladder lumen lined by **umbrella cells** whose apical surface is a **quasi-crystalline array of hexagonal uroplakin complexes** (four integral membrane proteins).
- **FimH binds mannosylated uroplakins; UPIa and UPIb** bind type 1-piliated UPEC; pilus tips sit in the **central cavity of the uroplakin hexamers**, and contact triggers **local actin rearrangement and membrane protrusion** around the bacterium (Fig. 26.8). **FimH binding is the initial step of the cascade.**
- FimH–UPIb binding → **phosphorylation of UPIII** → **raised intracellular calcium** (immediate urothelial response).
- **Host elimination mechanisms:** (1) **urothelial exfoliation**, (2) **exocytosis** from infected cells, (3) **autophagy**.
- **UPEC persistence in the bladder:** rapid internalisation into superficial cells (a niche protected from innate immunity) → **early IBC:** rods about 3 μm , loose random clusters → at **6–8 h** growth slows (doubling **>60 min**), rods shorten to about 0.7 μm , switch to a **biofilm-like community**.
 - IBC-like cells occur in about a fifth of voided urines from women with *E. coli* UTI, and such isolates recreate IBCs in mice.
- **Biofilms** raise antibiotic resistance by (1) **poor antibiotic penetration of matrix**, (2) **slow growth and altered physiology**, (3) **expression of factors that inhibit antibiotic activity**; they also **exclude neutrophils**.
- **Fluxing** — edge bacteria detach, resume rod form, become **motile** and escape into the lumen; may become **filamentous** ($\geq 70 \mu\text{m}$), by about 24 h; filaments may **evade immunity**. **Flagellar motility** is common in UPEC and aids adherence and dispersal.
- Escaped bacteria **reinvade** → **second and further IBC rounds** → after a few days, **quiescence**. Exfoliation exposes deeper layers → **QIRs in underlying transitional epithelium**; **urothelial disruption can reactivate them** → **rUTI from an intracellular bladder reservoir**.

FIG. 26.9 UPEC MODEL FOR ACUTE CYSTITIS, CHRONIC CYSTITIS AND QIR FORMATION (MODIFIED FROM HANNAN ET AL., 2012)

Phase	Step	Timing
Acute cystitis (IBC phase)	A Attachment (type 1 pili)	Immediate
	B Invasion	15 min–1 h
	C Intracellular replication	1–2 h
	D IBC formation	4–16 h
	E Fluxing and filamentation	16–24 h
Subacute–chronic (post-IBC phase)	F QIR formation	from 2 weeks
	G Chronic cystitis	from about 3–14 days

- Exfoliation liberates IBCs → **clonal recreation of IBCs**. QIRs are **metabolically inactive and provoke no measurable inflammation** — hence invisible between recurrences.

Natural Defences of the Urinary Tract

Periurethral and Urethral Region

- Normal flora of **introitus, periurethra and urethra** — **lactobacilli, CoNS, corynebacteria, streptococci** — forms a **barrier to uropathogen colonisation**.

- Colonisation shifts with **oestrogen and pH**, but acute change is most often due to **antimicrobials and spermicides** that **alter flora and increase epithelial receptivity**.
- Colonisation of periurethra/urethra is a **prerequisite for most infections**, but overcoming **urine and bladder defences** is pivotal; urethral defence relies on **regular urine flow** and **indigenous flora**.

Urine

- Antibacterial when the **inoculum is small**. Main inhibitory factors: **osmolality, urea, organic acids, pH**. Growth is deterred by **very dilute urine** or **high osmolality with low pH**; clinically these **do not separate susceptible from resistant patients**.
- **Uromodulin (Tamm-Horsfall protein)** — the **most abundant urinary protein**, made in the kidney; competes with uroplakins for **type 1 pili**, blocking adhesion.

Bladder

- Bacteria probably enter the bladder frequently; whether they establish infection depends partly on **the bladder's ability to empty**. Bacterial recognition **up-regulates antimicrobial peptides** from urothelium and resident immune cells.

Other Conditions That Increase Risk of Urinary Tract Infections

Obstruction

- **Obstruction at any level** (anatomical or high storage pressure) is a **key host susceptibility factor**: stasis impairs bladder and renal defences, **promotes bacterial growth and adherence**.
- Experimentally **the kidney resists haematogenous infection unless the ureter is ligated** — then **only the obstructed kidney is infected**. Obstruction converts cystitis or pyelonephritis into a **life-threatening infection**.

Vesicoureteral Reflux

- Association of **VUR + UTI** with **calyceal blunting and parenchymal scarring** first described by **Hodson and Edwards (1960)**.
- **Children** with high-grade VUR and UTI → **progressive scarring, proteinuria, renal failure**.
- **Adults** — VUR **does not reduce renal function** unless combined with **high storage pressure or stasis and concurrent UTI**.

VUR: CHILDREN VERSUS ADULTS

VUR in children	VUR in adults
High-grade VUR + UTI → progressive scarring, proteinuria, renal failure	No functional loss unless high pressure/stasis + UTI

Underlying Disease

- **High incidence of renal scarring** with causes of **chronic interstitial nephritis**, almost all of which **damage the renal papilla first**: **DM, sickle cell disorders, adult nephrocalcinosis, hyperphosphataemia, hypokalaemia, analgesic abuse, sulfonamide nephropathy, gout, heavy-metal poisoning, ageing**.

Diabetes Mellitus

- **Increased rUTI** from impaired local defence: **glycosuria, neutrophil dysfunction, increased uroepithelial adherence**; plus **incomplete emptying from diabetic neuropathy** and **chronic kidney disease** reducing anti-infective efficacy.
- **More severe infections**: **acute pyelonephritis, renal abscess, emphysematous pyelonephritis**; atypical organisms (yeast); sequelae include **papillary necrosis and perinephric abscess**.

Renal Papillary Necrosis

- **Role of infection is controversial.** Predisposing: **DM, analgesic abuse, sickle cell haemoglobinopathy, obstruction** (Box 26.2).
- **Spectrum:** acute fulminating illness ↔ chronic, incidental. Some **pass necrotic tissue** chronically; some never do. **Retained papillae may calcify** (especially with infection) and act as a **nidus for chronic infection**; **opportunistic fungal infection** reported.
- Management: **antimicrobials to control infection + early recognition and relief of ureteric obstruction by sloughed papillae** limit functional decline.

BOX 26.2 CONDITIONS ASSOCIATED WITH RENAL PAPILLARY NECROSIS

- **Diabetes mellitus; pyelonephritis; urinary tract obstruction; analgesic abuse; sickle cell haemoglobinopathies; renal transplant rejection; cirrhosis; dehydration, hypoxia and jaundice of infants.**
- Miscellaneous: **renal vein thrombosis, cryoglobulinaemia, renal candidiasis, contrast injection, amyloidosis, calyceal arteritis, necrotising angiitis, rapidly progressive glomerulonephritis, hypotensive shock, acute pancreatitis.**
- *Data from Eknayan et al., 1982.* Pattern: nearly all **compromise medullary blood flow or oxygenation of the papilla.** The classic mnemonic group is **DM, obstruction/pyelonephritis, analgesics, sickle cell.**

Human Immunodeficiency Virus

- **UTIs about fivefold commoner** in HIV-positive individuals; more *Acinetobacter* and *Salmonella*.
- **AIDS (AIDS-defining illness and/or CD4 <200 cells/mm³) → atypical infection:** fungal (*Candida*, *Aspergillus*, *Cryptococcus neoformans*, endemic fungi such as *Histoplasma*), mycobacterial (tuberculosis [TB] and non-tuberculous), viral (cytomegalovirus, adenovirus).
- **CD4 <200 → Pneumocystis pneumonia prophylaxis, typically TMP-SMX, which also lowers UTI risk** (unless sulfa allergy or glucose-6-phosphate dehydrogenase [G6PD] deficiency).

Renal Transplantation

- **UTIs are among the commonest infections after renal transplant** (more than a third of recipients have at least one) and **may worsen graft outcome.**
- **More resistant organisms and recurrent symptomatic episodes** (urinary tract colonisation).
- Risk factors: **age, female sex, long pre-transplant dialysis, urinary tract obstruction.**
- **Acute pyelonephritis of the graft is an independent risk factor for functional decline.**

Menopause

- **UTI frequency rises after menopause** (mechanism and treatment later).

Pregnancy

- See Bacteriuria in Pregnancy.

Spinal Cord Injury With High-Pressure Bladders

- **No bacteriuric group compares in severity and morbidity with SCI.** Nearly all need early catheterisation (spinal shock, poor contractility); long term → **poor compliance, high storage pressure → VUR, ureterectasis, hydronephrosis, renal impairment.**

KEY POINTS: PATHOGENESIS

- Most UTIs in women are **ascending**, from **bowel and skin flora.**
- **Virulence and host factors** determine which bacteria invade and how far.
- **Increased epithelial receptivity** predisposes to rUTI and may be **genetic.**

- **Obstruction and immunocompromise** are key host susceptibility factors.

Concept — *Pathogenesis is a **contest between adhesion and clearance**: every bacterial virulence factor (pili, biofilm, intracellular hiding) is a way of resisting being washed out or shed, and every host risk factor (obstruction, stasis, residual urine, altered flora) is a failure of washing out or shedding — which is why obstruction, not organism, most often makes an infection dangerous.*

EVALUATION

- Requires **thorough history** (urinary symptoms + infection risk factors); **examination necessary but often unrevealing**; diagnosis **depends on a properly collected urine sample**; extra tests guided by history.

Signs and Symptoms

PRESENTING SYNDROMES

Syndrome	Features
Cystitis	Dysuria, frequency, urgency ± suprapubic pain, haematuria, fever
Pyelonephritis	Fever, chills, flank pain, costovertebral angle tenderness (CVAT) ± nausea, vomiting, malaise; urinary symptoms may be absent
Fistula	Pneumaturia or faecaluria → vesicoenteric or vesicovaginal fistula
Renal/perirenal abscess	Indolent fever, flank mass and tenderness
Indwelling catheter	Usually ASB; but fever with bacteraemia can arise rapidly and become life-threatening

- **Elderly**: diagnosis uniquely difficult (see Elderly section).
- History must cover the items in Box 26.3.

BOX 26.3 IMPORTANT INFORMATION IN EVALUATION OF URINARY TRACT INFECTIONS

- Recent infections/antibiotic use; recent hospitalisation; comorbidities; history of paediatric voiding dysfunction; sexual and reproductive history; anatomical urological abnormalities; prior surgery of GU tract, reproductive organs or spine; family history; current medications.

Differential diagnosis

- **Voiding dysfunction not due to bacteria will not improve with antibiotics** — e.g. IC/PBS mimics UTI and patients receive repeated antibiotic courses despite negative cultures.

CYSTITIS VERSUS VAGINITIS VERSUS URETHRITIS

	Cystitis	Vaginitis	Urethritis
Onset	Acute	Subacute	Subacute
History	—	Vaginal discharge/odour; new or multiple partners	Discharge; new or multiple partners
Frequency/urgency	Prominent	Absent (also no haematuria, no suprapubic pain)	Present but less pronounced ; no fever or chills
Examination	—	Vaginal discharge; inflammatory cells in vaginal fluid	Urethral discharge with inflammatory cells, or initial-stream pyuria in men
Causes	<i>E. coli</i> etc.	HSV, gonorrhoea, <i>Chlamydia</i> , trichomoniasis, yeast, bacterial vaginosis	<i>N. gonorrhoeae</i> , <i>Chlamydia</i> , herpes simplex virus (HSV), trichomoniasis

- **Urethral injury** (intercourse, chemical irritants, allergy) causes dysuria with **trauma/irritant history and no discharge or pyuria**.

- **Rule: painless gross haematuria, or microhaematuria with a negative culture, always → upper and lower tract evaluation to exclude urological malignancy.**

Physical examination (abdomen and pelvis)

- **Suprapubic distension** → raised **postvoid residual (PVR)**; CVAT if pyelonephritis suspected.
- Women: **vaginal discharge**; palpate urethra for **urethral diverticulum** or **Skene's gland infection** (cystic structure, expressible fluid, tenderness); **atrophic epithelium** = **hypo-oestrogenic state** (a UTI risk factor); **pelvic organ prolapse (POP)** → incomplete emptying.
- Men: **digital rectal examination (DRE)** of prostate.
- **Focused perineal neurological examination** (sensation).

Diagnosis by Urine Testing

Urine Collection

- **Contamination reduction** improves accuracy; meatal cleansing lacks proven benefit but is **universally endorsed**.
- **Men**: cleanse glans (**retract foreskin** if uncircumcised) with **2% castile soap towelette** → **midstream sample in sterile cup**.
- **Women**: **spread labia** → cleanse **periurethral area** → **midstream sample**. **Vaginal epithelial cells and lactobacilli** = contaminated sample.

BOX 26.4 FACTORS THAT AFFECT ABILITY TO PROVIDE ADEQUATE MIDSTREAM CLEAN-CATCH SAMPLE

- **Increased body mass index (BMI)**; **vaginal atrophy**; **poor manual dexterity**; **inability to bear weight**; **intravaginal pessary**; **non-sterile collection receptacle**.
- With a **pessary**, a **catheterised sample** is preferable.
- **Bedpan or non-sterile cup samples should be disregarded** — yet patients are often treated on them regardless of symptoms.
- **Indwelling Foley catheter**: **remove it and obtain the sample from a newly, sterilely placed catheter** — otherwise you culture **catheter/bag colonisers**.
- **Stoma**: **never sample the appliance**; **catheterise (intubate) the stoma**.
- **Mid-catheter specimen** — more accurate than voided; use when **voided sample contaminated (many squamous cells)**, **pessary**, **cannot give clean catch**, or **cannot void**. **Iatrogenic infection risk is low**; in patients with equivocal symptoms it **roughly halves antibiotic prescribing** that a positive voided sample would have triggered.
- **Suprapubic aspiration** — **highly accurate**, samples bladder urine **without urethral bacteria**; some morbidity, so limited to those **unable to void on command (e.g. SCI)** — **even there, catheterised samples are preferred**.

URINE COLLECTION METHODS

Method	Accuracy	Use
Midstream clean catch	Contamination common in women	Default
Mid-catheter	More accurate	Contaminated voided sample, pessary, poor technique, cannot void, equivocal symptoms

Method	Accuracy	Use
Suprapubic aspirate	Most accurate	Rarely; cannot void on command

• **Key issues in analysis:**

- **Leucocyte esterase** — from WBC breakdown → indicates **pyuria, not bacteria**; common with **vaginal contamination**.
- **Nitrite** — bacterial **nitrate reductase** converts dietary nitrate. **All Enterobacteriaceae produce nitrite** (*E. coli*, *Klebsiella*, *Enterobacter*, *Proteus*, *Citrobacter*, *Morganella*, *Salmonella*). **Non-producers: all Gram-positives and Pseudomonas/Acinetobacter**. **Low-nitrate diet → false negative**.
- **Nitrite: high specificity, low sensitivity**. Esterase and nitrite are **more reliable at $>10^5$ CFU/mL**.
- **Cloudy urine** = pyuria or **phosphate crystals** in alkaline/concentrated urine.
- Analyse promptly; standing urine **alkalinises**, bacteria multiply, red cells lyse → **refrigerate if not analysed within a few hours** (unless preservative). **Hydration dilutes** constituents.

• **Urine dipsticks: most useful to rule out UTI**; each parameter has false positives and negatives.

- In strongly suggestive symptoms a positive dipstick supports starting antibiotics; **positive nitrite is the single variable most likely to rule in UTI** (with symptoms).
- **Best combination: positive nitrite, or positive leucocyte esterase plus blood**.
- Borderline sensitivity → **cannot replace microscopy** in symptomatic patients.
- Errors: **expired or badly stored strips**; **nitrite pad is air-sensitive** (open container → future false positives); **leucocyte reaction takes up to 5 min** (read early → false negative).

• **Urinalysis: automated analysis detects haematuria and bacteria more sensitively than dipstick**.

- Preferred tubes: **non-mercuric preservative** (chlorhexidine, ethylparaben, sodium propionate) → sample stable **up to 72 h without refrigeration**.
- Technique: **6–8 mL → centrifuge 1800–2000 rpm for 4 min** → microscopy for red blood cells (RBCs), WBCs, squamous cells, bacteria, crystals, casts. **Squamous cells = quality marker**.
- **No single constituent is pathognomonic**; interpret together. **Pyuria = at least 5–10 leucocytes per high-power field (HPF)**.
- Culture takes **48–72 h**, so urinalysis gives the early answer.
- **Strongest predictors: moderate bacteriuria, moderate pyuria and nitrite**. **Moderate pyuria (WBC >50) with typical symptoms justifies antibiotics**.
- **Other causes of pyuria: GU tuberculosis, urolithiasis, urothelial injury (incl. chlamydial urethritis), interstitial nephritis**.
- **No pyuria → question the diagnosis** until culture returns; **culture positive without pyuria → consider a catheterised sample**.
- **False-positive bacteria on microscopy** with negative culture — usually **vaginal lactobacilli and corynebacteria**.
- **A negative automated urinalysis reliably rules out UTI** (can obviate culture); **ruling in is harder** — abnormal parameters occur in **healthy asymptomatic women**, so urinalysis alone **overdiagnoses**.

• **Urine culture: the gold standard for bacteriuria**, supporting UTI in a **symptomatic** patient (technique from the 1950s). Collect midstream or catheterised into a **vacutainer with freeze-dried preservative** (practically eliminates false positives) → **blood agar and MacConkey plates**.

CULTURE THRESHOLDS FOR SIGNIFICANT BACTERIURIA

Threshold	Origin / use
$\geq 10^5$ CFU/mL	Kass (1956) — originally to confirm pyelonephritis , later extended to cystitis
$\geq 10^2$ CFU/mL of a known pathogen	Appropriate threshold in women with dysuria ; a fifth to two-fifths of symptomatic women have 10^2 – 10^4 CFU/mL

- **Why low counts in cystitis:** bacterial **doubling time in urine 30–45 min** versus **voiding every 15–30 min** from irritation — bacteria are **washed out before reaching 10^5** .
- Reports: **negative, commensal flora, or positive**. Commensal flora = CoNS, α - and non-haemolytic streptococci, diphtheroids, non-pathogenic *Neisseria*, yeast.
- **Susceptibility:** S (sensitive), I (indeterminate/intermediate), R (resistant), followed by **minimum inhibitory concentration (MIC)** — the **lowest concentration inhibiting growth**.

SUSCEPTIBILITY CATEGORIES

Category	Meaning
S	Inhibited by serum concentration achieved with the recommended dose
I	May still work where the drug is concentrated (urine) or at a higher dose
R	Resistant to standard achievable serum levels

- **Efficacy depends on urinary drug level and time above MIC**; resolution tracks **susceptibility to the urinary concentration**. Urine levels are often **several hundred times serum levels**; inhibitory urine levels are reached orally by **all common agents except macrolides (e.g. erythromycin)**.
- Choose on **site of infection and fluid (urine vs blood)** and **route**, not MIC alone — **the lowest MIC is not always the best drug**; route matters especially with **outpatient MDR organisms**.

FIG. 26.10 MIC REPORT FOR *E. coli* UTI

- Example report ($>10^5$ CFU/mL *E. coli*, clean catch): **susceptible to most agents; resistant to ampicillin, tetracycline and TMP-SMX; intermediate to ampicillin-sulbactam**. The legend groups ampicillin-sulbactam with the resistant agents.
- Reading point: an **"I" result is not "R"** — urinary concentration may still suffice.
- **Which organism matters:** Gram-negative rods in a symptomatic patient → treat (consensus). Gram-positives in women often represent periurethral contamination.
 - In symptomatic premenopausal women, *E. coli* in a midstream sample **strongly predicts bladder bacteriuria even at low counts**; enterococci and group B streptococci predict nothing at any count, and catheterised samples from such women usually grow *E. coli*. → Gram-positives rarely cause acute cystitis in isolation.
- **Symptoms + suggestive urinalysis + negative standard culture** → culture for atypical organisms (*U. urealyticum*, *M. hominis*).
- **Next-generation sequencing (NGS)** and **expanded quantitative urine culture (EQUC)** detect far more organisms, challenging **colony-count thresholds** and the dismissal of **polymicrobial growth as contamination**; some argue standard culture is **no longer the gold standard**. Clinical judgement is paramount to prevent overtreatment.

KEY POINTS: EVALUATION

- Urine results **cannot be interpreted without signs and symptoms**.
- Collect urine to **minimise contamination**.
- **Formal urinalysis is preferred over dipstick**.

- Culture provides **bacterial sensitivities**.

Concept — *Urine tests are better at excluding than confirming UTI: a negative urinalysis or dipstick is reassuring, but positives are diluted by contamination, colonisation and non-infective pyuria — so the quality of the sample and the presence of symptoms decide what a positive result means.*

LOCALISATION

Ureteral Catheterisation

- Used in **bacterial persistence** to locate the source: separates **upper vs lower tract, one kidney vs the other**, and can localise to **ectopic ureters or non-refluxing ureteric stumps** (saline irrigation) — **Stamey's technique**.
- Particularly important with **fever, flank pain or CVAT**, and in **renal transplant recipients** to decide whether **native kidneys harbour infection**.

Stone Cultures

- **Culture removed stones** to identify organisms and sensitivities in their interstices; analyse **separately from urine** — **results may be discordant**.
- **UTI is the commonest complication after stone procedures**; manipulating infected stones can release endotoxin → **systemic inflammatory response syndrome (SIRS) or fatal urosepsis**.
- **Stone culture predicts postoperative sepsis and SIRS better than periprocedural urine culture**; a systematic review supports it for **selecting antibiotics to prevent future UTI**, though its role in preventing postoperative UTI needs further study.

Concept — *When bladder urine and the infecting focus can differ, **sample the focus**: ureteric urine localises the side, and the stone — not the urine — tells you what will cause postoperative sepsis.*

IMAGING

- **Not required in most UTIs**; clinical and laboratory findings suffice.
- **Image** (to find abnormalities needing changed medical management or percutaneous/surgical intervention): **most men, compromised hosts, febrile infections, features of obstruction, failure to respond, recurrence pattern suggesting persistence** (Box 26.5).
- **UTI with possible obstruction must be evaluated urgently** (causes in Box 26.5).
- **DM** → **emphysematous pyelonephritis, papillary necrosis** (impacted papilla → acute obstruction). **Polycystic kidneys on dialysis** → prone to **perinephric abscess**.
- **Image if pyelonephritis persists after several days of appropriate antibiotics** (renal/perinephric abscess), and with **unusual organisms, especially urea-splitters (*Proteus*)** → look for **obstructing stones, strictures, fungal balls**.
- Second purpose: **find a focus of persistence** when bacteriuria **fails to resolve or recurs rapidly** — uncommon but often **surgically correctable** (Box 26.6).

BOX 26.5 INDICATIONS FOR RADIOLOGIC INVESTIGATION IN ACUTE CLINICAL PYELONEPHRITIS

- **Potential ureteric obstruction** (stone, stricture, tumour); **history of calculi, especially struvite**
- **Potential papillary necrosis** (sickle cell, severe DM, analgesic abuse)
- **GU surgery predisposing to obstruction** (reimplantation, diversion)
- **Poor response after 5–6 days** of appropriate antimicrobials

- **DM; polycystic kidneys on dialysis or with severe renal insufficiency; neuropathic bladder**
- **Unusual organisms: TB, fungi, urea-splitters (*Proteus*)**

BOX 26.6 CORRECTABLE UROLOGIC ABNORMALITIES THAT CAUSE BACTERIAL PERSISTENCE

- **Infection stones; chronic bacterial prostatitis; unilateral infected atrophic kidney; ureteric duplication and ectopic ureters; foreign bodies**
- **Urethral diverticula and infected periurethral glands; unilateral medullary sponge kidney; non-refluxing, normal-appearing infected ureteric stumps after nephrectomy**
- **Infected urachal cysts; infected communicating calyceal cysts; papillary necrosis; perivesical abscess with fistula to bladder**
- **Pattern: each is a nidus that antibiotics cannot sterilise, so each is cured only by removal or drainage.**

Radionuclide Studies

- **Rarely used now. Iodine-131 hippuran, technetium-99m (^{99m}Tc) glucoheptonate and dimercaptosuccinic acid (DMSA) show focal parenchymal damage, impaired function and reduced perfusion in acute infection.**
- **Gallium-67** — useful in pyelonephritis/abscess but **positive in non-infective disease too. Indium-111 WBC** — limited when the presentation is not suggestive of infection.

Ultrasonography

- **Non-invasive, quick, no radiation or contrast; good for calculi, hydronephrosis, pyonephrosis, perirenal abscess, PVR; can be paired with a plain radiograph for stones.**
- **Limits: operator-dependent; poor with obesity, dressings, drains, wounds; less sensitive than computed tomography (CT) and magnetic resonance imaging (MRI) for parenchymal change in pyelonephritis; inferior to CT for small renal and most ureteric stones and for ureteric obstruction.**

Computed Tomography and Magnetic Resonance Imaging

- **Best anatomical detail; more sensitive than excretory urography or ultrasound for acute focal bacterial nephritis (AFBN), renal/perirenal abscess and radiolucent calculi.**
- **Non-contrast CT = best test for urinary calculi. Contrast CT localises abscesses and guides (percutaneous) drainage.**
- **MRI has not replaced CT for renal inflammation but better shows extrarenal extension. Pelvic MRI with intravenous (IV) contrast = best test for urethral diverticulum.**

Voiding Cystourethrogram

- **Voiding cystourethrogram (VCUG) assesses VUR — perform in women with febrile UTIs, childhood VUR, or recurrent adult pyelonephritis; consider with rUTI plus hydronephrosis.**
- **Double-balloon VCUG for urethral diverticulum is historical; pelvic MRI is now standard.**

IMAGING MODALITY BY CLINICAL QUESTION

Question	Best test
Stones	Non-contrast CT
Abscess localisation / drainage	Contrast-enhanced CT
Extrarenal extension	MRI
Urethral diverticulum	Pelvic MRI with contrast (not double-balloon VCUG)
VUR / non-refluxing stump suspicion	VCUG

Question	Best test
Bedside: hydronephrosis, pyonephrosis, PVR	Ultrasound

Concept — *Imaging answers only two questions in UTI — is there obstruction or abscess needing drainage, and is there a focus sustaining persistence* — so it is indicated by features that raise either possibility (male sex, fever, non-response, rapid recurrence, unusual organisms), not by the infection itself.

MICROBIOME

- **Urine is not sterile.** 16S ribosomal RNA NGS shows many species in the urine of **healthy, asymptomatic** people; standard culture detects only **fast-growing, oxygen-tolerant** organisms and misses fastidious and anaerobic ones.
- **EQUC** — **larger urine volume, varied atmospheres, longer incubation** — grew bacteria from most samples that were **negative on standard culture**, identifying dozens of genera and species (Table 26.1) → confirms a **female urinary microbiome** of mostly **non-pathogenic** organisms.
- Implications: **bacteria will never be eradicated from urine, nor should that be the goal; many microbiome bacteria are protective.** Research asks whether **dysbiosis** underlies **urgency incontinence or painful bladder syndrome**.
- **Commercial NGS** (not FDA-approved): **no susceptibility testing, costly, risks inappropriate treatment, and reports go directly to patients** without clinician interpretation → confusion; **clinical utility still to be established.**

TABLE 26.1 LIST OF BACTERIAL SPECIES CULTURED IN DIFFERENT CONDITIONS (HILT ET AL., 2014)

Culture conditions tested	Aerobic 35 °C; aerobic 30 °C; CO ₂ 35 °C; anaerobic 35 °C; Campylobacter atmosphere (6% O ₂ , 10% CO ₂) 35 °C
Genera most often isolated	<i>Lactobacillus</i> (<i>gasseri</i> , <i>jensenii</i> , <i>crispatus</i> , <i>iners</i>), <i>Streptococcus</i> (esp. <i>anginosus</i>), <i>Corynebacterium</i> , <i>Actinomyces</i> , <i>Gardnerella</i> , <i>Staphylococcus</i> (<i>epidermidis</i> and others), <i>Micrococcus</i> , <i>Aerococcus</i> (<i>urinae</i> , <i>sanguinicola</i>), <i>Bifidobacterium</i>
Classic uropathogens also found	<i>E. coli</i> , <i>E. faecalis</i> , <i>K. pneumoniae</i> , <i>Enterobacter aerogenes</i> , <i>P. aeruginosa</i> , <i>Streptococcus agalactiae</i>
Fastidious / anaerobic organisms	<i>Actinobaculum schaalii</i> , <i>Alloscardovia</i> , <i>Facklamia</i> , <i>Finegoldia magna</i> , <i>Fusobacterium nucleatum</i> , <i>Peptoniphilus</i> , <i>Prevotella bivia</i> , <i>Propionimicrobium</i> , <i>Oligella</i> , <i>Trueperella</i> , <i>Campylobacter ureolyticus</i> ; <i>Candida glabrata</i> ; several unclassified isolates

- Purpose: to show that **many organisms need non-standard atmospheres or low-dilution plating**, so "**no growth**" on standard culture ≠ **sterile urine**. Pattern: **lactobacilli and other commensals dominate**. Trap: the presence of **Aerococcus, Gardnerella** or even **E. coli** in an asymptomatic woman does not, by itself, mean infection.

KEY POINTS: MICROBIOME

- **Urine is not sterile.**
- The **female urinary microbiome**, identified by sequencing, should inform UTI management and interpretation of **EQUC**.

Concept — *The microbiome turns the question from "are bacteria present?" to "is this organism causing harm?"* — the more sensitive the detection, the more the decision to treat must rest on symptoms rather than on the laboratory report.

ASYMPTOMATIC BACTERIURIA

- The microbiome renders the term ASB essentially obsolete, but it remains in use: **no signs or symptoms of UTI, yet bacteria in a non-contaminated sample.**

BOX 26.7 DIAGNOSIS OF ASYMPTOMATIC BACTERIURIA (INFECTIOUS DISEASES SOCIETY OF AMERICA [IDSA]: NICOLLE ET AL., 2005; NICOLLE, 2019)

Group	Specimen	Threshold
Asymptomatic women	Two consecutive voided specimens, same strain	$\geq 10^5$ CFU/mL
Asymptomatic men	Single voided specimen, one species	$\geq 10^5$ CFU/mL
Either sex	Single catheterised specimen, one species	$\geq 10^2$ CFU/mL

- Prerequisites: **no signs or symptoms**; specimen **collected to minimise contamination**.
- Trap: women need **two** voided samples (contamination risk), men only **one**; a catheter sample needs only **100 CFU/mL**.
- **Prevalence rises with age** — **<1% of schoolgirls** → **about a fifth of healthy community women over 80**; tracks **sexual activity** (married women far above age-matched nuns); rises after **menopause**; uncommon in young men, increasing with age.

TABLE 26.2 PREVALENCE OF ASYMPTOMATIC BACTERIURIA IN SELECTED POPULATIONS (DATA FROM NICOLLE ET AL., 2005)

Population	Prevalence (%)
Healthy premenopausal women	1–5
Pregnant women	1.9–9.5
Postmenopausal women 50–70 years	2.8–8.6
Diabetes — women / men	9–27 / 0.7–11
Elderly in community — women / men	10.8–16 / 3.6–19
Elderly in long-term care — women / men	25–50 / 14–50
SCI — intermittent catheterisation	23–89
SCI — sphincterotomy + condom catheter	57
Haemodialysis	28
Indwelling catheter — short term	9–23
Indwelling catheter — long term	100

- Purpose: shows **risk factors for ASB** — **advanced age, female sex, institutionalisation, DM, clean intermittent catheterisation (CIC) or indwelling catheter**. Pattern: prevalence climbs with **instrumentation and institutionalisation**, peaking at **100% with long-term catheters**. Trap: **high prevalence is exactly why these groups should not be treated** — only pregnancy, with a modest prevalence, is screened.
- The text gives **6–15%** for men over 75 in the community and **15–40%** for men in long-term care; the table gives 3.6–19% and 14–50% — **different source studies** for overlapping populations.
- **DM**: ASB more common in **women than men**; associated with **duration of disease and poor control**.
- **Organisms**: community elderly women — ***E. coli* predominates** (others: *K. pneumoniae*, *P. mirabilis*, *E. faecalis*). **ASB *E. coli* carry fewer virulence factors**; some are true commensals or **virulent strains turned commensal**. Men — Gram-negative bacilli, enterococci, CoNS. **Institutionalised/catheterised** → **polymicrobial**, with *P. aeruginosa*, *Morganella morganii*, *Providencia stuartii*.

- **Untreated ASB does not cause rUTI, hypertension, renal insufficiency or urinary tract cancer. Treating ASB is potentially harmful → more subsequent symptomatic UTIs, cost, adverse effects.**

GUIDELINE BODIES ON TREATING ASB

Body	Year	Position
IDSA (AUA-endorsed)	2005	Do not treat most adults; treat only as in Box 26.8; no recommendation for transplant/immunocompromised
IDSA update	2019	Reaffirms; extends “do not treat” to children, transplant recipients and non-urological surgery (Box 26.8)
US Preventive Services Task Force (USPSTF)	2008, reaffirmed 2019	Screen only pregnant women
Cochrane review (randomised/quasi-randomised trials; excluded pregnancy, catheters, stents, transplant, SCI, recent procedures, inpatients)	2015	Treatment eradicates bacteriuria more often but no difference in symptomatic UTI, complications or death; more adverse effects; no renal decline untreated

BOX 26.8 TREAT OR NOT TREAT ASYMPTOMATIC BACTERIURIA (IDSA 2005; 2019)

Do NOT treat	Treat
Premenopausal non-pregnant women; women with DM; older community dwellers; elderly institutionalised; SCI; indwelling catheters; pyuria with ASB ; infants and children; DM including men ; renal transplant >1 month ; non-renal solid organ transplant; non-urological surgery	Pregnancy; procedures with anticipated transmucosal bleeding
No consensus: neutropenia	

- **Landmark trial — Cai et al. (young women with rUTI and ASB, observe vs treat):** treating ASB increased symptomatic UTI over 1 year; **antibiotic use was an independent risk factor (hazard ratio [HR] about 3)**. Most untreated women who stayed well carried **asymptomatic *E. faecalis*** → **treatment removes a protective strain**. At about 3 years, recurrence was **roughly 38% untreated vs 70% treated**, with **more resistant *E. coli*** (amoxicillin-clavulanate, TMP-SMX, ciprofloxacin) in treated women **after ≥2 years**.
- **Barriers to guideline uptake:** lack of awareness, confusion, **fear of not prescribing**; **about half of practitioners across specialties are unaware of IDSA guidance** on ASB and cystitis.
- **Bacterial interference** — deliberately establishing **avirulent bacteriuria** to block virulent organisms; **efficacy not yet established**; merits study.
- **ASB before non-urological (joint) surgery:** in arthroplasty, **ASB was associated with more prosthetic infection**, but **preoperative treatment did not change it** and the **joint and urine organisms differed** → **ASB is a surrogate marker of infection risk, not the source. Do not treat ASB before orthopaedic surgery.** Dropping routine preoperative urine culture **did not change joint infection rates and saved money**.

KEY POINTS: ASYMPTOMATIC BACTERIURIA

- Prevalence varies with **age, sex and comorbidity**.
- Untreated ASB is **not associated with hypertension or renal insufficiency**.
- Guidelines advise **not screening or treating** except in specific groups.
- **Screen and treat in pregnancy and before urological procedures with anticipated mucosal bleeding.**

- **Treating ASB breeds MDR symptomatic UTI.**

Concept — *ASB is treated only where the bacteria are about to be given a route into blood or tissue — pregnancy (hydroureter and pyelonephritis risk) and mucosa-breaching urological procedures; everywhere else the resident strain is harmless or protective, and eradicating it selects resistance and invites a virulent replacement.*

PRINCIPLES OF ANTIMICROBIAL THERAPY

- Agent and duration depend on **spectrum against the known or probable pathogen** (by source of acquisition), **uncomplicated vs complicated**, **adverse effects**, **cost**.
- **Serum levels are irrelevant in uncomplicated UTI** (urine levels matter) but **critical in bacteraemia and febrile parenchymal infection** (kidney, prostate).

WHICH DRUG LEVEL MATTERS

Uncomplicated cystitis	Febrile / parenchymal infection
Urinary concentration matters	Serum and tissue concentration matter

- **Renal insufficiency:** adjust dose of **renally cleared** drugs; the failing kidney **may not concentrate the drug in urine** → eradication difficult. **Obstruction also lowers urinary drug levels.**
- Consider the drug's effect on **bowel and vaginal flora and the hospital ecology**; susceptibility varies by exposure and setting.
- **Do not treat a positive culture alone;** weigh the **likelihood of benefit vs harm**. **Uncomplicated cystitis rarely progresses even untreated — the goal is symptom relief.** In placebo arms of trials, a substantial minority of women were **symptom-free or improved within about 9 days**, and **progression to pyelonephritis was rare**.
- Finucane's framing: ask not whether it is a "real UTI" but **whether antibiotics directed at bacteriuria are more likely to benefit than harm**.
- **Non-steroidal anti-inflammatory drugs (NSAIDs) alone vs antibiotics** (meta-analysis of randomised trials, non-pregnant women): **antibiotics are more effective** (moderate evidence); NSAIDs cure a proportion but with **higher symptom burden, longer symptoms and more pyelonephritis**. Role of analgesics **combined** with antibiotics remains to be defined.

Ecological Impact and Collateral Damage

- **Collateral damage** = ecological adverse effects — **resistance, selection and MDR colonisation** — linked especially to **broad-spectrum cephalosporins and quinolones**.
- Particularly relevant to **uncomplicated cystitis** because (1) **minimal risk of invasion/sepsis** and (2) it is **one of the commonest antibiotic exposures in healthy people**.
- Resistance rises from **microbial traits, antibiotic selection pressure and spread; varies geographically**.

TYPES OF BACTERIAL RESISTANCE

Resistance type	Mechanism	Clinical relevance
Intrinsic chromosomal	Species lacks the drug's target/mechanism	Predictable from species
Acquired chromosomal	Resistant mutants selected during therapy; repeat culture at 24–48 h shows high counts of resistant mutants	Occurs when urine level $\approx$ or $<$ MIC: underdosing, non-compliance → choose drug with urine level $\gg$ MIC, dose fully, stress compliance

Resistance type	Mechanism	Clinical relevance
Extrachromosomal (plasmid)	Horizontal gene transfer , mostly plasmids ; arises in bowel flora ; far commoner than urinary mutant selection	All classes can induce it ; transferable within and across genera — e.g. tetracycline exposure → bowel strains resistant to tetracycline, ampicillin, sulfonamides and TMP → post-antibiotic UTIs often MDR

SPECIFIC B-LACTAMASE AND CARBAPENEMASE MECHANISMS

Mechanism	Encoded	Confers resistance to	Notes
ESBL (first reported 1983)	Plasmid	Most β-lactams incl. 3rd-generation cephalosporins	ESBL <i>E. coli</i> often co-resistant to fluoroquinolones and aminoglycosides
AmpC β-lactamase	Chromosomal in many Enterobacteriaceae; now also transmissible plasmids (e.g. into <i>E. coli</i>)	Broad-spectrum cephalosporins	—
<i>K. pneumoniae</i> carbapenemase (KPC) (first 2001)	—	Carbapenems (highly drug-resistant Gram-negatives)	Prevalent US mechanism; mostly <i>K. pneumoniae</i> but spread to other Gram-negative bacilli

- **Resistance scales with duration and quantity of use** (e.g. hospital fluoroquinolone use → *Pseudomonas* quinolone resistance); **reducing use can restore susceptibility**.
- **Fluoroquinolone overuse** → rising quinolone resistance across Europe, **higher still in ESBL isolates in institutions**, and **highest in transplant units using quinolone prophylaxis**. Prior quinolone use and **underlying urological disease** are the strongest determinants. **Fluoroquinolone resistance is a harbinger of MDR** and of resistance to first-line agents.
- **Commensal disruption**: gut → *Clostridioides difficile* and **vancomycin-resistant colonisation**, and susceptibility to enteric infection; vagina → **candidiasis**, related to **duration and breadth** of antibiotic use and **prior *Candida* episodes**. → **Narrow the spectrum and shorten the course, especially in uncomplicated UTI**.

Antimicrobial Formulary

- **Bactericidal** = irreversible killing; **bacteriostatic** = reversible inhibition of replication.

TABLE 26.3 BACTERIOSTATIC VERSUS BACTERICIDAL AGENTS

Bacteriostatic	Bactericidal
Chloramphenicol	Aminoglycosides
Clindamycin	Quinolones
Macrolides (azithromycin, erythromycin)	β-lactams
Sulfonamides	Vancomycin
Tetracycline	
Trimethoprim	

- **Nitrofurantoin is generally bacteriostatic but bactericidal at high doses/against some organisms**. Pattern: **cell-wall, DNA and aminoglycoside ribosomal agents kill**; other protein-synthesis and folate inhibitors only inhibit. Trap: aminoglycosides act on the ribosome yet are bactericidal; TMP and sulfonamides singly are static.

TABLE 26.4 MECHANISM OF ACTION OF COMMON ANTIMICROBIALS USED IN THE TREATMENT OF UTI

Drug/class	Mechanism of action	Mechanisms of resistance
β-lactams (penicillins, cephalosporins, aztreonam)	Cell wall synthesis inhibition	β-lactamase ; altered penicillin-binding protein ; porin change (reduced entry)
Aminoglycosides	Ribosomal protein synthesis inhibition	Reduced uptake ; aminoglycoside-modifying enzymes
Quinolones	DNA gyrase inhibition	Gyrase binding-site mutation ; porin change; active efflux
Fosfomycin	Cell wall synthesis inhibition	Amino acid substitutions or loss of transporters
Nitrofurantoin	Inhibits several bacterial enzyme systems	Not fully elucidated; develops slowly with prolonged exposure
TMP-SMX	Folate metabolism antagonism	Enterococci draw folate from the environment
Vancomycin	Cell wall synthesis inhibition (at a different point from β-lactams)	Enzymatic alteration of peptidoglycan target

- Pattern: **three different classes act on the cell wall** (β-lactams, fosfomycin, vancomycin) at different steps — hence **fosfomycin's limited cross-resistance**. Trap: **enterococci evade TMP-SMX by scavenging folate** even when tested "sensitive".

TABLE 26.5 RELIABLE COVERAGE OF ANTIMICROBIALS USED IN THE TREATMENT OF UTI OF COMMONLY ENCOUNTERED PATHOGENS

Agent/class	Gram-positive	Gram-negative
Amoxicillin/ampicillin	Streptococcus, enterococci	<i>P. mirabilis</i> only
Amoxicillin-clavulanate	Streptococcus, enterococci	<i>P. mirabilis</i> , <i>Klebsiella</i>
Ampicillin-sulbactam	Staphylococcus (not MRSA), enterococci	<i>P. mirabilis</i> , <i>H. influenzae</i> , <i>Klebsiella</i>
Antistaphylococcal penicillins	Streptococcus, Staphylococcus (not MRSA)	None
Antipseudomonal penicillins	Streptococcus, enterococci	Most, incl. <i>P. aeruginosa</i>
1st-gen cephalosporins	Streptococcus, Staphylococcus (not MRSA)	<i>E. coli</i> , <i>P. mirabilis</i> , <i>Klebsiella</i>
2nd-gen (cefamandole, cefuroxime, cefaclor)	Streptococcus, Staphylococcus (not MRSA)	<i>E. coli</i> , <i>P. mirabilis</i> , <i>H. influenzae</i> , <i>Klebsiella</i>
2nd-gen (cefoxitin, cefotetan)	Streptococcus	<i>E. coli</i> , <i>Proteus</i> incl. indole-positive, <i>H. influenzae</i> , <i>Klebsiella</i>
3rd-gen — ceftriaxone	Streptococcus, Staphylococcus (not MRSA)	Most, excluding <i>P. aeruginosa</i>
3rd-gen — ceftazidime	Streptococcus	Most, including <i>P. aeruginosa</i>
Aztreonam	None	Most, incl. <i>P. aeruginosa</i>
Aminoglycosides	Staphylococcus (urine)	Most, incl. <i>P. aeruginosa</i>
Fluoroquinolones	Streptococcus (agent-dependent)	Most, incl. <i>P. aeruginosa</i>
Nitrofurantoin	Staphylococcus (not MRSA), enterococci	Many Enterobacteriaceae (not <i>Providencia</i>, <i>Serratia</i>, <i>Acinetobacter</i>), <i>Klebsiella</i>
Fosfomycin	Enterococci	Most Enterobacteriaceae (not <i>P. aeruginosa</i>)
Pivmecillinam	None	Most, excluding <i>P. aeruginosa</i>

Agent/class	Gram-positive	Gram-negative
TMP-SMX	Streptococcus, Staphylococcus	Most Enterobacteriaceae (not <i>P. aeruginosa</i>)
Vancomycin	All, incl. methicillin-resistant <i>S. aureus</i> (MRSA)	None

- Pattern: antipseudomonal activity is confined to antipseudomonal penicillins, ceftazidime, aztreonam, aminoglycosides and fluoroquinolones. Enterococcal cover comes from aminopenicillins, antipseudomonal penicillins, nitrofurantoin and fosfomycin — never cephalosporins. Traps: ceftriaxone misses *Pseudomonas* but ceftazidime covers it; aztreonam and pivmecillinam have no Gram-positive cover; vancomycin has no Gram-negative cover.

TABLE 26.6 COMMON ADVERSE REACTIONS, PRECAUTIONS AND CONTRAINDICATIONS FOR ANTIMICROBIAL AGENTS USED IN TREATMENT OF UTI (EDITED FROM AMERICAN HOSPITAL FORMULARY SERVICE, 1995; CARBAPENEM DATA FROM LEE AND BRADLEY, 2019)

Drug/class	Common adverse reactions	Precautions and contraindications
Amoxicillin/ampicillin	Hypersensitivity (immediate or delayed); diarrhoea (esp. ampicillin); antimicrobial-associated pseudomembranous colitis (AAPMC); non-allergic maculopapular rash; reduced platelet aggregation	Rash more likely with concurrent viral illness or allopurinol
Amoxicillin-clavulanate	More diarrhoea/gastrointestinal (GI) upset than amoxicillin	—
Ampicillin-sulbactam	As amoxicillin	—
Antistaphylococcal penicillins	As amoxicillin; GI upset (oral); acute interstitial nephritis (esp. methicillin)	—
Antipseudomonal penicillins	As amoxicillin; hypernatraemia (sodium salts: carbenicillin, ticarcillin); injection-site reactions	Caution with sodium loading
Cephalosporins	Hypersensitivity (less than penicillins); GI upset; injection-site reactions; AAPMC; positive Coombs test; reduced platelet aggregation (cefotetan, cefamandole, cefoperazone)	Do not use after immediate penicillin hypersensitivity; caution after delayed reactions
Carbapenems	Injection-site reactions, diarrhoea, nausea, vomiting, rash, pruritus	Cross-sensitivity with penicillin <1% (positive skin test); with cephalosporins limited data but low
Aztreonam	Hypersensitivity (less than penicillins)	<1% cross-reactivity with penicillin/cephalosporin allergy; use with caution
Aminoglycosides	Ototoxicity (vestibular and auditory); nephrotoxicity (non-oliguric azotaemia); neuromuscular blockade at high levels	Avoid in pregnancy except pyelonephritis; avoid in severe renal impairment, DM, hepatic failure; caution in myasthenia gravis and with other oto/nephrotoxins
Fluoroquinolones	Mild GI; dizziness; photosensitivity; CNS effects (tremor, confusion, mood disorder, hallucinations); tendon rupture	Avoid in children and pregnancy (arthropathy); antacids, iron, zinc, sucralfate block absorption — separate by ≥2 h; hydrate; raise theophylline (ciprofloxacin, enoxacin > norfloxacin, ofloxacin); lower seizure threshold; dysglycaemia with antidiabetics; potentiate warfarin
Fosfomycin	Headache, GI upset, vaginitis	Hypersensitivity to formulation

Drug/class	Common adverse reactions	Precautions and contraindications
Pivmecillinam	Rash, GI upset	Caution in penicillin hypersensitivity
Nitrofurantoin	GI upset; peripheral polyneuropathy (renal impairment, anaemia, DM, electrolyte imbalance, vitamin B deficiency, debility); haemolysis in G6PD deficiency ; pulmonary hypersensitivity (acute to chronic: cough, dyspnoea, fever, interstitial change)	Table: avoid if creatinine clearance (CrCl) <50 mL/min (inadequate urine levels — see text for the revised threshold); monitor long-term use; avoid probenecid (blocks excretion); avoid magnesium or quinolones (antagonistic)
TMP-SMX	Hypersensitivity, rash; GI upset; photosensitivity; haematological toxicity (AIDS)	All reactions commoner in AIDS and elderly ; avoid in pregnancy ; avoid with warfarin (raises prothrombin time)
Vancomycin	" Red man syndrome " (flushing, fever, chills, rash, hypotension — histamine); oto/nephrotoxicity with other such drugs ; injection-site reactions	Caution with other oto/nephrotoxins

- Pattern: **β -lactam toxicity is mostly allergic and colitic**; aminoglycoside and vancomycin toxicity is oto/nephrotoxic; fluoroquinolone toxicity is musculoskeletal, neurological, metabolic and interaction-driven. Traps: TMP-SMX and fluoroquinolones both potentiate warfarin; nitrofurantoin's renal cut-off differs between this table (50 mL/min) and the 2015 Beers revision (30 mL/min).

Nitrofurantoin

- Active against most common uropathogens; **rapidly excreted in urine, but no therapeutic tissue levels (including gut) → useless for upper tract, complicated or bloodstream infection.**
- **Minimal effect on bowel and vaginal flora; acquired resistance exceedingly low.**
- Adverse: **GI upset**; rare **pulmonary fibrosis with chronic use**; haemolytic anaemia in G6PD deficiency.
- **Renal threshold** (table below): caution in significant impairment, especially **long courses** — in a large cohort CrCl 30–50 did not reduce efficacy, but CrCl <50 predicted pulmonary adverse events needing admission.

NITROFURANTOIN RENAL CUT-OFFS BY SOURCE

Source	Nitrofurantoin avoid below
Historical recommendation	CrCl <60 mL/min
American Geriatrics Society Beers Criteria (2015)	CrCl <30 mL/min
Table 26.6 (older formulary data)	CrCl <50 mL/min

- **Beers Criteria strongly advise against long-term suppressive use, but not against short courses** in selected elderly with normal renal function. The authors keep it **first-line for acute uncomplicated cystitis in selected elderly**, though the blanket long-term warning creates **insurance barriers** even to short courses.

Trimethoprim–Sulfamethoxazole

- Widely used; **therapeutic levels in most tissues**; covers most uropathogens **except *Enterococcus* and *Pseudomonas***; cheap; minimal bowel flora effect.
- Adverse: common **rash and GI upset**; rare **Stevens–Johnson syndrome and toxic epidermal necrolysis** — TMP-SMX is the antibiotic most often implicated.

Fosfomycin

- **Bactericidal**, oral and IV, phosphonic acid-like; active against most uropathogens.

- Advantages: **limited cross-resistance**, activity against **most Gram-negatives and vancomycin-resistant *Enterococcus* (VRE)**, **single-dose efficacy** in empirical uncomplicated cystitis → **excellent oral option for MDR organisms that would otherwise need IV therapy**.
- **Well tolerated** (low GI upset, headache). **Traditionally not for urosepsis**, though emerging data suggest a role in **bacteraemic MDR *E. coli* UTI**. **Cost** (3 g sachet) may limit use.

Fluoroquinolones

- Descend from **nalidixic acid**; **inhibit DNA gyrase**; **broad spectrum** — ***Enterobacteriaceae* and *P. aeruginosa***, high activity against ***S. aureus* and *S. saprophyticus***, **marginal streptococcal cover**; **most anaerobes resistant** → **vaginal and bowel flora spared**.
- Resistance has **risen with indiscriminate use**.
- **Not nephrotoxic**, but renal insufficiency prolongs half-life → **adjust dose at CrCl <30 mL/min**.
- Adverse: **GI (commonest; commensal disruption)**, hypersensitivity, skin reactions, **serious central and peripheral nervous system effects**, acute renal failure; **anxiety, depression, hallucinations, confusion, suicidal thoughts**.
- **Achilles tendinopathy/rupture** — class effect; **stop at the first sign of tendon pain**. Risk: **advanced age, chronic renal failure, male sex, normal BMI, corticosteroids**. Mechanism: ciprofloxacin **stimulates fibroblast matrix-degrading proteases** and **inhibits matrix synthesis**. Also paraesthesiae, myalgia, arthralgia.
- **Cartilage damage in immature animals** → **contraindicated in children, adolescents, pregnancy and breastfeeding**.
- Interactions (warfarin — a World Health Organization [WHO] warning — antacids, theophylline): see Table 26.6.

FDA FLUOROQUINOLONE WARNINGS — TIMELINE

Year	Warning
2008	Boxed warning: tendon rupture
2013	Irreversible peripheral nerve damage added
2015	Fluoroquinolone-associated disability (FQAD) recognised (chronic pain commonest)
2016	Boxed warning revised: reserve for patients with no other options for uncomplicated UTI — risk outweighs benefit
2018	Mental health effects (agitation, disorientation, delirium, memory, attention) and dysglycaemia
2018 (later)	Aortic dissection

- The **European Commission** has imposed similarly stringent restrictions. Mechanism of serious toxicity (**mitochondrial damage** proposed) is unproven.

Cephalosporins

- **Five generations**; as a group **good against *Enterobacteriaceae***, **poor against enterococci**.
- **1st** — more Gram-positive activity plus *E. coli*, *K. pneumoniae*; **2nd** — **anaerobic activity**; **3rd** — **most reliable β -lactams against community and nosocomial Gram-negatives**; **4th/5th** — usable in complicated UTI.
- **β -lactams are for uncomplicated cystitis only when first-line agents cannot be used**.

Carbapenems

- β -lactams, broad spectrum; bind and inactivate **penicillin-binding proteins**; **stable to most β -lactamases including AmpC and ESBLs**; cover many Gram-positive, Gram-negative and anaerobic organisms.

- **Gaps: *E. faecium* and MRSA.** Ertapenem additionally lacks *P. aeruginosa* and *Enterococcus* cover.

Aminoglycosides

- **With TMP-SMX or ampicillin**, effective for febrile UTI, especially in hospital; nephro- and ototoxicity → monitor renal and auditory function.
- **Once-daily dosing** exploits concentration-dependent killing (peak:MIC), time-dependent toxicity and a prolonged post-antibiotic effect.
- **Regimen: fixed 5–7 mg/kg gentamicin or tobramycin**; interval from one serum level plotted on a nomogram (Fig. 26.12) — e.g. 7 mg/L at 10 h → redose 7 mg/kg every 36 h. Effective, less nephrotoxic, cheaper (fewer levels and staff time).

FIGS. 26.11–26.12 ONCE-DAILY AMINOGLYCOSIDE DOSING (NICOLAU ET AL., 1995)

- **Fig. 26.11:** once-daily 7 mg/kg/24 h gives a **high peak then a long trough**; conventional 1.5 mg/kg/8 h gives **low, repeated peaks** — once-daily maximises peak:MIC and minimises time at toxic trough levels.
- **Fig. 26.12 nomogram** (gentamicin/tobramycin 7 mg/kg): the level drawn **6–14 h after the start of infusion** falls in a zone → **redose every 24, 36 or 48 h**.

Aminopenicillins

- **Resistance in 40–60% of urinary isolates** has limited ampicillin/amoxicillin; they disturb bowel and vaginal flora → **resistant reinfection and candidal vaginitis**.
- **Clavulanate** restores activity against β -lactamase producers, but **cost and GI effects** limit it.
- **Extended-spectrum penicillins** (pivmecillinam, piperacillin, mezlocillin, azlocillin) keep **enterococcal activity** and add many **ampicillin-resistant Gram-negatives** → suited to **nosocomial UTI** and **initial parenteral therapy of community pyelonephritis**, though cheaper agents may be equal.

Aztreonam

- **Spectrum similar to aminoglycosides, not nephrotoxic** (as all β -lactams), but narrower than 3rd-generation cephalosporins; use **mainly in penicillin allergy**.

Pivmecillinam

- **Prodrug of mecillinam**, penicillin-like β -lactam; **high Gram-negative activity**; used mainly in **Nordic countries** for empirical uncomplicated cystitis; **not available in the USA**; very low *E. coli* resistance (about 2%), safe and effective.

Concept — The formulary is organised by a single trade-off: **spectrum and tissue penetration buy efficacy in complicated infection but cost collateral damage** — so the narrow, urine-confined, flora-sparing agents (nitrofurantoin, fosfomycin, pivmecillinam) lead in cystitis, and the broad, tissue-penetrating agents (fluoroquinolones, cephalosporins, carbapenems) are held back for infections that invade.

Choice of Antimicrobial Agents

- Empirical choice: **complicated vs uncomplicated, spectrum vs probable pathogen, hypersensitivity, renal/hepatic toxicity, ecological impact, cost**; use **regional antibiograms**.
- ***C. difficile* infection (CDI)**: increasingly common and resistant; large US burden and mortality.
 - **Most likely culprits:** cephalosporins, clindamycin, carbapenems, TMP-SMX, fluoroquinolones, penicillin combinations. **Not associated:** macrolides, tetracyclines, aminoglycosides.
 - **Diagnosis requires both** (1) diarrhoea (≥ 3 unformed stools in 24 h) or radiological ileus/toxic megacolon, and (2) stool positive for toxigenic *C. difficile* or its toxins, or histological/colonoscopic pseudomembranous colitis.

- Treatment by **severity and recurrence risk**. The text's older statement (metronidazole for mild, vancomycin for severe, as metronidazole fails more with severity) is **superseded by the 2021 IDSA/ Society for Healthcare Epidemiology of America (SHEA) focused update** (Table 26.7), which adds **fidaxomicin** (macrolide) and **bezlotoxumab** (monoclonal antibody to toxin B).

TABLE 26.7 RECOMMENDATIONS FOR THE TREATMENT OF *Clostridioides difficile* INFECTION IN ADULTS (JOHNSON ET AL., IDSA/SHEA 2021)

Presentation	Preferred	Alternatives / adjuncts	Comments
Initial episode	Fidaxomicin 200 mg twice daily × 10 days	Oral vancomycin 125 mg four times daily × 10 days; if unavailable and non-severe: metronidazole 500 mg three times daily × 10–14 days	Non-severe = WBC ≤15,000/μL and creatinine <1.5 mg/dL
First recurrence	Fidaxomicin 200 mg twice daily × 10 days, or twice daily × 5 days then alternate days × 20 days	Tapered/pulsed vancomycin; standard vancomycin × 10 days; adjunct bezlotoxumab 10 mg/kg IV once	Taper example: 125 mg four times daily 10–14 d → twice daily 7 d → daily 7 d → every 2–3 days for 2–8 weeks; standard vancomycin if metronidazole was used first
Second or later recurrence	Fidaxomicin (either regimen); tapered/pulsed vancomycin; vancomycin then rifaximin 400 mg three times daily × 20 days; faecal microbiota transplantation (FMT)	Adjunct bezlotoxumab	Try antibiotics for ≥2 recurrences (3 episodes) before FMT
Fulminant (hypotension/ shock, ileus, megacolon)	Vancomycin 500 mg four times daily orally/nasogastric + IV metronidazole 500 mg 8-hourly, especially with ileus	Rectal vancomycin if ileus	—

- Bezlotoxumab: also for other recurrence risks (age >65, immunocompromise, severe CDI at presentation); FDA caution in congestive heart failure — reserve where benefit outweighs risk.
- Purpose: severity- and recurrence-staged escalation. Pattern: **fidaxomicin first line throughout, vancomycin always acceptable, metronidazole only a fallback for non-severe first episodes (and IV in fulminant disease)**. Trap: oral metronidazole is no longer first-line; FMT only after ≥2 recurrences.

Duration of Therapy

- Determined by **extent and duration of tissue invasion, urinary bacterial concentration, achievable urinary drug level, and host risk factors**.

KEY POINTS: PRINCIPLES OF ANTIMICROBIAL THERAPY

- Antimicrobial resistance is increasing.
- Selection weighs **efficacy, safety, cost and compliance**.
- **Agent and duration** are critical to limiting resistance and adverse events.

Concept — Resistance is manufactured mainly in the bowel, not the bladder: plasmid transfer among gut flora dwarfs mutant selection in urine, so the ecological footprint of an antibiotic — breadth, duration, repetition — matters more than its MIC against the organism in front of you.

BLADDER INFECTIONS

Uncomplicated Cystitis

- **Acute uncomplicated cystitis is the overwhelming majority of UTIs** and is often managed by non-urologists.
- Annually about one in ten women reports a UTI, and **most women have at least one in their lifetime**; incidence is low before puberty and **rises sharply in late adolescence and the second to fourth decades**; about a third of women have had a physician-diagnosed UTI **by age 26**.
- ***E. coli* causes 75–90% of acute cystitis in young women**; *S. saprophyticus* (a skin commensal) is the next common cause; less often *Klebsiella*, *Proteus*, *Enterococcus*. In men, *E. coli* and other Enterobacteriaceae.

BOX 26.9 RISK FACTORS FOR URINARY TRACT INFECTIONS

Mechanism	Risk factors
Reduced urine flow	Outflow obstruction (prostatic hyperplasia, prostatic carcinoma, urethral stricture, foreign body/calculus); neurogenic bladder ; inadequate fluid intake (dehydration)
Promote colonisation	Sexual activity (↑ inoculation); spermicide (↑ binding); oestrogen depletion (↑ binding); antimicrobials (↓ indigenous flora)
Facilitate ascent	Catheterisation ; urinary incontinence ; faecal incontinence ; residual urine with bladder wall ischaemia

- Purpose: sorts risk factors by **which host defence they defeat** — washout, flora barrier or the distance to ascend. Trap: **spermicide and oestrogen loss act by increasing bacterial binding**, not by increasing inoculum.

Clinical Presentation

- The validated **UTI Symptom Assessment questionnaire** scores seven symptoms — **frequency, urgency, painful urination, incomplete emptying, suprapubic pain/pressure, low back pain, haematuria** — for presence, **severity and bother**.
- **Cystitis is a superficial mucosal infection** → **fever and chills are usually absent**.
- **Malodorous or turbid urine alone is not a reliable sign of UTI** → advise **hydration and reassess**.
- **More than one irritative symptom raises the likelihood**, but **dysuria alone is the most reliable single indicator** and drives the highest severity and bother scores.
- In **baseline voiding dysfunction**, establish how symptoms **differ from baseline**. Distinguish **dysuria (burning on voiding)** from **constant vaginal/pelvic burning** (painful bladder syndrome, **genitourinary syndrome of menopause [GSM]**).
- **Dysuria + frequency without vaginal discharge or irritation predicts UTI in about 90%**; probability rises further with **haematuria, back pain and CVAT**.
- Symptom meta-analysis: **frequency, dysuria, haematuria, nocturia, urgency increase probability**; **vaginal discharge decreases it**; **haematuria + dipstick nitrite** → **probability ≥90%**.
- **Bottom line: acute-onset dysuria and a change from baseline voiding symptoms** best indicate acute uncomplicated cystitis.

Laboratory Diagnosis

- As in Evaluation (urine testing).

Management

- **Antimicrobial selection** (IDSA/European Society for Microbiology and Infectious Diseases [ESCMID] guideline, Fig. 26.13): individualise by **allergy, compliance, local antibiogram/practice, availability, cost**.

FIG. 26.13 IDSA-RECOMMENDED TREATMENT OF ACUTE UNCOMPLICATED CYSTITIS IN WOMEN

- **Entry:** woman with acute uncomplicated cystitis, **no fever or flank pain or other suspicion of pyelonephritis, able to take oral medication** — if not, treat as pyelonephritis/complicated UTI.
- **Can a first-line agent be used (availability, allergy, tolerance)?** → Yes → first-line; No → second-line.

UNCOMPLICATED CYSTITIS REGIMENS: FIRST-LINE VERSUS SECOND-LINE

Tier	Agent and regimen	Caveat
First-line	Nitrofurantoin monohydrate/macrocrystals 100 mg twice daily × 5 days	Avoid if early pyelonephritis suspected; efficacy ≈ 3-day TMP-SMX
First-line	TMP-SMX 160/800 mg (one double-strength [DS] tablet) twice daily × 3 days	Avoid if local resistance >20% or used for UTI in previous 3 months
First-line	Fosfomycin trometamol 3 g single dose	Lower efficacy ; avoid if early pyelonephritis
First-line	Pivmecillinam 400 mg twice daily × 5 days (3–7 days in text)	Lower efficacy ; not available in North America; avoid if early pyelonephritis
Second-line	Fluoroquinolones × 3 days	High collateral damage; reserve as last resort (FDA 2016); resistance high in some areas
Second-line	β-lactams (amoxicillin–clavulanate, cefdinir, cefaclor, cefpodoxime-proxetil) × 3–7 days; EAU: cefadroxil 500 mg twice daily × 3 days	Avoid ampicillin/amoxicillin alone ; lower efficacy; close follow-up

- **TMP-SMX/TMP** achieve bacteriological cure within 7 days in **>90%**; use where *E. coli* resistance is **≤20%**. **TMP alone** suits **postcoital prophylaxis** and **sulfa allergy** — but **TMP itself causes rash/hypersensitivity** that may be wrongly blamed on sulfa.
- Pattern: **first-line agents share low resistance and low collateral damage**; the fluoroquinolones are demoted **not for inefficacy but for ecology and toxicity**.
- **Duration of therapy:**
 - **Three days is the preferred duration for uncomplicated cystitis in women.** Across hundreds of trials, **3 days beats single dose** for cure, and **matches 7 days for cure with adverse effects as low as a single dose; 7 days only with complicating factors.**
 - Exceptions to "3 days": **nitrofurantoin 5 days; fosfomycin single dose.**
 - **Men — landmark trial (Drekonja et al., 2021):** in afebrile men with suspected UTI, **7 days of ciprofloxacin or TMP-SMX was non-inferior to 14 days** for symptom resolution. **European Association of Urology (EAU):** **TMP-SMX 160/800 twice daily × 7 days first-line for cystitis in men;** fluoroquinolones per local susceptibility.

DURATION OF THERAPY FOR CYSTITIS

Group	Duration
Women — TMP-SMX, fluoroquinolone, β-lactam	3 days (β-lactams 3–7)
Women — nitrofurantoin	5 days
Women — fosfomycin	Single dose
Women with complicating factors	7 days
Afebrile men	7 days (= 14 days)

- **Cost of therapy:** cost-effectiveness depends on **efficacy against *E. coli*** (failures → revisits, pyelonephritis, follow-up); **drug price is a poor predictor.**

Follow-Up

- **No follow-up visit or culture in asymptomatic women after treatment**; no urological evaluation if they respond.
- **UTI in most men is complicated** and linked to functional/anatomical pathology — in a small cohort of men evaluated after UTI, **about half had an abnormality, commonest incomplete emptying**.
- **Non-response → microbiological and urological evaluation** for unresolved/complicated UTI.
- **Compliance with recommended agents**: about half of surveyed physicians **did not know the IDSA guideline** and nearly a third **did not choose a first-line agent** → education and outreach needed.

Complicated Urinary Tract Infections

- Occur in a **compromised urinary tract** or with a **resistant pathogen** (Table 26.8); complicating factors may be **obvious** from severity/history or **revealed only by treatment failure**.

TABLE 26.8 HOST FACTORS CLASSIFYING A URINARY TRACT INFECTION AS COMPLICATED (DASON ET AL., CANADIAN UROLOGICAL ASSOCIATION, 2011)

Complication	Examples
Anatomical abnormality	Cystocele, fistula, diverticulum
Iatrogenic	Nosocomial infection, surgery, indwelling catheter
Urinary tract obstruction	BOO, pelviureteric junction obstruction, ureteric stricture
Voiding dysfunction	Neurological disease, pelvic floor dysfunction, VUR, incontinence, high PVR
Other	Pregnancy, urolithiasis, diabetes, other immunosuppression

- Purpose: a checklist of **host features that should change the default short oral course**. Pattern: each **creates stasis, a foreign surface or immune compromise**. Trap: **VUR and high PVR are classified as voiding dysfunction**, and **pregnancy and stones** sit under "other".
- Spectrum: **mild cystitis → life-threatening pyelonephritis and urosepsis**; often **MDR → urine culture is mandatory**.
- Empirical guidance is limited (heterogeneous hosts, few trials).
 - **Mild-moderate, oral outpatient therapy**: traditionally **fluoroquinolones for 10–14 days** (urinary concentration and tissue penetration) — now tempered by **resistance and toxicity**.
 - **Hospitalised**: IV agents chosen by local susceptibility.
- **Correct the underlying abnormality (e.g. obstruction) and host factors** — otherwise therapy fails.
- **Total 10–14 days of culture-directed therapy**; switch IV → oral once afebrile and stable; repeat culture if no response.

Emphysematous Cystitis

- **Rare, potentially life-threatening** complicated cystitis; **mortality up to 7%**. Pathognomonic: **gas within the bladder wall on cross-sectional imaging**.
- Typical patient: **elderly woman (60–70 years) with poorly controlled DM**; also **incomplete emptying/stasis (NGB, BOO)**.
- Organisms: ***E. coli* (about 60%), *K. pneumoniae* (10–20%)**.
- **Clinical presentation**: asymptomatic (up to 7%) to overt sepsis; **abdominal pain commonest (about 80%)**, then **gross haematuria (about 60%)**, obstructive symptoms; **fever in only 30–50%**.
- **Radiological findings**: plain film — **curvilinear radiolucency outlining the bladder wall**, separate from rectal gas; **"cobblestone" or "beaded necklace" pattern of submucosal blebs**. **CT is required to confirm**

and **exclude other pelvic gas (fistula, trauma, gangrene of adjacent organs)** (Fig. 26.14: intramural and intraluminal gas).

- **Management:** about 90% medical alone — (often IV) antibiotics with broad Gram-negative cover, bladder drainage, control of DM → narrow by culture. Gram-positive cocci on Gram stain → add ampicillin/amoxicillin (enterococcus). Surgery rare — poor response or severe necrosis: **debridement, partial or total cystectomy**.

Concept — *Emphysematous cystitis is **diabetic stasis cystitis with gas as its signature** — it is managed like any severe complicated cystitis (drain, treat, control glucose), and CT's real job is to prove the gas is intramural rather than from a fistula or gangrene that would need surgery.*

Unresolved Urinary Tract Infections

Clinical Presentation

- **Unresolved** = initial therapy failed to eliminate symptoms and/or bacterial growth. If symptoms persist at the end of treatment or recur shortly after → **urinalysis and culture with susceptibilities**; if symptoms are significant and not relieved by **urinary analgesics**, start **broad-spectrum empirical therapy** pending culture.

BOX 26.10 CAUSES OF UNRESOLVED BACTERIURIA, IN DESCENDING ORDER OF IMPORTANCE

1. Bacterial resistance to the drug selected
2. Development of resistance in initially susceptible bacteria
3. Two species with mutually exclusive susceptibilities
4. Rapid reinfection with a new resistant species during therapy
5. Azotaemia
6. Papillary necrosis from analgesic abuse
7. Giant staghorn calculi — "critical mass" of susceptible bacteria too great
8. Self-inflicted infection or deception about taking drugs (Munchausen variant)

MECHANISMS OF UNRESOLVED BACTERIURIA

Mechanism	Explanation	Clue
1. Pre-existing resistance (commonest)	Recent antibiotics → bowel colonisation with resistant bacteria; β-lactams, tetracycline, sulfonamides notorious for plasmid R factors carrying multi-resistance	Organism resistant from the outset
2. Emergent resistance	Low-count resistant subpopulation selected when urine level < killing concentration	On-treatment culture: same species, now resistant
3. Unsuspected second pathogen	Treating the dominant strain unmasks a resistant second strain	New species present from the start
4. Rapid reinfection	New resistant species introduced during therapy	Suspect enterovesical fistula
5–7. Organism still susceptible	Drug can't reach adequate urine level (azotaemia; papillary necrosis dilutes drug), or bacterial mass overwhelms it (staghorn — even susceptible bacteria cannot be inhibited beyond a critical density, especially on a foreign body)	On-treatment culture: same species, still sensitive
8. Non-compliance/deception	—	—

Laboratory Diagnosis

- **Urinalysis and culture mandatory. Causes 1–4 (resistant bacteria) need no further work-up.** If the organism is **still sensitive** to the current drug → **renal function and imaging** for renal/urinary tract abnormalities.

Management

- Significant symptoms → broad-spectrum empirical therapy **assuming resistance** → **choose a different agent from the original.**

Concept — The on-treatment culture sorts every unresolved infection into two bins: if the organism is now resistant, the problem is microbiological and needs only a new drug; if it is still sensitive, the problem is the host (drug delivery or bacterial mass) and needs investigation.

Recurrent Urinary Tract Infections

- **Recurrent UTI (rUTI) = 2 UTIs in 6 months or ≥3 in 12 months.**
- Common in healthy women — about a fifth of college women have another culture-proven UTI **within 6 months**; 20–30% recurrence overall. **A recent prior UTI is one of the strongest risk factors.**

BACTERIAL PERSISTENCE VERSUS REINFECTION

Bacterial persistence	Reinfection
Same strain from a focus in the tract	Same organism >2 weeks after treatment, or a different organism
Recurr quickly	Recurr later
Uncommon	Commoner — ascent from faecal flora or re-emergence from intracellular urothelial reservoirs

- In men, reinfection raises suspicion of an anatomical abnormality.

Evaluation

- Aim: **prevent recurrence** by identifying risk factors and modifiable behaviour; exclude external anatomical or functional abnormality (if present → **complicated**).
- **History:** number, frequency, **culture results**, symptoms, triggers; **pneumaturia, faecaluria, obstipation, diverticulitis, pelvic surgery or radiotherapy → fistula; bowel symptoms** (constipation, diarrhoea, faecal incontinence); prior treatments and response.
- **Risk factors** in Box 26.11. **Nonoxynol-9 kills lactobacilli but not *E. coli*; spermicide-coated condoms increase UTI.**

BOX 26.11 RISK FACTORS FOR RECURRENT URINARY TRACT INFECTIONS

- **Sexual activity;** new partner within past year; family history of UTI in a first-degree female relative; recent antimicrobial use; spermicide use; UTI before menopause; menopause; incontinence, raised PVR, cystocele (postmenopausal).
- **Examination:** suprapubic palpation (**distended bladder; PVR drives bacteriuria**); pelvic examination — vaginal epithelium (postmenopausal), POP, urethral diverticulum.
- **Urinalysis and culture are imperative:** many are treated for symptoms that are really **irritative storage symptoms**; document **bacteriuria, burden, species and resistance**. **Culturing in uncomplicated rUTI reduced hospitalisation and IV antibiotic use.**
- **PVR and uroflow:** essential with emptying symptoms or a distended bladder; in postmenopausal women with rUTI, **higher PVR and lower flow** are associated with recurrence; abnormal values may **reclassify the UTI as complicated**.

- **Imaging and cystoscopy are not routine** — low yield in uncomplicated rUTI, and not recommended by the AUA, American College of Radiology, Canadian Urological Association or EAU. With complicated risk factors → imaging and cystoscopy (Table 26.9).
 - Preferred imaging: renal and bladder ultrasound ± plain abdominal film; CT when suspicion of complicated infection remains high despite normal imaging, or for stones/obstruction.
 - Pregnancy test before ionising imaging in premenopausal women if uncertain.
 - Cystoscopy adds nothing beyond imaging in uncomplicated rUTI; useful in complicated UTI.

TABLE 26.9 INDICATIONS FOR FURTHER INVESTIGATION OF RECURRENT URINARY TRACT INFECTION (DASON ET AL., 2011)

- Previous urinary tract trauma or surgery; previous bladder or renal calculi; gross haematuria after infection resolves; obstructive symptoms, low flow or high PVR
- Urea-splitting bacteria on culture; previous abdominopelvic malignancy; persistence despite sensitivity-based therapy; diabetes or other immune compromise
- Pneumaturia, faecaluria, anaerobes, or diverticulitis history; repeated pyelonephritis; asymptomatic microhaematuria after infection resolves
- Pattern: each indication signals stone, obstruction, fistula, malignancy or persistence. Trap: recurrence frequency alone is not on the list.

Bacterial Persistence

- Failure of an appropriate course for culture-proven acute infection → suspect persistence — rare in rUTI but critical to find; requires cystoscopy and imaging.
- Usually a correctable urological abnormality (Box 26.6) where antibiotics cannot penetrate or eliminate the nidus; surgical removal can cure (Stamey).
- Work-up: cystoscopy, ultrasound, CT; VCUG for VUR or non-refluxing ureteric stump; pelvic MRI = standard for urethral diverticulum; retrograde urethrography for suspected stricture in men.
- Struvite (infection) stones are one of the commonest causes. Urease-producing organisms (*P. mirabilis*) alkalise urine → precipitation of calcium, magnesium, ammonium and phosphate → branched struvite stones. Bacteria shelter within the stone — post-treatment urine may be sterile, misleadingly. Oxalate and apatite stones may be secondarily colonised. Risk: chronic catheters, urinary diversion, stasis.
 - AUA (Medical Management of Kidney Stones): surgically remove infection stones in all patients fit for intervention. Non-surgical candidates with residual/recurrent struvite → acetohydroxamic acid + antibiotic suppression.
- Non-obstructing stones with rUTI — controversial: half of patients remain infection-free a year after complete stone extraction; recurrence more likely with type 2 DM, hypertension, Black race. AUA permits observation of asymptomatic non-obstructing stones ≤10 mm. For stones <6 mm with UTI, minimise recurrences before considering removal. Whether clearing small stones reduces rUTI is unknown.

Management

- Most women with rUTI have no correctable abnormality. Options in order: behavioural modification → non-antibiotic therapy → antibiotic treatment/prophylaxis as a last resort.
- AUA 2019 guideline on recurrent uncomplicated UTI in women (revalidated 2021); EAU, National Institute for Health and Care Excellence (NICE) and other societies' guidelines conflict on prevention → confusion.
- Quality of life (QoL): patients fear the next UTI; anxiety, fragmented care from providers unaware of guidelines. Clinicians should educate and set a protocol for future episodes; patients increasingly

understand antibiotic risks and want **stewardship and mental-health acknowledgement**. **Education/ counselling on risk factors reduced symptomatic recurrence and improved QoL.**

- **Antibiotic resistance:** the Centers for Disease Control and Prevention (CDC) lists it among the **most serious health threats** — **millions of resistant infections and tens of thousands of US deaths annually**; similar European burden; **hospital-acquired MDR infections rose during COVID-19.**
- **Antibiotic stewardship programmes** improve prescribing, outcomes, adverse effects and susceptibility; a Cochrane review showed **better guideline compliance and shorter courses** → **UTI management must follow stewardship** (Fig. 26.15).

FIG. 26.15 ALGORITHM FOR MANAGEMENT OF RECURRENT UTI IN WOMEN

- **Entry:** UTI-specific symptoms (**acute dysuria + change from baseline**) with **positive urinalysis and culture** (midstream clean-catch [MSCC] or catheterised) and **≥3 episodes in 12 months or 2 in 6 months.**
 - **Not met:** negative culture or atypical symptoms → **consider another diagnosis**; true UTI but below the threshold → **treat each episode.**
- **Same organism recurring at short intervals?** → **Yes:** evaluate for bacterial persistence (urological evaluation incl. **upper tract imaging and cystoscopy**) → nidus found → **remove it**; nothing found → manage as reinfection.
- **Reinfection** → **behavioural modification** (hydration, frequent voiding — not evidence-based) → **modifiable risk factors?**
 - **Emptying symptoms** → check PVR; intercourse-related → **avoid spermicide, void pre- and postcoitally** (not evidence-based).
- → **Cranberry, ideally ≥36 mg PAC** → then:
 - **Coitus-related** → **postcoital antibiotics**; postmenopausal → **vaginal oestrogen**; neither → **methenamine with vitamin C** if renal function normal.
- Note that **continuous daily antibiotic prophylaxis does not appear** in the algorithm.
- **Behavioural modification** — low risk, supports bladder health:
 - **Hydration:** flushes adherent bacteria and sheds urothelial cells. **Landmark trial (Hooton et al., 2018):** premenopausal women with rUTI who **added ≥1.5 L water daily** had **about half as many cystitis episodes** over a year (1.7 vs 3.2). Meta-analysis: **fewer patients with rUTI at ≤6 months** with increased fluids (water, juices, D-mannose in water).
 - **Frequent, complete voiding; void after (and, where possible, before) intercourse** — intercourse causes transient bacteriuria.
 - **Avoid spermicides (nonoxynol-9)** — deplete vaginal flora and raise pH.
 - **Pubic shaving and thong underwear: no evidence.**
- **Self-start therapy** — patient starts antibiotics at symptom onset, often **without a sample**:
 - Older studies suggested patients diagnose accurately; originally required a sample and follow-up — **that expectation has eroded.**
 - **Fluoroquinolones were once "ideal"** self-start agents; the practice **perpetuates overuse**; anxious patients **treat minor changes**; in the authors' view **risks outweigh benefits** — **not sustainable.**
 - **Preferred approach:** at symptom onset → **urinary analgesic + urine sample + observation** → **IDSA-recommended agent (per antibiogram) only if symptoms persist**, tailored to culture. **Many episodes are transient and non-bacterial.**

- **Low-dose continuous prophylaxis:**

- Rationale: eradicates uropathogens from **introital and bowel reservoirs** without (it was thought) fostering resistance. Older trials (TMP-SMX, TMP, nitrofurantoin, cephalexin, fluoroquinolones) for **6–12 months** reduced UTIs; some noted **more non-*E. coli* UTIs** afterwards.
- **Cochrane 2004** (randomised controlled trials [RCTs]; latest 1995): **6–12 months of prophylaxis reduces clinical and microbiological recurrence while taken, with more adverse effects; after stopping, recurrence reverts to placebo levels. Postcoital prophylaxis is effective and preferable to daily in intercourse-related rUTI.**
- Caveats: **poor-quality, decades-old trials**, inconsistent definitions, **no agreed duration, pre-dating current resistance**; long-term suppression **drives MDR at a societal level.**
- **AUA: clinicians "may prescribe" prophylaxis** after weighing risks, benefits and alternatives. **The authors (and the EAU) regard long-term daily prophylaxis as a last resort**; patients wanting indefinite prophylaxis must be told there are **no long-term data** and of **toxicity and MDR risks.**
- **Postcoital prophylaxis is acceptable** with a **clear intercourse–UTI link** when other measures fail; choose by **prior cultures, allergy, local resistance, ecology.**
- **Nitrofurantoin** — popular (urine-concentrated, broad, little resistance) but **pulmonary fibrosis, renal toxicity and neurotoxicity with long-term use are rarely discussed**; a meta-analysis found it as effective as norfloxacin, TMP, TMP-SMX, cefaclor, oestriol or methenamine but with **more (mainly GI) adverse effects.**

ANTIBIOTIC PROPHYLAXIS STRATEGIES COMPARED

Prophylaxis strategy	Evidence	Position
Continuous daily low-dose	Effective only while taken ; old trials; adverse effects; MDR	AUA: "may prescribe"; authors/EAU: last resort
Postcoital	Effective; preferable to daily for intercourse-related rUTI	Acceptable when other options fail
Self-start	Old data only	Discouraged ; analgesic + sample + observe instead

- **Non-antibiotic management** — the focus of the last decade's research; a meta-analysis found **OM-89, Urovac, vaginal oestrogen, cranberry and acupuncture** of varying benefit; **oral oestrogen and lactobacilli ineffective; vitamin C ineffective.** Head-to-head trials against **antibiotic prophylaxis (the gold standard)** and standardised products are needed.

- **Cranberry:**

- Active ingredient **type A proanthocyanidin (PAC)** — **blocks P-fimbrial adhesion of *E. coli* to uroepithelium**; may also act via **gut microbiota.**
- **Cochrane reviews conflict:** 2008 — **effective in women with rUTI** (not in older people or catheterised patients); 2012 — **no significant overall benefit**, including rUTI subgroup, though **equivalent to antibiotic prophylaxis** in two trials; 2023 (largest) — **supports use in women with rUTI, children, and after interventions.**
- Problems: **heterogeneous products/doses, high dropout with juice, PAC content rarely reported.**
- **Dose: 36 mg PAC daily effective; 72 mg may be better** (dose-dependent antiadhesion in urine 6 h after dosing, Fig. 26.16); **antiadhesion is short-lived → take twice daily**; high-dose PAC (2 × 18.5 mg) helped women with **<5 UTIs/year.**
- **Nursing-home trial of 72 mg PAC: no benefit** — but most had **no prior bacteriuria/rUTI**, samples were voided not catheterised, and **PVR/incontinence unmeasured → not grounds to abandon cranberry in all women.**

- Other meta-analyses: benefit, especially after **rUTI and gynaecological surgery; combinations with lactobacilli, D-mannose, propolis add nothing.**
- **Warfarin: FDA warning removed 2011;** international normalised ratio (INR) interaction not substantiated → **safe with warfarin.**
- **AUA supports offering cranberry;** authors recommend **≥36 mg PAC.**
- **Oestrogen:**
 - Postmenopausal reinfection linked to **residual urine (prolapse)** and **oestrogen loss → atrophic epithelium, raised vaginal pH, loss of lactobacilli, E. coli colonisation.**
 - **Landmark trial — Raz and Stamm (1993), intravaginal oestriol vs placebo:** UTI rate reduced about **90% (0.5 vs 5.9 episodes per patient-year);** mechanism **return of lactobacilli and fall in vaginal pH;** **more dropouts on oestriol** (local burning, itching, irritation). **Oestradiol vaginal ring** also reduced UTIs.
 - **Cochrane 2008: vaginal oestrogen effective; oral oestrogen not** (AUA concurs). **Benefit takes ≥12 weeks.** Adverse: **breast tenderness, discharge, spotting, local irritation.** All forms (cream, tablet, ring) improve atrophy; recent meta-analysis and a multicentre placebo RCT confirm **vaginal (not oral) oestrogen reduces rUTI.**

VAGINAL VERSUS ORAL OESTROGEN

Vaginal oestrogen	Oral oestrogen
Reduces rUTI (RCTs, meta-analyses)	No reduction
Restores lactobacilli, lowers pH	—
Minimal systemic absorption — levels within postmenopausal range	Systemic exposure

- **Oestrogen safety:**
 - **Boxed warnings apply to all routes** and are **not supported for vaginal use;** low-dose vaginal preparations keep plasma levels **within normal postmenopausal range.**
 - Observational data (incl. **Women's Health Initiative Observational Study**): **no increased fracture, breast cancer, other cancer or cardiovascular disease.**
 - **Breast cancer survivors** (incl. on tamoxifen): **no increased recurrence;** American College of Obstetricians and Gynecologists (ACOG) and American Society of Clinical Oncology (ASCO) support use for GSM in selected cases → **consider for rUTI with oncologist endorsement.**
 - Yet **providers and patients remain hesitant.**
- **Laser therapy for GSM** (fractional CO₂, erbium:YAG; collagen stimulation): small observational studies; efficacy and safety for GSM unestablished; no data for rUTI.
- **Probiotics:**
 - **Lactobacilli are the dominant vaginal/periurethral commensal;** some strains (e.g. **hydrogen peroxide producers**) interfere with uropathogens.
 - Early products (Lactobacillus GC drink; *L. casei* v *rhamnosus* suppository) **failed;** *L. crispatus* vaginal suppositories and oral *L. rhamnosus* GR-1/*L. reuteri* RC-14 more promising.
 - **Cochrane 2015 and later meta-analyses: not recommended** (small, flawed trials). Strain, dose and route unknown.
- **Methenamine:**
 - Used for about a century; **70–90% renally excreted;** in **acid urine (pH <6)** converts to **ammonia and formaldehyde** (bactericidal/bacteriostatic by concentration). Optimal formaldehyde in vitro at **pH <5.85;** whether acidification (**vitamin C 1–4 g**) is needed in vivo is unknown.

- **Two FDA-approved salts: hippurate (1 g twice daily) and mandelate.** Early placebo RCTs: **substantial (56–75%) reduction in rUTI.**
- Acid salts are **non-specifically bacteriostatic** → **ineffective against urease producers** (which alkalise urine).
- Adverse: **GI upset, dysuria, rash, raised liver enzymes.** Contraindicated: **CrCl <50 mL/min, severe liver disease, hypersensitivity; avoid with sulfonamides, antacids, carbonic anhydrase inhibitors.**
- **Cochrane 2012 (hippurate):** heterogeneous; **benefit in normal urinary tracts, especially short-term (≤1 week) therapy; no benefit with anatomical abnormalities;** adverse effects infrequent (nausea, diarrhoea).
- Recent: high success in a tertiary rUTI cohort; **RCT: methenamine hippurate non-inferior to daily antibiotic prophylaxis over 12 months, cheaper and with less resistance;** another small RCT: **as effective as TMP.**
- **Formaldehyde carcinogenicity** relates to **nasopharyngeal, sinonasal and lymphohaematopoietic** cancers, **not urothelial** → bladder cancer risk is **theoretical.**

TABLE 26.10 METHENAMINE DRUG INTERACTIONS (LO ET AL., 2014)

Interacting agent	Effect
Antacids	May diminish methenamine effect
Carbonic anhydrase inhibitors	May diminish effect
Sodium picosulfate	Its effect may be diminished by methenamine
Urine-alkalinising foods/diets (pH >5.5) — almonds, chestnuts, flax/pumpkin/sunflower seeds, berries, cherries, grapes, citrus, cantaloupe, figs, kiwi	May diminish effect

- Purpose: everything that **raises urinary pH blocks formaldehyde generation.** Trap: **citrus fruit alkalises urine** (metabolised to bicarbonate) despite being acidic to taste; **sodium picosulfate is the one interaction running the other way.**
- **D-mannose:** simple sugar that **binds FimH (tip of type 1 pili)** → **competitive inhibitor of adhesion;** used in animals. Trials: **as effective as nitrofurantoin with fewer adverse effects;** reduced acute symptoms; a meta-analysis says it "**appears protective**" vs placebo; **Cochrane 2022: evidence can neither support nor refute it.** Well tolerated; **optimal dose unknown.**
- **Immunoactive prophylaxis:**

IMMUNOACTIVE PROPHYLAXIS PRODUCTS

Product	Route	Content	Evidence
OM-89 (Uro-Vaxom)	Oral, daily	Extract of 18 heat-killed UPEC serotypes; ↑ neutrophil/macrophage phagocytosis	Meta-analysis: fewer women with ≥1 UTI; about half the mean UTI count; adverse effects = placebo; manufacturer funding, variable quality, short follow-up
Urovac	Vaginal suppositories (primary ×3 weekly ± boosters ×3 monthly)	10 heat-killed species: 6 UPEC serotypes, <i>P. vulgaris</i> , <i>K. pneumoniae</i> , <i>M. morganii</i> , <i>E. faecalis</i> ; ↑ IgA and IgG	Boosters helped; meta-analysis: slight reduction; small trials from one group
Uromune (MV 140)	Sublingual	Inactivated <i>E. coli</i> , <i>K. pneumoniae</i> , <i>P. vulgaris</i> , <i>E. faecalis</i>	Small studies: efficacy and tolerability

- Recent reviews: **promising but better trials needed** before universal recommendation.

- **Future directions:** acupuncture, intravesical hyaluronic acid (glycosaminoglycan-layer analogue), inoculation with low-virulence *E. coli*, herbal products (*uva ursi*) — the field should focus on non-antibiotic prevention.

NON-ANTIBIOTIC PROPHYLAXIS — SUMMARY

Agent	Mechanism	Evidence	AUA stance
Vaginal oestrogen	Restores lactobacilli, lowers pH	Strongest (landmark RCT, meta-analyses)	Recommended (postmenopausal)
Cranberry (≥ 36 mg PAC, twice daily)	Blocks P-fimbrial adhesion	Conflicting; latest Cochrane favourable	May be offered
Methenamine hippurate	Formaldehyde in acid urine	RCT non-inferior to antibiotics	—
D-mannose	Blocks FimH (type 1 pili)	Promising; Cochrane inconclusive	—
Increased water ≥ 1.5 L/day	Washout	RCT positive (premenopausal)	—
OM-89 / Urovac / Uromune	Immunostimulation	Some benefit; bias, small trials	—
Probiotics	Flora restoration	Not effective	—
Oral oestrogen; vitamin C	—	Not effective	—

KEY POINTS: RECURRENT URINARY TRACT INFECTIONS

- **Thorough history:** prior symptoms, urinalysis/culture results, triggers.
- **Identify and eradicate sources of persistence.**
- **Imaging and cystoscopy are not recommended for recurrent uncomplicated UTI.**
- **Prevention should focus on non-antibiotic interventions.**

Concept — Recurrent UTI in women is **a problem of reinfection in a receptive host, not of a hidden focus** — so investigation is reserved for features of persistence or complication, and prevention targets the introital and bladder environment (flora, pH, adhesion, washout) rather than repeatedly sterilising the urine.

KIDNEY INFECTIONS

Acute Pyelonephritis

- Defined as **inflammation of kidney and renal pelvis**, but **diagnosed clinically**; upper tract infection can be proven by **ureteric catheterisation or bladder washout**, which are **impractical and unnecessary** in most; **no non-invasive localisation test is fully reliable**.

Clinical Presentation

- Spectrum: **Gram-negative sepsis** ↔ cystitis with mild flank pain.
- **Classic:** abrupt chills, fever ≥ 38 °C (100.4 °F), unilateral or bilateral flank/costovertebral angle pain or tenderness; lower urinary tract symptoms (LUTS) often absent.
- **Symptoms correlate poorly with the true site of infection** (localisation studies by ureteric catheterisation and bladder washout).
- **Deep CVAT** on examination. May **mimic GI disease** (abdominal pain, nausea, vomiting, diarrhoea).
- **Asymptomatic progression to chronic pyelonephritis** can occur in compromised hosts; **acute renal failure is rare**.

Laboratory Diagnosis

- **Neutrophil leucocytosis**; urinalysis — **numerous WBCs, often clumped**, bacterial rods or chains of cocci.
- **Granular or leucocyte casts suggest pyelonephritis**; a cast containing bacteria in its matrix is specific (Fig. 26.21).
- Blood: raised **erythrocyte sedimentation rate (ESR)**, **C-reactive protein (CRP)**, creatinine (if renal failure), reduced CrCl; **blood cultures may be positive**.

Bacteriology

- **Urine cultures positive**, though a minority have **<10⁵ CFU/mL** and a negative urine Gram stain.
- ***E. coli* (a special virulent subgroup) causes most cases**. Without VUR, patients of **P blood-group phenotype** are susceptible to **recurrent pyelonephritis from P-piliated *E. coli*** binding P antigen receptors. Much community pyelonephritis comes from **a few resistant clonal groups**.
- **Suspect resistant species (*Proteus*, *Klebsiella*, *Pseudomonas*, *Serratia*, *Enterobacter*, *Citrobacter*)** with rUTI, hospitalisation, catheters, recent instrumentation.
- **Gram-positives rarely cause pyelonephritis except *E. faecalis*, *S. epidermidis*, *S. aureus***.
- **Blood cultures if systemic toxicity, hospitalisation, or risk factors such as pregnancy**.

Renal Ultrasonography, Computed Tomography and Magnetic Resonance Imaging

- **Ultrasound: focal swelling, increased or decreased echogenicity** (Fig. 26.22: echogenic, swollen upper pole).
- **CT/MRI: focal swelling, diminished, heterogeneous enhancement** (Fig. 26.23: patchy); unenhanced CT — reduced density; MRI — **restricted diffusion**.
- **Changes usually resolve**, though **chronic scarring** can follow.

Differential Diagnosis

- **Appendicitis, diverticulitis, pancreatitis** — similar pain intensity, **different location**, usually **normal urine**.
- **Herpes zoster** — superficial loin pain **without urinary symptoms**; obvious once the rash appears.

Management

- **Initial management** — three groups:

ACUTE PYELONEPHRITIS: MANAGEMENT GROUPS

Group	Setting
1. Uncomplicated, not needing admission	Oral outpatient ; imaging usually deferred (ultrasound if indicated)
2. Uncomplicated, normal tract, ill enough for parenteral therapy	Inpatient IV
3. Complicated (hospitalisation, catheter, urological surgery, tract abnormality)	Inpatient IV; CT to define tract and extent

- **Outpatient first line: oral ciprofloxacin × 7 days where community fluoroquinolone resistance ≤10%** (regimens, including extended-release [XR] ciprofloxacin, in Table 26.11), ± an **initial one-off parenteral dose**; **TMP-SMX only if the organism is known to be sensitive**. Future guidance must address **cephalosporins and alternatives** given quinolone resistance and collateral damage.
- **Parenteral therapy if high fever, high WBC, vomiting, dehydration, sepsis, complicated pyelonephritis, or outpatient failure**. IDSA inpatient options: **fluoroquinolone**; **aminoglycoside ± ampicillin**; **extended-spectrum cephalosporin ± aminoglycoside**; **carbapenem** (doses in Table 26.11). **Gram-positive cocci → ampicillin-sulbactam ± aminoglycoside**. Plus IV fluids and antipyretics.

- **Obstruction:** unilateral obstruction **does not reduce overall renal function**, but a **solitary obstructed kidney causes acute renal failure** → dose by glomerular filtration rate (GFR); relieve substantial obstruction promptly by the simplest, safest route — ureteric stent or percutaneous nephrostomy.
- **Urine Gram stain** guides initial choice; therapy must reach **renal tissue and urine**.

TABLE 26.11 TREATMENT REGIMENS FOR ACUTE COMPLICATED AND UNCOMPLICATED PYELONEPHRITIS IN FEMALES (IDSA)

Circumstance	Route	Drug	Dose	Frequency	Days
Outpatient — moderately ill, no nausea/vomiting	Oral ^a	TMP-SMX DS^b	160/800 mg	12-hourly	14
		Ciprofloxacin^c	500 mg	12-hourly	7
		Ciprofloxacin XR ^c	1000 mg	24-hourly	7
		Levofloxacin^c	750 mg	24-hourly	5
Inpatient — severely ill, possible sepsis	Parenteral ^d	Ampicillin + gentamicin	1–2 g + 1–1.5 mg/kg	6-hourly + 8-hourly	10–14
		Levofloxacin ^c	500–750 mg	24-hourly	10–14
		Ceftriaxone	1 g	24-hourly	10–14
		Carbapenem	Varies	—	—
Pregnant	Parenteral ^d	Ampicillin + gentamicin	1–2 g + 1–1.5 mg/kg	6-hourly + 8-hourly	10–14
		Aztreonam	1 g	8-hourly	10–14
	Oral	Cephalexin	500 mg	6-hourly	10–14

- ^a May start with **one IV dose of a long-acting agent** (ceftriaxone 1 g or consolidated 24-h aminoglycoside). ^b **Only if organism known susceptible**; otherwise give that initial parenteral dose. ^c **Only where community fluoroquinolone resistance ≤10%**. ^d **Parenteral until afebrile, then oral per sensitivities**. All doses adjusted for renal function.
- Purpose: matches **severity and pregnancy to route and agent**. Pattern: **fluoroquinolones shorten courses (5–7 days) vs 14 for TMP-SMX**. Traps: **no fluoroquinolone in pregnancy** (ampicillin/gentamicin, aztreonam, cephalexin instead); **TMP-SMX needs proven susceptibility**; the **10% quinolone threshold for pyelonephritis differs from the 20% TMP-SMX threshold for cystitis**.
- **Subsequent management:** urine sterilises within hours, yet **fever, chills and flank pain may persist for several days** despite successful therapy → observe closely. Adjust to response and susceptibility; **swap toxic drugs (aminoglycosides) for less toxic ones (aztreonam, cephalosporins)**.
 - **Complicated pyelonephritis with positive blood cultures** → parenteral until clinically stable; **negative blood cultures** → 2–3 days parenteral suffices.
 - Then **oral agent (fluoroquinolone, TMP, TMP-SMX; amoxicillin or amoxicillin-clavulanate for Gram-positives)** at full dose for a further 10–14 days.
- **Unfavourable response:** immediate re-evaluation — repeat **urine and blood cultures**, adjust by susceptibility, and CT for **obstruction, abscess, stones, anatomical abnormality, renal or perinephric abscess**. **Fever >72 h** → CT is most helpful. Radionuclide imaging shows **reduced flow, delayed peak and excretion**, and **VUR-related cortical defects**.

- **Follow-up:** selected patients need **VCUG, cystoscopy, localisation studies** and correction of abnormalities. **DMSA 10–20 years after adult pyelonephritis shows scars in about half, but renal function change is minimal and unrelated to scarring.**

FIG. 26.24 MANAGEMENT OF ACUTE PYELONEPHRITIS

- **Pyelonephritis (fever, flank pain, leucocytosis) → split by sepsis:**
 - **No sepsis, nausea or vomiting** → urine culture; **optional imaging** to exclude obstruction/stones → **outpatient oral fluoroquinolone 7–10 days.**
 - **Sepsis** → **imaging for complicating factors**, blood and urine cultures → **inpatient parenteral antibiotics.**
- **Improvement within 72 h** → **oral step-down**; urine culture **4 days on and 10 days off** treatment; urological evaluation if indicated.
- **No improvement or deterioration** → **admit outpatient**; review cultures; **drain obstruction or abscess.**

Concept — *In pyelonephritis the 72-hour mark is the pivot: fever may lag behind a sterilised urine, but persistence beyond three days means an undrained collection or obstruction until CT proves otherwise — antibiotics treat the infection, drainage treats the complication.*

Acute Focal or Multifocal Bacterial Nephritis

- **Uncommon, severe renal infection with dense leucocyte infiltrate in one lobe (focal) or several (multifocal).**
- **Clinical presentation:** like pyelonephritis but **more severe**; about half **diabetic**; **sepsis common**; leucocytosis, **Gram-negative UTI**, about half **bacteraemic**. **AFBN lies midway between pyelonephritis and renal abscess.**
- **Radiological findings:** diagnosis is **radiological**; lesion slightly **less dense** than surrounding parenchyma.
 - **Ultrasound:** **poorly marginated, relatively sonolucent** mass with low-amplitude echoes **disrupting the corticomedullary junction** (the Fig. 26.25 example is slightly more echogenic than normal parenchyma; lesions **resolve after antibiotics**).
 - **CT requires contrast** (barely visible unenhanced): **wedge-shaped zones of reduced enhancement, no wall, no liquefaction.**
 - **Gallium uptake larger than the mass.** Multifocal disease — **same findings in several lobes.**

AFBN VERSUS RENAL ABSCESS ON IMAGING

AFBN	Renal abscess
Wedge-shaped, poorly marginated	Round, well defined
No wall, no liquefaction	Liquid centre; chronic → enhancing ring
Seen only after contrast	Visible before and after contrast
Medical treatment	Size-dependent: antibiotics or drainage

- **Management:** AFBN is probably an **early phase of abscess formation**; progression to frank abscess despite appropriate therapy is documented. **Hydration + IV antibiotics ≥7 days, then 7 days oral.** Usually responds; follow-up shows **resolution of wedge zones**. **Non-response → exclude obstruction, renal/perirenal abscess, renal carcinoma, renal vein thrombosis.** Long-term multifocal cases may show **reduced renal size and calyceal deformity suggesting papillary necrosis.**

Emphysematous Pyelonephritis

- **Urological emergency** — acute necrotising parenchymal and perirenal infection by gas-forming organisms.

- Pathogenesis unclear: **high tissue glucose fermented to CO₂** (e.g. by *E. coli*) is proposed, but does not explain its rarity among the many diabetic Gram-negative UTIs or non-diabetic cases.
- Besides DM, many have **obstruction (calculi, papillary necrosis)** and **significant renal impairment**. **Mortality about 19–43% historically.**

Clinical Presentation

- **Adults** (juvenile diabetics not at risk); **women more than men.**
- **Severe acute pyelonephritis**, sometimes after chronic infection; **classic triad: fever, vomiting, flank pain.**
- Occurs in **insulin-dependent and non-insulin-dependent diabetics**, even without obstruction; non-diabetics usually have **ureteric obstruction** and **less extensive disease.**
- **Pneumaturia absent unless the collecting system is involved.** Urine culture invariably positive — *E. coli* **commonest**; *Klebsiella*, *Proteus* less so.

Radiological Findings

- **Plain film: mottled gas over the kidney** (often mistaken for bowel gas); a **crescent of gas over the upper pole is more distinctive**; later gas extends to **perinephric space and retroperitoneum** (Fig. 26.26).
- **Ultrasound: strong focal echoes** of intraparenchymal gas.
- **CT is the imaging of choice** — defines extent, guides management (Fig. 26.27).

EMPHYSEMATOUS PYELONEPHRITIS: CT PATTERN AND PROGNOSIS

CT pattern	Prognosis
No fluid; streaky or mottled gas (± bubbly/loculated gas)	Rapid parenchymal destruction; 50–60% mortality
Renal/perirenal fluid; bubbly or loculated gas, or gas in collecting system; no streaky/mottled gas	<20% mortality

- **Huang–Tseng classification (2000): four CT classes of progressively greater gas extent** in renal and adjacent tissue — used to **guide management.**
- **Obstruction in about a quarter.** Nuclear renal scan to assess function of both kidneys.

EMPHYSEMATOUS PYELONEPHRITIS VERSUS EMPHYSEMATOUS PYELITIS

Emphysematous pyelonephritis	Emphysematous pyelitis
Gas in parenchyma ± perinephric space	Gas in the collecting system
Mostly diabetic	Often non-diabetic
Emergency; high mortality	Less serious; usually responds to antibiotics

Management

- **Most are septic** → fluid resuscitation, glucose and electrolyte control, broad-spectrum antibiotics; relieve obstruction by nephrostomy or stent.
- **Adverse prognostic factors:** hypoalbuminaemia, shock at presentation, bacteraemia, need for haemodialysis, thrombocytopenia, altered mental status, polymicrobial infection.
- **Evolution:** until the late 1980s **emergency nephrectomy ± drainage** (still 40–50% mortality) → now **percutaneous catheter drainage** as definitive treatment, with **nephrectomy** for extensive diffuse gas with renal destruction.
- **Early diagnosis and treatment of UTI in diabetics** may prevent progression and nephrectomy.

Concept — *Fluid is a good sign and streaky gas a bad one: a contained, liquefied collection can be drained, whereas diffuse streaky gas without fluid means parenchyma is necrosing faster than the host can wall it off*

— the same principle that moved treatment from routine nephrectomy to drainage with nephrectomy reserved for diffuse destruction.

Renal Abscess

- Collection of pus confined to the renal parenchyma.

RENAL ABSCESS: THEN VERSUS NOW

Pre-antibiotic era	Current era
About 80% haematogenous staphylococcal	Mostly Gram-negative (since about 1970): <i>E. coli</i> , <i>Klebsiella</i> , <i>Proteus</i> , <i>Pseudomonas</i>
Young men, no prior renal disease	Prior renal disease or obstruction; no sex or side predilection
Haematogenous seeding of normal kidney	Ascending infection with tubular obstruction (prior infection, calculi)

- **Gram-negative haematogenous pyelonephritis is almost impossible to produce experimentally** unless the kidney is **traumatised or completely obstructed**; **normal and partially obstructed kidneys reject blood-borne Gram-negatives** → ascent is the main Gram-negative route (haematogenous Gram-negative seeding occurs with **IV drug use and skin infection**).
- **Risk factors: UTI, calculi, DM** (DM and **longer hospital stay** very significant); link with **VUR less clear** than for pyelonephritis.

Clinical Presentation

- **Fever, chills, abdominal or flank pain**, sometimes **weight loss and malaise**, ± cystitis symptoms; may be **vague**, delaying diagnosis (historically to exploration or autopsy).
- History may reveal a **Gram-positive source 1–8 weeks earlier** (skin carbuncles, **IV drug use**) or recent UTI/pyelonephritis. Complicated UTI (**stasis, calculi, pregnancy, NGB, DM**) predisposes.
- Even with **normal tracts**, nearly all have a predisposing factor: **DM, HIV, steroids, IV drug use, liver disease, UTI, pregnancy**.

Laboratory Diagnosis

- **Marked leucocytosis**. Blood cultures positive in only about a quarter to a third; urine, blood and abscess isolates rarely match (few concordant).
- **Pyuria and bacteriuria may be absent** unless the abscess communicates with the collecting system. Haematogenous Gram-positive abscesses → urine sterile or a different organism.
- **Ultrasound** — **quickest, cheapest**: **echo-free or low-echo space-occupying lesion with increased through-transmission** (Fig. 26.28); margins indistinct acutely with **oedematous surrounding parenchyma**, later **well-defined**; internal echoes vary with **debris**; **air** → **strong echo with shadow**. **Abscess vs tumour may be hard to separate**.
- **CT** — **diagnostic procedure of choice**: **well defined before and after contrast**. Early: **renal enlargement, rounded low-attenuation foci** (Fig. 26.29); after days, **thick fibrotic wall**; chronic: **obliterated tissue planes, thickened Gerota's fascia, low-attenuation mass with an enhancing "ring sign"** (vascular wall) (Fig. 26.30).

Management

- Classic treatment **percutaneous or open drainage**, but **IV antibiotics and observation are appropriate for small abscesses (≤3 cm) in stable patients**; early antibiotics may **obviate intervention**.
- **CT/ultrasound-guided aspiration** to distinguish **abscess from cystic tumour**; culture the aspirate.
- **Start IV antibiotics immediately**:

RENAL ABSCESS: EMPIRICAL THERAPY BY SOURCE

Presumed source	Organism	Empirical agent
Haematogenous	Penicillin-resistant <i>Staphylococcus</i>	Penicillinase-resistant penicillin; vancomycin if penicillin-allergic
Ascending (abnormal tract)	Gram-negatives	IV 3rd-generation cephalosporin, antipseudomonal penicillin, or aminoglycoside

- **Serial ultrasound or CT until resolution**; worsening → suspect **misdiagnosis, perinephric extension or resistant organism**.
- **Size governs management:**

RENAL ABSCESS: MANAGEMENT BY SIZE

Size	Management
≤3 cm	Antibiotics alone (also in obstructed/anomalous tracts if stable)
3–5 cm	No consistent pathway — antibiotics alone or drainage; ≤5 cm with normal tracts resolved on antibiotics alone in one series
>5 cm	Percutaneous drainage first line ; may need several drains, manipulations, surgical washout or nephrectomy
Any size — immunocompromised or non-responding	Evaluate for percutaneous drainage

TABLE 26.12 REPORTED PRIMARY TREATMENT OF RENAL OR PERINEPHRIC ABSCESES BY SIZE (LEE ET AL., 2010)

Series (country, period)	<3 cm	3–5 cm	>5 cm
USA 1984–1993; USA 1991–2002	Antibiotics	Percutaneous drainage (PCD)	PCD
Italy 1984–1997	Antibiotics	Antibiotics	—
Brazil 1992–2002	Antibiotics	PCD or surgical drainage (SD)	SD
Taiwan 2001–2006	Antibiotics	Antibiotics	PCD or SD
Korea 2001–2008	Antibiotics	Antibiotics	PCD

- Purpose: shows the **size-stratified consensus**. Pattern: **universal agreement below 3 cm (antibiotics) and above 5 cm (drainage)**; the **3–5 cm band is where series diverge**. Trap: **drainage is not mandatory for every abscess**. Mean hospital stays ranged about 11–24 days.

Concept — *The modern renal abscess is an ascending Gram-negative complication of a damaged kidney, not a haematogenous staphylococcal lesion of a normal one — so the source dictates the antibiotic, and size dictates whether antibiotics alone can reach the pus.*

Infected Hydronephrosis and Pyonephrosis

- **Infected hydronephrosis** = bacterial infection of a hydronephrotic kidney. **Pyonephrosis** = infected hydronephrosis with **suppurative parenchymal destruction and total or near-total loss of function** (Fig. 26.31: thinned cortex/medulla, pus-distended pelvicalyceal system, distal ureteric obstruction). The boundary is **clinically indistinct**; **rapid diagnosis and treatment prevent permanent loss of function and sepsis**.
- **Clinical presentation**: usually **very ill** — high fever, chills, flank pain and tenderness; occasionally only **fever and vague GI discomfort**. Prior calculi, infection or surgery common. **Bacteriuria may be absent if the ureter is completely obstructed**.
- **Radiological findings**: ultrasound — **internal echoes in the dependent part of a dilated collecting system**; **fluid–debris levels**; **focal hypoechoic areas in the parenchyma suggest pyonephrosis**. CT non-specific — **thick pelvis, perirenal stranding, striated nephrogram** (Fig. 26.32: echogenic septa, debris-

filled calyces; antegrade pyelogram shows pus-filled calyces, scarred pelvis, stone — kidney did not recover).

- **Management:** antibiotics + drainage of the infected pelvis by ureteric stent or percutaneous nephrostomy; once stable, identify and treat the cause of obstruction.

Concept — An infected, obstructed kidney is an abscess with a collecting system — antibiotics cannot sterilise it until it is drained, and a sterile urine may simply mean the ureter is completely blocked.

Perinephric Abscess

- **Beyond the renal capsule but within Gerota's fascia.** Sources: rupture of a cortical abscess, extravasation of infected urine from obstruction, haematogenous seeding (about a third, usually skin). Pyonephrosis with calculus predisposes; about a third are diabetic. Infected perirenal haematoma (haematogenous or direct).
- **Rupture through Gerota's fascia → paranephric abscess;** paranephric abscess also arises from bowel, pancreas or pleura; conversely perinephric/psoas abscess may follow bowel perforation, Crohn disease, thoracolumbar osteomyelitis.
- **Organisms:** *E. coli*, *Proteus*, *S. aureus*.

Clinical Presentation

- **Insidious; symptoms >5 days in most;** resembles pyelonephritis but more than a third are afebrile; flank or abdominal mass in about half; CVAT usual. Limp with flexed, externally rotated hip → psoas abscess.
- **Labs:** leucocytosis, raised creatinine, pyuria. Urine culture predicts the abscess organism in only about a third; polymicrobial blood culture suggests perinephric abscess but identifies all organisms in under half; another series found urine the most sensitive → **treat culture results with caution.**
- **Pyelonephritis responds in 4–5 days; perinephric abscess does not → suspect with UTI plus flank mass or fever persisting after 4 days of antibiotics.** Often coexists with renal abscess.
- **CT shows the primary abscess, extension to flank or psoas (Fig. 26.33), route of spread and obstruction (Fig. 26.34: pararenal abscess involving adrenal) → plans drainage. Ultrasound:** anechoic mass displacing kidney ↔ echogenic collection blending with perirenal fat; can delineate collections outside Gerota's fascia.
- **Mortality fell from 40–50% to about 12% with imaging, yet diagnosis still lags by about 3–4 days and only about a third are correctly diagnosed at presentation — delay contributes to nearly all deaths → keep a low threshold for imaging.**

Management

- **Immediate antibiotics, guided by Gram stain:** aminoglycoside + antistaphylococcal penicillin (methicillin/oxacillin); penicillin allergy → cephalosporin or vancomycin.
- **Small (<3 cm) abscesses in immunocompetent patients may resolve with antibiotics alone; delay and size >3 cm predict failure and death → prompt diagnosis and drainage of collections >3 cm.**
- **Larger or non-responding → intervention:** classic surgical drainage or nephrectomy (non-functioning/severely infected kidney), now largely CT/ultrasound-guided percutaneous drainage; the threshold is lower than for renal abscess.
- **Then treat the underlying cause — cortical abscess or enteric communication promptly; nephrectomy for pyonephrosis at the same time if feasible, otherwise staged.** A proportion of large drained abscesses still need nephrectomy for non-functioning kidneys — but far fewer than historically.

RENAL VERSUS PERINEPHRIC ABSCESS

	Renal abscess	Perinephric abscess
Site	Parenchyma	Beyond capsule, within Gerota's fascia
Antibiotics alone	≤3 cm (up to 5 cm in some series)	<3 cm, immunocompetent
Drainage	>5 cm first line; 3–5 cm variable	>3 cm — lower threshold
Empirical cover	Haematogenous: antistaphylococcal; ascending: Gram-negative	Aminoglycoside + antistaphylococcal penicillin

- **Perinephric abscess versus acute pyelonephritis:** delay in diagnosis is the main obstacle; commonest misdiagnosis is **pyelonephritis**. Two discriminators:

ACUTE PYELONEPHRITIS VERSUS PERINEPHRIC ABSCESS

	Acute pyelonephritis	Perinephric abscess
Symptoms before admission	≤5 days	>5 days
Fever after appropriate antibiotics	Never >4 days	≥5 days (median 7)

Concept — Time is the diagnostic test for perinephric abscess — a pyelonephritis that has lasted too long before admission or will not defervesce by day four is a collection until imaged, and the mortality of this condition lies almost entirely in that delay.

Xanthogranulomatous Pyelonephritis

- **Rare, severe, chronic infection** → **diffuse renal destruction**; mostly **unilateral, enlarged, non-functioning kidney with obstruction from stones**; **lipid-laden foamy macrophages**; starts in **pelvis and calyces**, then destroys parenchyma and adjacent tissue.
- **Imitates almost every renal inflammatory disease and renal cell carcinoma (RCC)** radiologically; on **frozen section mimics clear cell carcinoma** → unnecessary radical nephrectomy.
- **Under 1% of histologically documented chronic pyelonephritis**, but **up to about a fifth of nephrectomies for pyelonephritis**.

Pathogenesis

- **Triad: nephrolithiasis, obstruction, infection.** Stones in **up to about 80%**, about **half staghorn**.
- **Obstruction then *E. coli* infection** reproduces tissue destruction and lipid collection by macrophages (**xanthoma cells**) — sheets around abscesses and calyces with **lymphocytes, giant cells, plasma cells**.
- **Bacteria of low virulence** (spontaneous bacteraemia rare). Other factors: **venous occlusion and haemorrhage, abnormal lipid metabolism, lymphatic blockage, failed antibiotic therapy, altered immunity, ischaemia**.
- **No single cause:** an **inadequate acute inflammatory response** in an **obstructed, ischaemic or necrotic kidney**.

Pathology

- **Massively enlarged kidney with normal contour.** **Diffuse (>80%)** or **segmental** (parenchyma around one or more calyces, or one pole of a duplex).
- **Cut surface:** **stones, peripelvic fibrosis, dilated pus-filled calyces, pelvis not dilated (encased by fibrosis), papillary necrosis**; advanced — **multiple abscesses with viscous pus lined by yellow tissue**, thin cortex replaced by xanthogranulomatous tissue, **thickened capsule**, frequent **perinephric/paranephric extension** (Fig. 26.37A).

- Microscopy: yellow nodules = **sheets of foamy histiocytes (small dark nuclei, clear cytoplasm)** with lymphocytes, giant cells, plasma cells (Fig. 26.37B). **Xanthoma cells are not specific** — seen wherever inflammation and obstruction coexist. Lipid (cholesterol esters) may derive from **lysed red cells after haemorrhage**.

Clinical Presentation

- Suspect with UTI + unilateral enlarged, non- or poorly functioning kidney + stone or mass indistinguishable from tumour.
- Flank pain, fever and chills (about two-thirds each), persistent bacteriuria (about half); malaise; flank mass in about 60%, prior calculi in about a third; less often hypertension, haematuria, hepatomegaly. History of UTI and instrumentation; DM a risk factor.
- Peak fifth to seventh decades; women more often; no side predilection.

Bacteriology and Laboratory Diagnosis

- **Proteus is the commonest organism** (linked to stone formation), *E. coli* also common; tissue cultures almost always positive; anaerobes occur; about 10% mixed. About a third have sterile urine (prior antibiotics) — organism may be found **only in operative tissue culture**.
- Urinalysis: pus and protein. Blood: anaemia; hepatic dysfunction in up to half.
- Almost always unilateral → azotaemia/renal failure uncommon.

Imaging

- CT is the most useful (Fig. 26.38): classic triad in 50–80% — unilateral enlargement, little or no function, large renal pelvic calculus.
 - Reniform mass; pelvis tightly contracted around a central calcification, without pelvic dilatation; parenchyma replaced by water-density masses (dilated calyces/abscesses).
 - Cavity walls blush on contrast (vascular granulation tissue) but cavities do not enhance — tumours and other inflammatory lesions usually do.
 - Shows extent and involvement of adjacent organs/abdominal wall.
- Ultrasound: global enlargement; multiple hypoechoic fluid-filled masses (debris-filled calyces, destruction); focal (tumefactive) XGP = solid segmental mass ± stone → RCC in differential.
- ^{99m}Tc-DMSA quantifies lost function. MRI: better for extrarenal extension; lesions intermediate on T1, hyperintense on T2. Imaging, though distinctive, often cannot separate XGP from RCC.

Differential Diagnosis

XGP: DIFFERENTIAL DIAGNOSIS

Entity	Distinguishing point
Segmental XGP without calculi	Difficult to diagnose
XGP with massive pelvic dilatation	Indistinguishable from pyonephrosis
XGP in small contracted kidney	Non-specific findings
Renal parenchymal malakoplakia	Enlargement and multiple inflammatory masses but usually no calculi
Renal lymphoma	Hypoechoic masses around a non-dilated pelvis; clinically obvious, usually bilateral, no calculi

Management

- Main obstacle is misdiagnosis; with CT, about 90% are diagnosed preoperatively.
- Antibiotics to stabilise; rarely, long-term antibiotics eradicate infection and restore function.

- **Nephrectomy usually** (diffuse disease or cannot exclude tumour); **localised XGP → partial nephrectomy**.
- **Frozen section may mimic clear cell carcinoma**, and XGP coexists with RCC, **papillary urothelial carcinoma of pelvis or bladder**, and **squamous cell carcinoma of the pelvis** → if malignancy cannot be excluded, perform nephrectomy.
- **Extensive retroperitoneal disease** → **remove kidney and perinephric fat**; may require dissection from diaphragm, great vessels and bowel. **Remove the entire inflammatory mass** — the tissue is infected in about three-quarters; **incision and drainage alone** → **protracted illness and renocutaneous fistula**, making later nephrectomy harder.
- **Laparoscopic nephrectomy**: early series found no benefit; later, larger series deem it **reasonable** (retroperitoneal, or transperitoneal hand-assisted) but **difficult with high conversion rates**.

Concept — XGP is **chronic obstruction plus low-grade infection in a host that cannot mount an acute response** — hence the stone, the non-functioning kidney, the fat-laden macrophages and the resemblance to cancer; because the tissue itself is infected and may harbour or mimic carcinoma, the answer is excision of the whole mass, not drainage.

Malakoplakia

- **"Soft plaque"** — unusual inflammatory disease first described in the **bladder**, also in **GU and GI tracts, skin, lungs, bones, mesenteric nodes**. Described by **Michaelis and Gutmann (1902)**; characterised by **von Hansemann (1903)** as **soft yellow-brown plaques** with histiocytes containing **basophilic lysosomal inclusions** (Michaelis–Gutmann bodies).
- **Abnormal macrophage function in response to bacterial** (mostly *E. coli*) **infection**.

Pathogenesis

- Bacteria or fragments form the **nidus for calcium phosphate crystals** that laminate Michaelis–Gutmann bodies. Leading theory: **defective phagocytic/degradative function of histiocytes** → chronic inflammation, often in **immunocompromised hosts**.

Pathology

- **Diagnosis by biopsy**. Grossly: **confluent tumour-like masses** and **firm plaque-like cortical lesions** (Fig. 26.39). **Large histiocytes (von Hansemann cells)** and **small basophilic intra- or extracytoplasmic calculospherules** — **Michaelis–Gutmann bodies, pathognomonic** (Fig. 26.39C: incompletely destroyed bacteria with lipoprotein membrane).
- **Electron microscopy: intact coliforms and fragments within phagolysosomes**.
- **Michaelis–Gutmann bodies may be absent early and are not required for diagnosis**.
- **Macrophages contain abundant α 1-antitrypsin**, unchanged through early, granulomatous and fibrosing phases; lookalike processes lack it (**except a few macrophages in TB and XGP**) → **α 1-antitrypsin immunostaining aids early diagnosis**.

Clinical Presentation

- **Most >50 years; urinary tract female:male 4:1** (not in other organs). Any part of the GU tract, **frequently bladder**.
- Patients often **debilitated, immunosuppressed, chronically ill**; notably after **solid organ (especially renal) transplantation** — graft kidney or ureter, GI and skin.
- **Bladder**: irritative LUTS, haematuria; cystoscopy — **mucosal plaques or nodules** → **fungating, sessile masses** causing filling defects; **distal ureteric stricture** → **obstruction or non-function**.
- **Renal parenchymal**: one or more masses with **chronic *E. coli* infection**; may cause **renal vein and inferior vena cava thrombosis**. **Testis** → **epididymo-orchitis**. **Prostate** — rare, **mimics carcinoma**.

- **Mortality can exceed 50%.**

Radiological Findings

- **Multifocal:** enlarged kidneys with multiple filling defects; no calcification, lithiasis or hydronephrosis; best seen on **ultrasound** (confluent masses raise parenchymal echogenicity, distort central echo complex), CT (foci less dense than enhanced parenchyma) or arteriography.
- **Unifocal:** non-calcified mass indistinguishable from other inflammatory or neoplastic lesions; solid or cystic depending on necrosis. **CT best shows extrarenal extension.**

Differential Diagnosis

- Consider **malakoplakia** with renal mass(es) ± collecting-system filling defects in women with recurrent *E. coli* UTI or altered immunity, and in a renal transplant recipient with persistent UTI despite appropriate therapy.
- Exclude **cystic disease** (ultrasound/CT); **metastases/lymphoma** (late disease); **multifocal RCC** (von Hippel–Lindau context); **XGP** (enlarged kidney but calculi and obstruction common); **multiple abscesses** (haematogenous).

XGP VERSUS MALAKOPLAKIA

	XGP	Malakoplakia
Cell	Lipid-laden foamy macrophages (xanthoma cells)	Von Hanseman histiocytes with Michaelis–Gutmann bodies
Stones / obstruction	Usually present (staghorn)	Absent
Organism	<i>Proteus</i> , <i>E. coli</i>	<i>E. coli</i>
Laterality	Almost always unilateral	Uni- or multifocal; may be bilateral
Sex	Women more	Women 4:1 (urinary tract)
Treatment	Nephrectomy (or partial)	Control infection with intracellularly active antibiotics; nephrectomy for symptomatic unilateral disease

Management

- **Control the UTI** to stabilise disease. Agents with **intracellular bactericidal activity** preferred: **sulfonamides, rifampicin, doxycycline, TMP; fluoroquinolones** (concentrated in macrophages) effective.
- Bladder disease: **ascorbic acid and cholinergics (bethanechol)** with antibiotics — both **raise intracellular cyclic guanosine monophosphate (cGMP)**, the postulated macrophage defect.
- **Reduce immunosuppression** where possible; **surgery if progression despite medical therapy; nephrectomy for symptomatic unilateral renal disease.**
- **Prognosis by extent:** bilateral or transplant-kidney parenchymal disease → death usually within 6 months; unilateral disease → long-term survival after nephrectomy.

Concept — *Malakoplakia is a failure to digest what has been swallowed* — macrophages ingest *E. coli* but cannot degrade it, leaving mineralised remnants (Michaelis–Gutmann bodies); treatment therefore uses antibiotics that work inside macrophages and agents that correct the degradative defect, not merely drugs that sterilise urine.

Renal Echinococcosis

- Available online only; not included in the source provided.

KEY POINTS: KIDNEY INFECTIONS

- **Acute pyelonephritis:** abrupt chills, fever, flank/CVA tenderness — spectrum from **cystitis-like to sepsis**.
- **Emphysematous pyelonephritis:** life-threatening; **diagnosed radiologically by parenchymal gas; percutaneous drainage or surgery**.
- **Renal abscess:** best delineated by CT; **IV antibiotics ± drainage; small abscesses → antibiotics alone**.
- **Pyonephrosis:** infected hydronephrosis — **prompt diagnosis; IV antibiotics plus drainage**.
- **XGP:** chronic infection of a **poorly functioning, stone-obstructed kidney; mimics tumour**.
- **Malakoplakia:** **abnormal macrophage function; Michaelis–Gutmann bodies** (lysosomal inclusions).

Concept — Across renal infections the governing axis is **diffuse inflammation versus contained collection**: pyelonephritis and AFBN are treated with antibiotics; pyonephrosis, abscess and perinephric abscess add drainage once a collection is large or obstructed; and XGP and extensive emphysematous disease, where the kidney itself is the infected tissue, end in nephrectomy.

OTHER INFECTIONS

Fournier's Gangrene

- **Potentially life-threatening necrotising fasciitis of the external genitalia, perineum and/or perianal region.** Synonyms: **idiopathic gangrene of the scrotum, streptococcal scrotal gangrene, perineal phlegmon, spontaneous fulminant gangrene of the scrotum**.
- Described by **Baurienne (1764)** and **Fournier (1883)** as abrupt, **idiopathic** fulminant genital gangrene in healthy young men.

FOURNIER'S GANGRENE: ORIGINAL VERSUS CONTEMPORARY

Fournier's original description	Contemporary disease
Idiopathic	A source is identified in about 95%
Healthy young men	Mostly 50–79 years; male:female about 10:1
Abrupt, fulminant	Broader age range; may be indolent

- **Sources:** skin, urethra, rectum. Well-documented link with **urethral stricture/obstruction with extravasation, and instrumentation**.
- **Predisposing factors:** DM, local trauma, paraphimosis, periurethral urinary extravasation, perirectal/perianal infection, surgery (circumcision, herniorrhaphy).
- **Fascial spread from a urethral source:** Buck's fascia (penis) → dartos fascia (scrotum and penis) → Colles' fascia (perineum) → Scarpa's fascia (anterior abdominal wall).
- **Polymicrobial — aerobic–anaerobic synergy:** facultative organisms (*E. coli*, *Klebsiella*, enterococci) with anaerobes (*Bacteroides*, *Fusobacterium*, *Clostridium*, microaerophilic streptococci). Other necrotising fasciitis classes are defined by microbial composition.

Clinical Presentation

- History of **perineal trauma, instrumentation, sexually transmitted disease (STD)–related urethral stricture, urethrocutaneous fistula**.
- **Clues to source:** rectal — pain, rectal bleeding, anal fissures; **dermal** — acute/chronic scrotal infection, recurrent **hidradenitis suppurativa, balanitis**.

- **Sequence: cellulitis at the portal of entry** → swollen, erythematous, tender as deep fascia is involved; **pain prominent, fever and systemic toxicity marked** → **scrotal swelling and crepitus increase rapidly** → **dark-purple areas** → **extensive gangrene**. **Abdominal wall involvement in an obese diabetic** → **very rapid spread** (Fig. 26.41).
- GU symptoms: **dysuria, urethral discharge, obstructed voiding**. **Sepsis markers: altered mental state, tachypnoea, tachycardia, temperature >38.3 °C (101 °F) or <35.6 °C (96 °F)**.

FIG. 26.41 FOURNIER'S GANGRENE OF THE SCROTUM (HAGEDORN AND WESSELLS, 2017)

- **Necrotic skin along the median raphe; erythema spreading over both groins signals progressive disease deep to the skin** — the skin underestimates the fascial extent.

Laboratory Diagnosis and Radiological Findings

- **Anaemia** (thrombosis and ecchymosis reduce functioning red cell mass; sepsis reduces production); **raised creatinine, hyponatraemia, hypocalcaemia** — **bacterial lipases free fatty acids that chelate ionised calcium**.
- **Laboratory Risk Indicator for Necrotising Fasciitis (LRINEC) score** — **CRP, WBC, haemoglobin, sodium, creatinine, glucose; ≥6 raises suspicion, ≥8 strongly predictive**; validated adjunct to clinical assessment.
- **Crepitus (subcutaneous gas) is early** → **plain film** may show air. **CT has the highest specificity** and delineates **soft tissue and fascial gas** for surgical planning (Fig. 26.42: fluid pocket with gas in scrotum tracking into perineum).
- **Biopsy of ulcer base: intact epidermis, dermal necrosis, vascular thrombosis, polymorphonuclear invasion, subcutaneous necrosis**.

Management

- **Prompt diagnosis is critical**. Early signs (pain, oedema, erythema) resemble cellulitis; **systemic toxicity out of proportion to local signs, and gas on imaging**, are the warnings.
- **Sequence: fluid resuscitation and electrolyte correction + broad-spectrum IV antibiotics as soon as suspected** → **immediate debridement** → **re-look at 24–48 h if debridement in doubt** → **hyperbaric oxygen (adjunct)** → **reconstruction**.
- **Antibiotics must cover aerobes, anaerobes, Gram-positives and Gram-negatives:**

FOURNIER'S GANGRENE: ANTIBIOTIC REGIMENS

Regimen	Note
Piperacillin-tazobactam	Especially if <i>Pseudomonas</i> suspected
Ampicillin-sulbactam	β-lactam/β-lactamase inhibitor
Vancomycin or carbapenem + clindamycin or metronidazole	—

- **Clinically obvious disease (deep pain with patchy surface hypoaesthesia or crepitus, bullae, skin necrosis)** → **straight to theatre**.
- **Debridement**: extensive incision through skin and subcutaneous tissue **beyond the involved area until normal fascia**; excise necrotic fat and fascia; **leave open**.
- **Orchidectomy almost never needed** — **testes have an independent blood supply** from the compromised scrotal fascial/cutaneous circulation.
- **Suprapubic diversion** if urethral trauma or extravasation; **colostomy** if colonic or rectal perforation.
- **Hyperbaric oxygen** (with debridement and antibiotics) may **shorten stay, improve healing and limit spread**. **Myocutaneous flap reconstruction** after healing improves cosmesis.

Outcome

- **Mortality about 20–40%** (wide range); higher with **liver failure, chronic alcoholism, DM, advanced age, congestive heart failure, coagulopathy, renal failure**.
- **A true urological emergency:** early recognition, aggressive antibiotics and surgical debridement.

Periurethral Abscess

- Named in the section's Key Points (secondary to **urethral stricture or catheterisation**; treatment **surgical debridement, suprapubic urinary drainage and antibiotics**); the body text of this section was not in the source provided.

KEY POINTS: OTHER INFECTION

- **Fournier's gangrene** = necrotising fasciitis of **external genitalia, perineum and/or perianal region**.
- **Emergent surgical debridement and broad-spectrum antibiotics** are the essentials.
- **Periurethral abscess** may follow **urethral stricture or catheterisation**; treat with **debridement, suprapubic drainage and antibiotics**.

Concept — In Fournier's gangrene **the skin is the last layer to show the disease**: infection runs along continuous fascial planes with thrombosis of the vessels that feed the skin, so systemic toxicity outruns the visible signs, debridement must chase normal fascia rather than normal skin, and the testes — perfused independently — are spared.

BACTERIURIA IN PREGNANCY

Pregnancy

- **Bacteriuria in pregnancy → pyelonephritis → adverse birth outcomes**; hence **pregnancy is one of the few indications to treat ASB (IDSA)**. American College of Obstetricians and Gynecologists (ACOG): screening urine culture at the first visit.
- **Prevalence of ASB is not raised by pregnancy** (about 2–7% in the text; 1.9–9.5% in Table 26.2 — different source series). Risk rises with **low socioeconomic status, multiparity, sickle cell trait**.
- **Non-pregnant women often clear ASB**; pregnant women tend to stay bacteriuric and become symptomatic.
- **Pyelonephritis: 1–4% of all pregnancies**; historically **15–45% of untreated bacteriuric pregnancies** (contemporary uncomplicated singleton data much lower); **60–75% in the third trimester** (peak hydronephrosis and stasis); **10–20% recur** before or soon after delivery.

BACTERIURIA: NON-PREGNANT VERSUS PREGNANT

	Non-pregnant	Pregnant
ASB prevalence	Similar	Similar
Natural history	Often clears spontaneously	Persists; progresses to pyelonephritis
Screen/treat ASB	No	Yes

Pathogenesis

- Anatomical and physiological changes of pregnancy **alter the natural history of bacteriuria** and may require altered therapy.

Anatomical and Physiological Changes During Pregnancy

- **Increase in renal size:** length + about 1 cm — vascular and interstitial volume, not true hypertrophy; biopsies show no histological change.
- **Smooth muscle atony of collecting system and bladder:** reduced ureteric peristalsis; most women have significant ureteric dilatation by the third trimester — from progesterone relaxation and uterine compression at the pelvic brim (the mechanical effect dominating later); progesterone also increases bladder capacity.
- **Bladder changes:** uterus displaces bladder superiorly and anteriorly; bladder hyperaemic/congested endoscopically; oestrogen → bladder hypertrophy and urethral squamous change.
- **Augmented renal function:** GFR rises 30–50% (raised cardiac output) with more proteinuria → "normal" non-pregnant creatinine may signify renal insufficiency in pregnancy; proteinuria abnormal only >300 mg/24 h.
- These changes (from the first trimester) cause stasis and mild hydroureteronephrosis → pyelonephritis.

FIG. 26.44 PROGRESSIVE HYDROURETER AND HYDRONEPHROSIS ON INTRAVENOUS PYELOGRAM DURING NORMAL PREGNANCY (HUNDLEY ET AL., 1935)

- Bilateral dilatation visible from 15 weeks, increasing through 39 weeks; predominantly right-sided; both tracts normal by 6 weeks postpartum.

TABLE 26.14 AVERAGE VALUES FOR SERUM CREATININE AND UREA NITROGEN (DAVISON AND LINDHEIMER, 1978)

	Non-pregnant women (mg/dL)	Pregnant women (mg/dL)
Serum creatinine	0.7	0.5
Urea nitrogen	13.0	9.0

- Purpose: shows the fall in creatinine and urea from raised GFR. Trap: a creatinine of 0.7–0.9 mg/dL, normal outside pregnancy, may indicate renal impairment in pregnancy.

Complications Associated With Bacteriuria During Pregnancy

Prematurity and Perinatal Mortality

- Pre-antibiotic era: symptomatic UTI/pyelonephritis → prematurity, low birth weight, fetal death.
- Cochrane 2019: treating ASB in pregnancy reduces pyelonephritis, preterm delivery and low birth weight — the basis for screen-and-treat protocols.

Laboratory Diagnosis

- Urinalysis and dipstick screening have significant false-negative rates → screening culture in the first trimester for all. USPSTF: culture at 12–16 weeks or first visit if later.
- Negative early culture → bacteriuria later is unlikely. Women with rUTI or VUR may benefit from prophylaxis.

Management

- Choose with maternal and fetal toxicity in mind; expanded volume, fetal distribution, raised renal blood flow and GFR may lower serum drug levels. Uropathogens are the same as in non-pregnant women.

ANTIBIOTICS IN PREGNANCY: SAFE VERSUS AVOID

Safe / preferred	Avoid or restrict
Aminopenicillins, cephalosporins — safe throughout	Fluoroquinolones — immature cartilage

Safe / preferred	Avoid or restrict
Nitrofurantoin — alternative in penicillin allergy; avoid near term (haemolysis in G6PD-deficient fetus/neonate)	Trimethoprim — folate antagonism; avoid especially in first trimester
Fosfomycin — lower UTI	Chloramphenicol — "grey baby" syndrome (neonate cannot metabolise/excrete it)
	Tetracycline — fetal malformation, maternal hepatic decompensation

- ACOG 2017: sulfonamides and nitrofurantoin remain first-line in the second and third trimesters; in the first trimester only when no suitable alternative exists.

TABLE 26.15 COMMON ANTIBIOTICS AND PREGNANCY CONSIDERATIONS (MODIFIED FROM SCHAEFFER, 2002; BADALATO AND SHAH, AUA CORE CURRICULUM 2022)

Agent	FDA category	Risk	Notes
Penicillins	B	Low; commonly used	—
Cephalosporins	B	Low; commonly used	Limited data on ceftazidime, cefepime
Monobactams	B	Low; limited data	—
Clindamycin	B	Low; commonly used	—
Macrolides	B	Low	Azithromycin commonest
Nitrofurantoin	B	Conflicting; commonly used	Contraindicated at/near term (neonatal haemolysis); not for pyelonephritis (poor tissue levels); avoid in G6PD deficiency
Fosfomycin	B	Low; limited data	Lower UTI only
Vancomycin	—	Limited data; likely safe	—
Daptomycin	B	Very limited data	—
Gentamicin	C	Conflicting; commonly used	Nephro- and ototoxicity; useful for pyelonephritis
Carbapenems	C imipenem; B meropenem	Very limited data	—
TMP-SMX	C	—	Avoid in first trimester (neural tube defects via folate); theoretical hyperbilirubinaemia near term — caution late third trimester
Linezolid	C	Very limited data	—
Fluoroquinolones	C	Conflicting; avoid	Teratogenicity risk
Tetracyclines	D	High risk; avoid	Tooth and bone discolouration (tetracycline; doxycycline less clearly, but avoided)

- FDA categories: **B** — no risk in animals, no adequate human studies; **C** — animal fetal harm, no controlled human studies; **D** — human fetal risk, benefit may justify use. **Phased out from 2015** but still on labels.
- Purpose: **agent selection by trimester and indication**. Pattern: **β-lactams are the backbone; trimester matters for TMP-SMX (early) and nitrofurantoin (term)**. Traps: **nitrofurantoin is category B yet contraindicated near term; gentamicin is category C yet used for pyelonephritis; meropenem B but imipenem C**.

- **After one pyelonephritis or two cystitis episodes → daily suppression with nitrofurantoin or cephalexin** (stop nitrofurantoin near delivery); **follow-up cultures**; positive → investigate.
- **Cochrane 2015 (rUTI prevention in pregnancy): close surveillance alone was as effective as daily nitrofurantoin plus surveillance.**
- **Single-dose vs 4–7-day ASB treatment (systematic review): longer courses cure more** (and possibly reduce low birth weight); no difference in pyelonephritis or preterm birth → **use a standard-length course.**
- **Pyelonephritis in pregnancy: admit, IV antibiotics until improvement; β -lactams (category B) commonly used; carbapenems for severe or MDR cases; tailor to culture, then oral. No improvement by 72 h → renal ultrasound (stone, obstruction, abscess).**

KEY POINTS: BACTERIURIA IN PREGNANCY

- **Screen by urine culture at the first prenatal visit or 12–16 weeks.**
- **Prevalence unchanged by pregnancy, but spontaneous resolution is unlikely.**
- **Treat all bacteriuria in pregnancy.**
- **Bacteriuria progresses to pyelonephritis more often and causes adverse birth outcomes.**

Concept — *Pregnancy does not make bacteriuria commoner — it makes it more dangerous: progesterone and uterine compression convert a harmless colonised bladder into a stagnant, dilated upper tract, which is why pregnancy is the one population in which ASB is screened for and treated.*

BACTERIURIA AND URINARY TRACT INFECTIONS IN THE ELDERLY

- **The US population aged ≥ 65 is projected nearly to double by 2050 → growing public health importance.**

Epidemiology

- **Bacteriuria rises with age; the female excess narrows in old age (still commoner in women); most elderly bacteriuric patients are asymptomatic.**
- **Prevalence rises with age and comorbidity, exceeding 50% in selected groups (figures in Table 26.2); functional impairment doubles it.**

Screening for Bacteriuria

- **Do not screen** community or long-term care elderly. **No link between ASB and deteriorating renal function; treating ASB does not improve incontinence;** the association of bacteriuria with **reduced survival is not shown to be causal.**

Pathogenesis

- **Multifactorial: declining cell-mediated immunity, incomplete emptying, NGB, perineal soiling from faecal/urinary incontinence, more catheterisation, oestrogen depletion (women).**
- ***E. coli* remains commonest (50–75%), but more *Proteus*, *Klebsiella*, *Enterobacter*, *Serratia*, *Pseudomonas* and enterococci. Enterococci often resolve untreated (colonisation). Polymicrobial bacteriuria commoner.**
- **The shift, polymicrobial pattern and resistance are driven by institutionalisation, hospitalisation, catheterisation and antibiotic use.**

Diagnosis

- **Subtle presentations** (abdominal discomfort, **delirium**); even upper tract infection may lack **fever or leucocytosis**.
- **New localised lower tract symptoms — especially dysuria — remain the key criteria.**
- **Bacteriuria + altered mental state ≠ UTI**; no data show antibiotics improve confusion in bacteriuric elderly.
- **Do not culture for non-specific symptoms (dizziness, confusion), chronic symptoms (incontinence, frequency) or asymptomatic patients.**
- **Confusion without urinary symptoms → hydrate and reassess**; test urine **only if altered mental state or change in urine persists after fluids**, or in dementia unresponsive to other measures.
- Meta-analysis: **delirium is associated with UTI but not with ASB**; strongest effect modifiers **retention, dehydration, catheters**; whether treating bacteriuria helps delirium is **uncertain**.
- **Never culture to document clearance.**
- **Thresholds: $\geq 10^5$ CFU/mL with symptoms; $\geq 10^2$ CFU/mL in catheterised specimens.**
- **Dipstick is better at ruling out than ruling in** (moderate sensitivity, lower specificity). **Pyuria is common and a poor predictor of bacteriuria in the elderly** — many women with pyuria have no bacteriuria — and pyuria alone is not an indication for treatment; **absence of pyuria reliably predicts absence of bacteriuria.**
- **Catheterised specimen may be the only clean sample** in institutionalised or dexterity-limited patients.
- **Urological evaluation** (creatinine, CT, ultrasound, **urodynamic studies [UDS]**, cystoscopy) for **renal dysfunction, calculi, hydronephrosis, retention, NGB**; timing by clinical setting.
- **Prompt therapy needed** when bacteriuria → UTI with **structural abnormality (obstruction/ hydronephrosis)**, severe DM, or urea-splitters (*Proteus*, *Klebsiella*) → **infection stones and renal damage.**
- **Sepsis increasingly common in the elderly: catheters and invasive devices, prostheses, chemotherapy, corticosteroids**, and longer survival with metabolic, neoplastic and immune disorders.

Management

- **Randomised trials in nursing-home residents: treating ASB gives no benefit, only adverse effects, resistant reinfection and cost → do not treat ASB in long-term care.**
- **IDSA 2019 recommends against screening for or treating ASB in functionally impaired community-dwelling older people and in long-term care residents.**
- **McGeer surveillance criteria** (Table 26.16) standardise UTI diagnosis in aged-care residents; **limited in cognitive impairment.**
- **Mild LUTS → await culture**; start empirical antibiotics only if immediate treatment needed.
- **Duration:** traditionally 7 days; **Cochrane: short (3–6 days) = long (7–14 days) for uncomplicated symptomatic UTI**; short courses better than **single dose** for persistence; **fever or systemic infection → 10–14 days. Goal: eliminate symptoms, not sterilise urine.**
- **Greater drug toxicity:** impaired metabolism/excretion → high serum levels and renal damage; interactions; **narrow therapeutic margin → dose and monitor carefully. Fluoroquinolones: tendon rupture risk greatest over 60** (boxed warning) and **CNS effects.**

TABLE 26.16 SURVEILLANCE DEFINITIONS FOR URINARY TRACT INFECTIONS — REVISED MCGEER CRITERIA (STONE ET AL., 2012; RYAN ET AL., 2018)

	No indwelling catheter (criteria 1 AND 2)	Indwelling catheter (criteria 1 AND 2)
1. Clinical (≥ 1)	(a) Acute dysuria, or acute pain/swelling/tenderness of testis, epididymis or prostate; (b) fever or leucocytosis + ≥ 1 localising sign: acute CVA pain/tenderness, suprapubic pain, gross haematuria, new/marked increase in incontinence, urgency or frequency; (c) no fever/leucocytosis $\rightarrow \geq 2$ of: suprapubic pain, gross haematuria, new/increased incontinence, urgency, frequency	(a) Fever, rigors or new hypotension without another source; (b) acute mental or functional decline without alternative diagnosis + leucocytosis; (c) new suprapubic or CVA pain/tenderness; (d) purulent pericatheter discharge, or acute testicular/epididymal/prostatic pain, swelling or tenderness
2. Microbiological	$\geq 10^5$ CFU/mL of ≤ 2 species (voided), or $\geq 10^2$ CFU/mL of any number (in-and-out catheter)	$\geq 10^5$ CFU/mL of any organism(s) from catheter specimen

- UTI can be diagnosed without localising symptoms if blood and urine isolates match and no other source exists.
- Purpose: stops bacteriuria alone being labelled UTI in residents. Pattern: non-catheterised residents need localising signs; catheterised residents can qualify on systemic signs. Traps: confusion counts only with leucocytosis and only in catheterised residents; voided specimens tolerate at most two species.

KEY POINTS: BACTERIURIA IN THE ELDERLY

- Do not screen; do not treat ASB — no link to uncomplicated UTI or renal deterioration.
- UTI may present subtly; a high index of suspicion is needed.
- Treatment must allow for age-related physiology and comorbidity.
- Never culture to document clearance.

Concept — In the elderly *bacteriuria is the background, not the signal*: because a positive culture is expected, diagnosis must rest on new localising symptoms (or, with a catheter, systemic signs without another source), and non-specific decline should be met with fluids and reassessment before antibiotics.

CATHETER-ASSOCIATED URINARY TRACT INFECTION

- **Definition (2015 guidelines):** UTI after an indwelling catheter has been in place >2 days, with ≥ 1 symptom (suprapubic or CVA tenderness, frequency, urgency, dysuria, fever $>38^\circ\text{C}$) and a single organism $\geq 10^5$ CFU/mL.
- Catheter-associated bacteriuria is the commonest hospital-acquired infection and inevitable — about 5–10% per catheter-day (text gives about 10% in one place and 5–10% in another). Intermittent catheterisation: 1–3% per catheterisation.
- Risk factors: duration, female sex, no systemic antibiotics, breaches of catheter care.
- Mostly asymptomatic: short term — only a minority of bacteriuric episodes are symptomatic; long term — about 1 febrile episode per 100 catheter-days.
- Cost is substantial and higher for resistant Gram-negatives (parenteral therapy). 2008: US Centers for Medicare and Medicaid Services (CMS) stopped reimbursing CAUTI costs.
- Perioperative catheter indications: intraoperative output monitoring, prolonged surgery, urological surgery — and limit duration.
- Bundled "6-C" approach (Mayo Clinic; large fall in CAUTI):

THE 6-C CAUTI BUNDLE

C	Action
Consider	Appropriate placement; daily review of need
Connect	—
Clean	Catheter care
Closed	Maintain closed system
Call	Irrigation only when necessary
Culture	Only when indicated

Pathogenesis

- **Entry:** at insertion (**urethral bacteria pushed in**, poor technique); thereafter **periurethral** (extraluminal — **predominant in women**, most had prior urethral colonisation with the infecting strain) or **intraluminal** (from the **drainage bag** — commoner when **catheterised patients are clustered**).

ROUTES OF CATHETER-ASSOCIATED INFECTION

Extraluminal (periurethral)	Intraluminal
Ascent along outer surface from meatus	Ascent from contaminated drainage bag/tubing
Commonest in women	Cross-infection among clustered catheterised patients

- **Two populations: planktonic (urine) and biofilm (catheter surface).** Biofilm — organisms in matrix of bacterial and host products → **encrustation**; **shields bacteria from antibiotics → CAUTI and resistance**; **forms within minutes**, grows with dwell time.
- ***Pseudomonas* and *Proteus* are adept biofilm formers** → prevalent here. ***E. coli* still commonest**, with *Pseudomonas*, *Proteus*, *Enterococcus*. **Long-term (>30 days): polymicrobial, often 4–5 organisms**; **populations dynamic**.

Clinical Presentation

- **Suprapubic discomfort, fever, chills or flank pain** suggest symptomatic UTI.

Laboratory Diagnosis

- **$\geq 10^2$ CFU/mL is significant** — even low counts progress to $\geq 10^5$ in almost all. **Pyuria does not discriminate infection** in catheterised patients.

Management

- **Aseptic insertion and a closed, dependent drainage system are essential.**
- **Clean the catheter–meatal junction daily with water**; **avoid antimicrobial agents** (select resistant organisms such as *Pseudomonas*).
- **Silicone = latex** for CAUTI rates.

CATHETER MODIFICATIONS

Catheter modification	Evidence
Silver oxide/alloy catheter; hydrogen peroxide in bag	Mixed; silver alloy reduces bacteriuria/CAUTI in short-term hospital catheterisation (Cochrane)
Nitrofurazone-impregnated	Small CAUTI reduction but more discomfort and cost (Cochrane)

- **Short-term catheter about to be removed (≤ 3 days) in an asymptomatic patient → dipstick can exclude bacteriuria.**

- **Systemic antibiotics during catheterisation:** delay bacteriuria only **3–4 days**, then rates equalise, with resistance and side effects → **not recommended**.
- **Bacterial interference in SCI:** colonisation with a **non-virulent strain reduced symptomatic UTI and improved QoL**.
- **Treat only symptomatic (e.g. febrile) CAUTI** → **culture first** → **stop antibiotics within 48 h of resolution** → **change the catheter** (encrustation shelters bacteria).

KEY POINTS: CATHETER-ASSOCIATED BACTERIURIA

- Aseptic insertion and closed, dependent drainage minimise bacteriuria.
- Catheter-associated bacteriuria is inevitable.
- Only symptomatic CAUTI needs treatment.

Concept — *The catheter, not the bladder, is the infected organ: a biofilm forms within minutes and is beyond the reach of antibiotics, so prevention is about minimising catheter-days and keeping the system closed, and treatment of a symptomatic episode includes removing the colonised surface by changing the catheter.*

MANAGEMENT OF URINARY TRACT INFECTIONS IN PATIENTS WITH SPINAL CORD INJURY

- All UTIs in SCI are complicated. No bacteriuric group compares in severity and morbidity. Early catheterisation is near-universal (**overactivity or flaccidity**); many develop **ureterectasis, hydronephrosis, VUR, calculi**. Bacteriological and urodynamic advances have **greatly reduced morbidity and mortality**.

Epidemiology

- Before CIC and modern antibiotics, UTI and renal failure caused most SCI deaths; both have fallen since.
- UTI is among the commonest urological complications — about a third bacteriuric at any time; **almost all eventually**; significant bacteriuria and febrile UTI recur several times per person-year on intermittent or condom drainage. **UTI is the commonest cause of fever in SCI**.
- Risk factors (National Institute on Disability and Rehabilitation Research Consensus Conference [NIDRRCC], 1992): impaired voiding, overdistension, raised intravesical pressure, obstruction, VUR, instrumentation, stones; also low fluid intake, poor hygiene, perineal colonisation, pressure sores and local trauma, reduced host defence.

Pathogenesis

- Bladder management method profoundly affects UTI incidence.

SCI: BLADDER MANAGEMENT METHOD AND INFECTION

Method	Bacteriuria / outcome
Indwelling urethral catheter	Most likely to cause UTI; most bacteriuric by 30 days; all cultures positive in one series
Suprapubic catheter	Eventually equivalent infection rate, but onset of bacteriuria delayed
Condom drainage	Bacteriuria in most to almost all

Method	Bacteriuria / outcome
Clean intermittent catheterisation (CIC) (Lapides, 1972)	Lowest risk of long-term complications

- **CIC** (never rigorously compared head-to-head) **keeps intravesical pressure low, reduces stones**, and reduces **UTI, fever, bacteraemia, epididymitis and prostatitis**; long-term follow-up shows **fewer complications than urethral catheterisation and no more than any other method**.
- **Sterile vs clean ("no-touch") CIC: conflicting**; an **introducer-tip** catheter bypassing the distal 1.5 cm of urethra reduced UTI. **Low-friction/hydrophilic catheters** may be **less traumatic** and **delay the first UTI**, but an effect on bacteriuria/UTI is **unproven**.

Clinical Presentation

- **Most bacteriuric SCI patients are asymptomatic → do not treat ASB; do not screen** asymptomatic patients on indwelling, condom or intermittent catheters.
- **No standard UTI definition in SCI. Sensory loss → frequency, urgency, dysuria usually absent**. Instead: **flank, back or abdominal discomfort; leakage between catheterisations; increased spasticity; malaise, lethargy; cloudy, malodorous urine; fever** (commonest cause).

Bacteriology and Laboratory Diagnosis

- **Bacteriuria and pyuria; pyuria is not diagnostic** (catheter irritation).

NIDRRCC CRITERIA FOR SIGNIFICANT BACTERIURIA IN SCI

Drainage	Significant count
Indwelling urethral or suprapubic catheter	Any detectable bacteria (low counts rise rapidly to $\geq 10^5$)
Intermittent catheterisation	$\geq 10^2$ CFU/mL
Catheter-free men (clean voided)	$\geq 10^4$ CFU/mL

- Organisms differ in **cause, complexity and susceptibility**, by catheter type and duration: *E. coli* frequent; **enterococci, *P. mirabilis*, *Pseudomonas* commoner than in intact cords**; also *Klebsiella*, *Serratia*, *Staphylococcus*, *Candida*. **Short-term catheter: single organism; >1 month: polymicrobial**.

Management

- **Culture before empirical therapy** (diverse, resistant flora).
- **Afebrile: β -lactams, TMP-SMX or nitrofurantoin pending culture; oral fluoroquinolone second-line**. **Change an indwelling catheter** (drainage, remove encrusted foci).
- **Duration (no consensus; expert):**

SCI: DURATION OF THERAPY

UTI type	Duration
New afebrile	5–7 days
Recurrent afebrile	7–10 days
Febrile	14 days

- **Fever or chills → admit; parenteral aminoglycoside + penicillin or 3rd-generation cephalosporin**; involve an antimicrobial expert, especially with recurrences.
- **No improvement in 24–48 h → repeat culture, adjust therapy, image for obstruction, stones, abscess**.
- **Post-treatment cultures unnecessary** (recolonisation common) — **except urea-splitters, which must be shown eradicated**.

- **Bladder irrigation** (saline/antibiotic) traditionally **not recommended**; a small series of **gentamicin instillations** in NGB showed **fewer symptomatic UTIs, fewer oral courses and less MDR**.
- **Methenamine ineffective in NGB** (Cochrane). **AP not supported in SCI/NGB** — no reduction in symptomatic UTI and **about a doubling of resistant bacteria**.
- **Recurrent symptomatic UTI** → **imaging, UDS, review of bladder management** (catheter drainage, CIC technique and frequency, voiding schedule); **lowering storage pressure may reduce symptomatic UTI**.
- **Bladder cancer: catheterisation >10 years raises risk**; in large SCI databases a small proportion developed bladder cancer, and **duration of catheterisation was the only significant predictor**; proposed carcinogens — **chronic infection/inflammation and nitrosamines in infected urine**.

KEY POINTS: MANAGEMENT OF UTIS IN PATIENTS WITH SCI

- Most bacteriuric SCI patients are **asymptomatic**; **treat only symptomatic patients**.
- Symptoms: **fever; flank, back or abdominal discomfort; leakage between catheterisations; spasticity; malaise, lethargy; cloudy, malodorous urine**.
- **Culture before empirical therapy** — diverse, resistant flora.
- **CIC carries the lowest risk of long-term complications**.
- **Chronic infection can be carcinogenic**.

Concept — *In SCI the bladder-management method is the main modifiable determinant of infection: low-pressure, intermittent emptying (CIC) prevents the stasis, high pressure and reflux that make infection dangerous, whereas antibiotics can neither prevent colonisation nor safely eradicate it — so they are reserved for symptoms, which must be recognised through non-classical signs.*

KIDNEY TRANSPLANT RECIPIENTS

- **No consensus on treating ASB after renal transplant.**

RENAL TRANSPLANT: GUIDANCE ON ASB

Source	Recommendation
IDSA 2005	No recommendation (insufficient evidence)
IDSA 2019	Do not screen or treat ASB >1 month after transplant
American Society of Transplantation (AST) 2019	Do not routinely culture or treat ASB (even MDR) >2 months after transplant; if screened and positive → repeat culture; observe if cleared or a different organism; consider treatment with unexplained creatinine rise; do not treat asymptomatic candiduria except before an invasive urological procedure
Cochrane 2018	ASB in about a fifth to a half of recipients; insufficient evidence for routine treatment ; unclear whether to screen

- **Immunosuppression (especially early months)** → **recognising and treating UTI is paramount and may be critical for graft survival**. Recipients have **higher VUR risk**.
- **Risk factors: female sex, DM, cadaveric graft, two episodes of ASB, prolonged pre-transplant haemodialysis**.
- **Duration (early, first 6 months): uncomplicated UTI 5–10 days — not single-dose or 3-day; pyelonephritis/complicated UTI 14–21 days**.
- **Recurrent UTI** — **AST stepwise evaluation: PVR + upper tract imaging (ultrasound/CT) + evaluation for prostatitis** → **correct structural abnormality** → if none, **VCUG for VUR** (surgery or endoscopic injection if

present) → if no VUR, **UDS** → **medical management of bladder dysfunction; surgery for outflow obstruction.**

- **Ureteric stents** at transplantation **reduce leak and obstruction** but **raise UTI risk by about half** (Cochrane 2013); **removal within 2 weeks lowers risk.** **AST: remove catheters and stents within 4 weeks; consider early stent removal in severe UTI.**

Concept — *After transplantation the balance tips with time: **early on, graft and host are vulnerable** — longer courses, low threshold, early removal of hardware — **but once the graft is established (beyond 1–2 months), ASB behaves as in other hosts and treating it only selects resistance.***

ANTIMICROBIAL PROPHYLAXIS FOR COMMON UROLOGIC PROCEDURES

Principles of Surgical Antimicrobial Prophylaxis

- **AUA Best Practice Statement (BPS): 2008 (validated 2011) → 2019 current.**
- **Prophylaxis (prevent postprocedural local/systemic infection) vs treatment (eradicate suspected or documented infection).**
- **2019 BPS: consider periprocedural AP for all urological procedures that breach a tissue barrier.**

PRINCIPLES OF SURGICAL PROPHYLAXIS

Principle	Rule
Timing	Within 60 min of the procedure; vancomycin and fluoroquinolones within 120 min
Duration	Single dose for most uncomplicated procedures; maintain levels through the procedure; ≤24 h only in special circumstances
Agent	Target likely local organisms; guided by local antibiogram , cost and safety
Non-traditional prophylaxis	Periprocedural treatment of the urinary tract before manipulating colonised hardware (stents, catheters)

- Two key questions: **can the host respond to bacteriuria/bacteraemia**, and **what are the consequences of infection** — plus **likelihood of invasion at the surgical site.**
- **Fluoroquinolone boxed warning** must be weighed when choosing AP. **AP has its own morbidity** — rash, GI upset, **allergic nephritis, anaphylaxis.**

BOX 26.13 HOST FACTORS THAT INCREASE THE RISK OF INFECTION (CRUSE, 1992; MANGRAM ET AL., 1999)

- **Advanced age; anatomical anomalies; poor nutrition; smoking; chronic corticosteroids; immunodeficiency** (e.g. untreated HIV)
- **Chronic indwelling hardware; infected endogenous/exogenous material** (e.g. stones); **distant coexistent infection; prolonged hospitalisation**
- The last four act by **raising local bacterial load or altering flora**; the first six by **impairing host response.**

ASB AND FUNGURIA BEFORE PROCEDURES: GUIDELINE BODIES

Body	Position
AUA BPS 2019	ASB and asymptomatic funguria need no treatment before procedures that do not breach the urothelium; low-risk patients may not need AP for such procedures (except pregnancy); single-dose antifungal prophylaxis for endoscopic, robotic or open surgery on the urinary tract; no repeat urinalysis to prove clearance
EAU 2017 (updated 2022)	ASB not a risk factor for procedures not entering the urinary tract; a definite risk factor for procedures breaching the mucosa (especially endoscopic) → culture and treat ASB preoperatively; do not treat before arthroplasty

Body	Position
IDSA 2016 (candidiasis)	Treat all candiduria before any urological procedure (oral fluconazole or IV amphotericin B for several days before and after)
AUA BPS (disagrees with IDSA)	Antifungal prophylaxis only for high-pressure-irrigant endoscopy (transurethral resection of bladder tumour [TURBT] or prostate [TURP], ablative outlet procedures, therapeutic ureteroscopy) and entry into the urinary tract ; full treatment for symptomatic fungal UTI at stent/tube change; neutropenia + funguria + obstruction → longer course; fungal balls (mycetoma) → preoperative fungal culture and 5–7 days periprocedural antifungal

- No benefit from treating ASB before arthroplasty or open-heart surgery.
- Consider AP at voiding trials and catheter, drain, stent or nephrostomy removal, especially with risk factors — a meta-analysis showed fewer symptomatic UTIs with AP at catheter removal.

AUA BPS: PROPHYLAXIS BY WOUND CLASS

Class	AP
I — clean	Not necessary, except groin or perineal incisions
II — clean-contaminated (incl. prostate biopsy)	Recommended ; transurethral and therapeutic upper tract endoscopy → single dose covering Gram-negatives and enterococci (1st/2nd-generation cephalosporin or TMP-SMX cited)
III — contaminated	Strongly recommended ; colorectal → aerobic + anaerobic cover (1st-generation cephalosporin + metronidazole)
IV — dirty/infected	Empirical treatment until cultures return
GU prosthetics	Not specifically addressed; cover skin flora

TABLE 26.17 RECOMMENDED ANTIMICROBIAL PROPHYLAXIS FOR UROLOGIC PROCEDURES (LIGHTNER ET AL., AUA 2019)

Procedure (wound class)	Likely organisms	AP indicated	Agent(s) of choice	Alternative(s)	Duration
LOWER TRACT					
Cystourethroscopy with minor manipulation (mucosal break, biopsy, fulguration) — clean-contaminated	Gram-negative rods (GNR), rarely enterococci	Uncertain ; consider host factors; invasiveness ↑ surgical site infection (SSI) risk ; none needed if preprocedure culture sterile	TMP-SMX, amoxicillin-clavulanate	1st/2nd-gen cephalosporin; aminoglycoside (aztreonam) + ampicillin	Single dose
Transurethral (TURP, TURBT, laser enucleation/ablation) — clean-contaminated	GNR, rarely enterococci	All	Cefazolin, TMP-SMX	Amoxicillin-clavulanate; aminoglycoside (aztreonam) + ampicillin	Single dose
Prostate brachytherapy / cryotherapy — clean-contaminated	<i>S. aureus</i> , skin; GNR	All	Cefazolin	Clindamycin	Single dose

Procedure (wound class)	Likely organisms	AP indicated	Agent(s) of choice	Alternative(s)	Duration
Transrectal prostate biopsy — contaminated	GNR, anaerobes; MDR cover if antibiotics within 6 months, international travel, health care worker	All	Fluoroquinolone; 1st/2nd/3rd-gen cephalosporin (ceftriaxone commonly) ± aminoglycoside	Aztreonam; may need infectious disease consultation	Single dose
UPPER TRACT					
Percutaneous renal surgery (percutaneous nephrolithotomy [PCNL]) — clean-contaminated	GNR, rarely enterococci; skin, <i>S. aureus</i>	All	1st/2nd-gen cephalosporin; aminoglycoside (aztreonam) + metronidazole; or clindamycin	Ampicillin-sulbactam	≤24 h
Ureteroscopy, all indications — clean-contaminated	GNR, rarely enterococci	All; undetermined benefit for uncomplicated diagnostic	TMP-SMX; 1st/2nd-gen cephalosporin	Aminoglycoside (aztreonam) + ampicillin; amoxicillin-clavulanate	Single dose
OPEN / LAPAROSCOPIC / ROBOTIC					
Not entering urinary tract (adrenalectomy, lymphadenectomy) — clean	<i>S. aureus</i> , skin	Consider; may not be required	Cefazolin	Clindamycin	Single dose
Penile surgery (circumcision, biopsy) — clean-contaminated	<i>S. aureus</i>	Likely not required	—	—	—
Urethroplasty / anterior urethral reconstruction — controlled entry	GNR, rarely enterococci, <i>S. aureus</i>	Likely required	Cefazolin	Cefoxitin, cefotetan, ampicillin-sulbactam	Single dose
Controlled entry into urinary tract (nephrectomy, pyeloplasty, radical prostatectomy, partial cystectomy) — clean-contaminated	GNR (<i>E. coli</i>), rarely enterococci	All	Cefazolin, TMP-SMX	Ampicillin-sulbactam; aminoglycoside (aztreonam) + metronidazole or clindamycin	Single dose
Small bowel (ileal conduit) — clean-contaminated	Skin, <i>S. aureus</i> , GNR, rarely enterococci	All	Cefazolin	Clindamycin + aminoglycoside; cefuroxime; aminopenicillin/β-lactamase inhibitor + metronidazole	Single dose

Procedure (wound class)	Likely organisms	AP indicated	Agent(s) of choice	Alternative(s)	Duration
Large bowel (colon conduit) — clean-contaminated	GNR, anaerobes	All	Cefazolin + metronidazole; cefoxitin; cefotetan; ceftriaxone + metronidazole; ertapenem — plus mechanical bowel preparation and oral neomycin + erythromycin or neomycin + metronidazole	Ampicillin-sulbactam, ticarcillin-clavulanate, piperacillin-tazobactam	Single parenteral dose
Implanted prostheses (artificial urinary sphincter [AUS], inflatable penile prosthesis [IPP], sacral neuromodulator) — clean	GNR, <i>S. aureus</i>; increasing anaerobes and fungi	All	Aminoglycoside (aztreonam) + 1st/2nd-gen cephalosporin or vancomycin	Aminopenicillin/β-lactamase inhibitor (ampicillin-sulbactam, ticarcillin, tazobactam combinations)	≤24 h
Inguinal and scrotal (orchidectomy, vasectomy/reversal, varicocelectomy, hydrocelectomy) — clean	GNR, <i>S. aureus</i>	Increased-risk cases	Cefazolin	Ampicillin-sulbactam	Single dose
Vaginal surgery (slings, fistula repair, urethral diverticulectomy) — clean-contaminated	<i>S. aureus</i> , streptococci, enterococci, vaginal anaerobes , skin	All	1st/2nd-gen cephalosporin (cefoxitin/cefotetan better anaerobic cover; cefazolin equivalent for slings)	Ampicillin-sulbactam; aminoglycoside (aztreonam) + metronidazole or clindamycin	Single dose
OTHER					
Shock wave lithotripsy (SWL) — clean	GNR, rarely enterococci	Only if risk factors	TMP-SMX; 1st/2nd-gen cephalosporin; amoxicillin-clavulanate	Ampicillin + aminoglycoside (aztreonam); clindamycin	Single dose

- Footnotes: **GU organisms** = *E. coli*, *Proteus*, *Klebsiella*, enterococci; **intestinal** = *E. coli*, *Klebsiella*, *Enterobacter*, *Serratia*, *Proteus*, enterococci, anaerobes; **skin** = *S. aureus*, CoNS, group A streptococci. **Duration** may instead be a full culture-directed course for documented infection (treatment, not prophylaxis). Aztreonam substitutes for aminoglycosides in renal insufficiency. "Transurethral" includes any biopsy, resection, fulguration, foreign-body removal, urethral dilatation/urethrotomy, ureteric instrumentation incl. catheterisation or stent placement/removal. Penicillin allergy: clindamycin, or aminoglycoside + metronidazole/clindamycin as general alternatives. Colorectal: add oral neomycin + erythromycin base or metronidazole. Routine vancomycin AP is not recommended — it is less effective than β-lactams against methicillin-sensitive *S. aureus*.
- Purpose: matches procedure to flora, indication, agent and duration. Pattern: single dose is the default; only PCNL and prostheses allow up to 24 h; bowel and transrectal procedures add anaerobic/MDR

cover. Traps: **SWL needs AP only with risk factors; penile surgery likely needs none; diagnostic ureteroscopy has undetermined benefit; vancomycin is not a routine prophylactic.**

Urethral Catheterisation and Removal

- AP depends on **health, sex, living circumstances and indication**. **One-off catheterisation: infection 1–2% in healthy domiciliary women, much higher in hospital** → with risk factors (Box 26.13), oral TMP-SMX lowers post-procedural infection.
- **Evidence on AP during short-term catheterisation conflicts:**

PROPHYLAXIS DURING SHORT-TERM CATHETERISATION: CONFLICTING REVIEWS

Review	Finding
Cochrane 2013 (short-term ≤14 days, surgical patients; AP 3 days postoperatively or day 2 to removal)	Less bacteriuria, pyuria, fever, Gram-negative cultures — interpret with caution (limited evidence, adverse effects)
Cochrane 2005 (AP until removal or first 3 postoperative days)	No reduction in bacteriuria or infection

- **Indwelling catheter: bacteriuria 5–10% per catheter-day; routine AP during catheterisation not recommended** (rapid resistance).
- **Long-term intermittent catheterisation: no routine prophylaxis** (comparable UTI rates, low overall risk).
- **After removal:** a quarter of asymptomatic bacteriuric women catheterised 4–6 days developed UTI within 14 days; **1-day TMP-SMX = 10-day course**; no equivalent data in men.
- **Chronic catheter changes: AP not indicated** (occasional low-grade bacteraemia, no systemic sequelae).

Urodynamics

- **Minimally traumatic** — low infection risk in normal hosts. **AUA: no AP for UDS in "index patients" (no risk factors).**
- Systematic review of RCTs: **AP reduced bacteriuria** (symptomatic UTI not assessed; only a small minority with ASB become symptomatic within a week). **Cochrane 2012: no significant reduction in UTI.**
- **Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction (SUFU) 2017 BPS (non-index patients):** screen for UTI symptoms and dipstick all; suspected UTI → postpone and treat.

SUFU 2017: PROPHYLAXIS FOR URODYNAMICS

No AP needed	AP recommended
No GU anomaly; DM; prior GU surgery or prior UTI; postmenopausal ; recent hospitalisation; cardiac valve disease ; nutritional deficiency; obesity	Neurogenic lower urinary tract (LUT) dysfunction; raised PVR; ASB; immunosuppression (incl. renal transplant); age >70; indwelling catheter, external collection device or intermittent catheterisation; total joint implant within 2 years or high joint-infection risk

- **Agent: single oral TMP-SMX; alternatives 1st/2nd-gen cephalosporin, amoxicillin-clavulanate, fluoroquinolone, parenteral aminoglycoside + ampicillin.** The only nitrofurantoin study (UDS + cystoscopy) showed **no reduction in bacteriuria**; the authors nevertheless use **nitrofurantoin as the alternative to TMP-SMX** (fluoroquinolone warnings). Mostly **Level IV evidence**; judgement prevails.
- **EAU: no AP for cystoscopy or UDS.**
- **Trap: DM, prior UTI, valve disease and obesity do not warrant AP for UDS; age >70, raised PVR and ASB do.**

Transrectal Ultrasound-Guided Prostate Biopsy

- **Infectious complications rising** (overall up to several per cent; **sepsis 0.3–3.1%**). **Commonest cause: fluoroquinolone-resistant *E. coli*. Commonest risk factor: antibiotics within 6 months.**

- **Fluoroquinolone AP reduces infection substantially vs placebo, but fluoroquinolone (FQ)-resistant rectal colonisation is common (up to about a fifth), most resistant strains are more virulent, yet about 90% of urologists still use empirical fluoroquinolones.**

TRANSRECTAL ULTRASOUND (TRUS) BIOPSY PROPHYLAXIS STRATEGIES

Strategy	Evidence	Issues
Empirical fluoroquinolone	Standard; single dose = 3 days	Rising FQ-resistant infection
Augmented (aminoglycoside + fluoroquinolone)	Meta-analysis of RCTs: fewer UTIs and admissions	Resistance will evolve
Targeted (rectal swab culture)	Meta-analysis: more effective than empirical; one study eliminated infections and saved cost (about 38 swabs per infection prevented); a very large US series found no difference in sepsis	Cost, extra visits, special media, laboratory
Fosfomycin	As effective as fluoroquinolone	—

- **Minimum 1 day of AP.**
- **AUA White Paper update (2017):** assess risk factors for resistance; urine culture and treat infection in men with LUTS; fluoroquinolone within 1 h; rectal swab → culture-specific agent (gentamicin, ceftriaxone commonest); use alternatives if local *E. coli* fluoroquinolone resistance >20%.
- **EAU:** AP plus povidone-iodine rectal cleansing.

Concept — *Transrectal biopsy inoculates the patient's own rectal flora into the prostate, so the prophylactic must match that flora — which, after decades of fluoroquinolone use, is increasingly resistant; hence the move from empirical to augmented or swab-targeted prophylaxis.*

Shock Wave Lithotripsy

- **AUA Surgical Management of Stones:** no AP if preoperative urine microscopy is benign. Meta-analysis of RCTs: no reduction in bacteriuria, fever or UTI; prospective series: very low ASB and UTI rates.
- **Recent UTI or infection stones → full treatment course before SWL** (treatment, not prophylaxis). **EAU:** no AP for SWL.

Endoscopic Procedures: Lower Urinary Tract

Simple Cystoscopy (Without Manipulation)

- Older trials (sterile urine, no AP): **UTI in a few per cent** (higher with **prior UTI**; one series much higher); **single-dose TMP or ciprofloxacin reduced infection to 1–5%; no serious systemic infections.**
- Recent evidence: **AP unnecessary for diagnostic cystoscopy** — large series without AP: **few febrile UTIs, all settled on oral antibiotics, none needing admission** (including patients with ASB); meta-analysis: **no difference; ciprofloxacin or fosfomycin not indicated for flexible cystoscopy.**
- **Risk factors:** recent antibiotics or hospitalisation; a **risk-plus-antibiogram strategy reduced AP without more UTIs.**
- **EAU 2017:** strongly recommends against AP for cystourethroscopy (flexible or rigid) — low absolute risk, huge volume, resistance risk.

Transurethral Resection of the Prostate and Bladder (Cystourethroscopy With Manipulation)

- **Higher risk than diagnostic cystoscopy** — mucosal trauma, longer/harder procedures, pressurised irrigation, resection of infected material.
- **TURP (most studied)** — meta-analysis: **AP markedly reduced postoperative bacteriuria and septicaemia;** most effective: **fluoroquinolones, aminoglycosides, cephalosporins, TMP-SMX.** Short courses (2–5 days

while the catheter remains) beat single doses, and continuing the agent briefly with the catheter **does not increase resistance**.

- **TURBT**: no recent trials; TURP data suggest benefit, but **EAU (two RCTs) finds no benefit — AP only in high-risk TURBT**; a prospective study found **no benefit**.
- **Other manipulations (laser prostatectomy, bladder biopsy)**: extrapolated from TURP/TURBT; **office biopsy without AP = theatre biopsy with AP** for UTI and febrile UTI — yet **CMS mandates AP** for cystoscopy with biopsy in theatre, and IDSA and AUA concur. IDSA: **fluoroquinolone, TMP-SMX or cefazolin** for lower tract instrumentation with risk factors.
- **Risk factors**: obesity, DM, smoking, immunosuppression, malnutrition; urological — anomalies, obstruction, stones, indwelling or externalised catheters.
- **Known preoperative UTI → eradicate first** (this is treatment); **operating on unsterilised urine → bacteraemia in about half**.
- **Pressurised-irrigant diagnostic/therapeutic upper tract studies → AP covering uropathogens**.

DIAGNOSTIC CYSTOSCOPY VERSUS TURP

Diagnostic cystoscopy	Transurethral resection (TURP)
No AP (EAU strongly against; recent AUA-aligned evidence)	AP in all (single dose or short course with catheter)
Minimal urothelial injury	Mucosal trauma, pressurised irrigant, infected tissue

Endoscopic Procedures: Upper Urinary Tract

Ureteroscopy

- **Higher risk than cystoscopy** (mucosal trauma, duration, **irrigant pressure**, infected material). **RCT: fluoroquinolone AP reduced post-ureteroscopy bacteriuria** in healthy patients with sterile urine.
- **Suspected infection → culture and full treatment before**; some use **furosemide diuresis** intraoperatively.
- Meta-analysis (ureteroscopic lithotripsy): **no difference in febrile UTI**, but **single preoperative dose lowers bacteriuria and pyuria → single oral preoperative dose recommended**.

Ureteral Stenting

- **Give AP at stent removal**; **culture the stent** in high-risk patients to guide later treatment.
- **Stent colonisation far exceeds urine colonisation**; **dwell time is the main determinant and is not reduced by AP**; urine culture identifies the stent organism in only a small minority.

URINE VERSUS STENT COLONISATION

Urine	Stent
Colonised in a minority	Colonised in about half or more
Poor surrogate for stent organism	Culture it directly when risk is high

Percutaneous Renal Surgery (Prophylaxis Indicated in All Patients)

- For **large stones, pelviureteric junction obstruction, urothelial carcinoma surveillance**. **Pyrexia and bacteraemia are frequent — parenchymal injury, pressurised irrigation, infected stones**; risk rises with procedure duration and irrigant volume. No RCTs.
- **Positive preoperative culture → treat first**; **negative → AP against common uropathogens**.

Open, Laparoscopic and Robotic Surgery

- Wound classes (Table 26.18): **AP for clean-contaminated and contaminated**; **treatment for dirty**. **Laparoscopy lowers SSI** (general surgery data).

- **Clean** — e.g. radical nephrectomy without entering the urinary tract; elective opening of the urinary tract = Class II; entering an infected tract = Class III (higher SSI).
- **Give within 1 h; stop by 24 h** — longer courses show no benefit.
- **First-generation cephalosporins** standard for clean-contaminated (**few allergies, long half-life, cheap**); β -lactam allergy → **vancomycin or clindamycin** (National Surgical Infection Prevention Project [NSIPP] 2004).
- **Intestinal reconstruction** → 1st-generation cephalosporin + metronidazole.
- **Colon/appendix**: oral neomycin + erythromycin or neomycin + metronidazole 18–24 h before + IV cefotetan or cefoxitin 30–60 min before incision (NSIPP, WHO); β -lactam allergy → clindamycin + gentamicin, aztreonam, or ciprofloxacin.
- **Dirty wounds** (all abscesses, traumatic GU perforation) → **broad empirical cover + intraoperative cultures**, then tailor.

TABLE 26.18 WOUND CLASSIFICATIONS (MODIFIED FROM MANGRAM ET AL., 1999)

Class	Definition	Urological examples
I — clean	Uninfected; no entry into respiratory, GI or GU tract	Inguinal/scrotal procedures for non-infective indications; retroperitoneal lymph node dissection (RPLND)
II — clean-contaminated	Controlled entry into respiratory, GI or GU tract; no other contamination	Opening the urinary tract: nephrectomy, cystectomy, prostatectomy, endoscopic procedures
III — contaminated	Infected stone procedures; use of bowel segments	PCNL for struvite/infected stones; transrectal prostate biopsy
IV — dirty	Open trauma, abscesses	Debridement — organisms present before surgery
Prosthesis implantation	—	IPP, AUS: cover skin organisms; MRSA colonisation → many surgeons add vancomycin

- Purpose: class dictates prophylaxis vs treatment. Traps: any endoscopic procedure is Class II; transrectal biopsy is Class III; this table places bowel segments in Class III, whereas Table 26.17 labels bowel procedures clean-contaminated.

Skin Preparation for Surgery

- **Pre-admission bathing, mechanical bowel preparation, hair removal and skin preparation** all help prevent SSI; details in the AUA BPS.

Vaginal Surgery

- No urological RCTs; vaginal hysterectomy data: AP markedly reduces pelvic infection; ≤ 24 h as effective as longer courses.
- AUA BPS (2010, 2019): AP for all vaginal procedures, including midurethral sling (MUS) — expert opinion. MUS studies (uncontrolled series, an RCT stopped early for low event rates, decision analysis) suggest AP may be unnecessary because infection is rare.

Special Considerations

Patients With Risk of Endocarditis

- Risk of infective endocarditis (IE) after urological procedures is low.
- American Heart Association (AHA) 2007: AP solely to prevent IE is not recommended for GU procedures, though GU instrumentation can cause transient enterococcal bacteraemia. American College of Cardiology (ACC)/AHA 2017: no benefit; IE is likelier from random bacteraemia — not even for high-risk patients (prosthetic valve, previous IE, cardiac transplant with valvulopathy).

- **Exception (Class IIb): high-risk cardiac condition + active urinary infection or colonisation before GU manipulation (incl. elective cystoscopy) → treat to sterilise the urine — amoxicillin/ampicillin first line (enterococcus), vancomycin if intolerant, or culture-directed.**

Patients With Indwelling Orthopaedic Hardware

ORTHOPAEDIC HARDWARE: STATEMENTS COMPARED

Statement	Position
AUA/American Academy of Orthopaedic Surgeons (AAOS)/infectious disease specialists advisory 2003	No AP if joint replacement, pins, plates or screws ≥2 years old; <2 years → oral quinolone, or IV ampicillin 2 g + gentamicin 1.5 mg/kg (vancomycin if ampicillin-allergic) 30–60 min before
AAOS 2009	Consider AP for all total joint replacements before any bacteraemia-risk procedure (cost and consequences of infected joints)
AUA BPS 2019	AP to prevent hip/knee infection for high-bacteraemia-risk procedures, joints <2 years, high-risk patients — weak evidence, but potentially catastrophic harm

- Procedures warranting AP for joint seeding: stone manipulation, transmural incision of the urinary tract, upper tract endoscopy, bowel segments, transrectal prostate biopsy.
- Patient factors: recent prosthesis or compromised host with urinary diversion, indwelling stent or catheter, recent retention, or UTI → AP before urinary tract procedures.

TABLE 26.19 ANTIMICROBIAL REGIMENS FOR PATIENTS WITH INDWELLING ORTHOPEDIC HARDWARE (AUA AND AAOS, 2003)

Patient type	Recommendation
Total joint >2 years ago; pins, plates, screws; no host risk factors	Not recommended empirically
Total joint <2 years ago, or aberrant host factor(s)	Oral quinolone, or ampicillin 2 g IV + gentamicin 1.5 mg/kg IV, 30–60 min before; vancomycin 1 g IV over 1–2 h if ampicillin-allergic

- Purpose: time since implantation and host factors decide prophylaxis. Trap: pins, plates and screws alone never need AP; the 2-year cut-off is the pivot.

KEY POINTS: ANTIMICROBIAL PROPHYLAXIS FOR COMMON UROLOGIC PROCEDURES

- AP = antimicrobial before and for a limited time after a procedure to prevent local/systemic infection.
- Procedure type and host defences determine need.
- Special considerations: TRUS biopsy, endocarditis risk with bacteriuria, orthopaedic hardware.

Concept — Prophylaxis is justified by the product of wound contamination and host vulnerability: where neither is raised (diagnostic cystoscopy, SWL with sterile urine, UDS in index patients) the ecological cost outweighs the benefit; where either is (bowel, infected stones, transrectal biopsy, prostheses, immunocompromise) a single, well-timed, flora-matched dose is the default, and anything longer is treatment, not prophylaxis.